

Introduction to Artificial Intelligence for Data Analysis

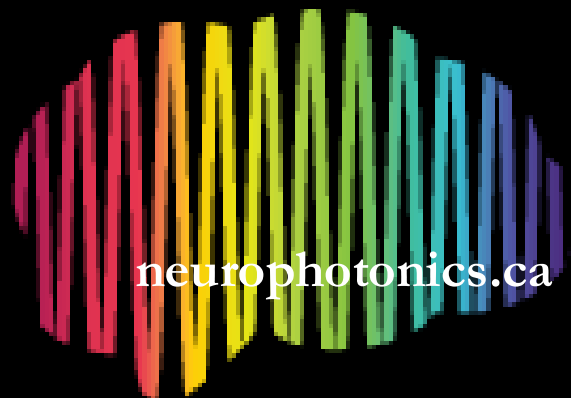
Neurasmus Neurocomputation Workshop 2026

Flavie Lavoie-Cardinal

Canada Research Chair in Intelligent Nanoscopy of Cellular Plasticity

Department of Psychiatry and Neuroscience

CERVO Brain Research Center, Institute for Intelligence and Data



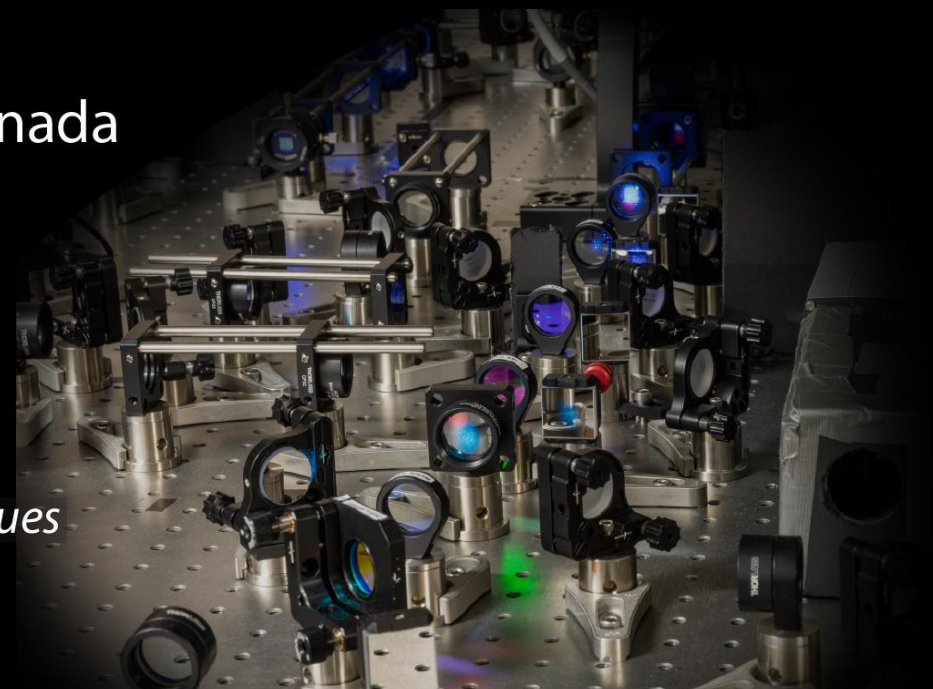
- ❖ International Summer School
Frontiers in Neurophotronics
- ❖ Canadian Neurophotronics
Center

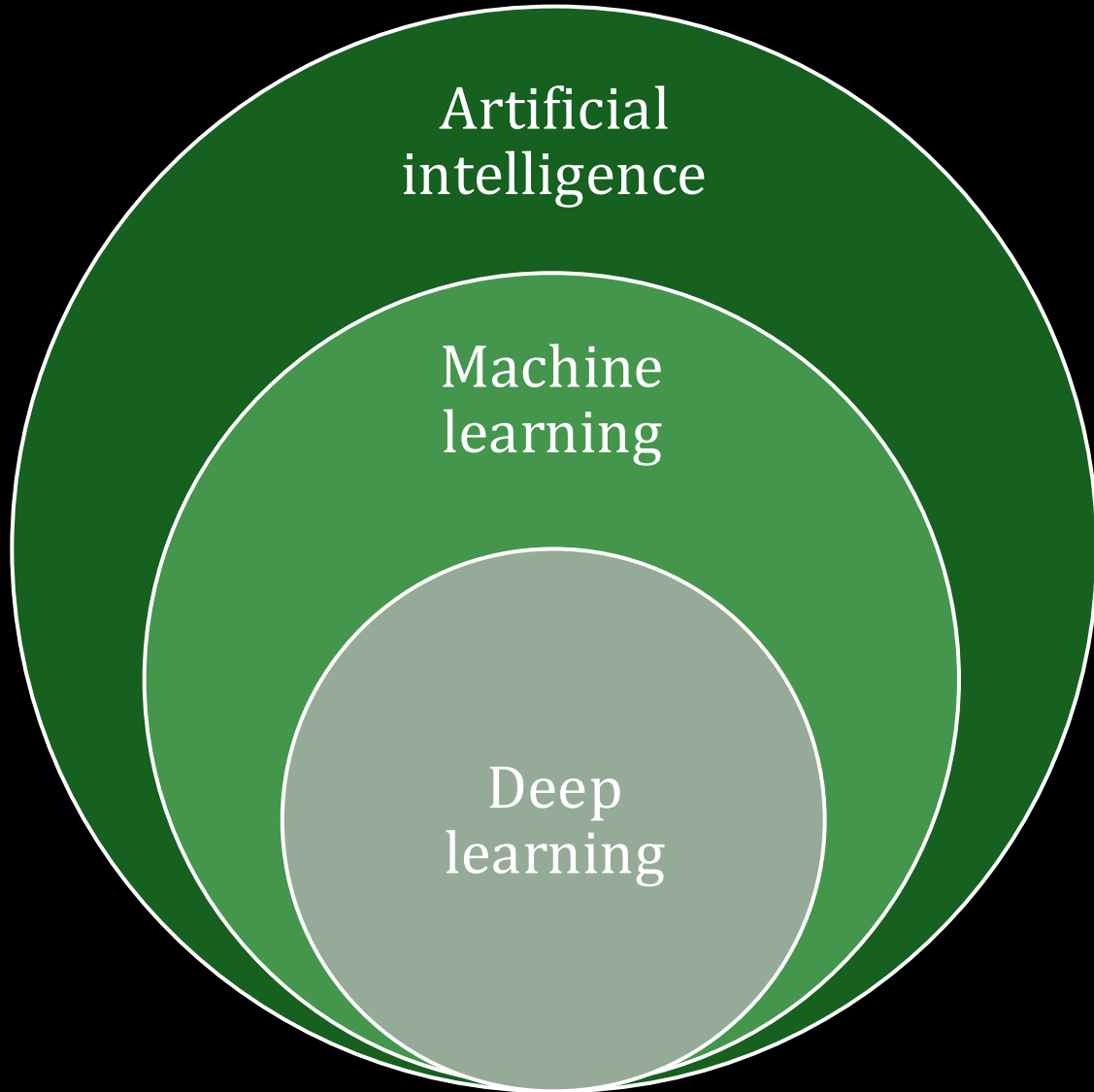
17th edition

May 28th - June 7th 2024 | Québec City | Canada

Frontiers in Neurophotronics Summer School

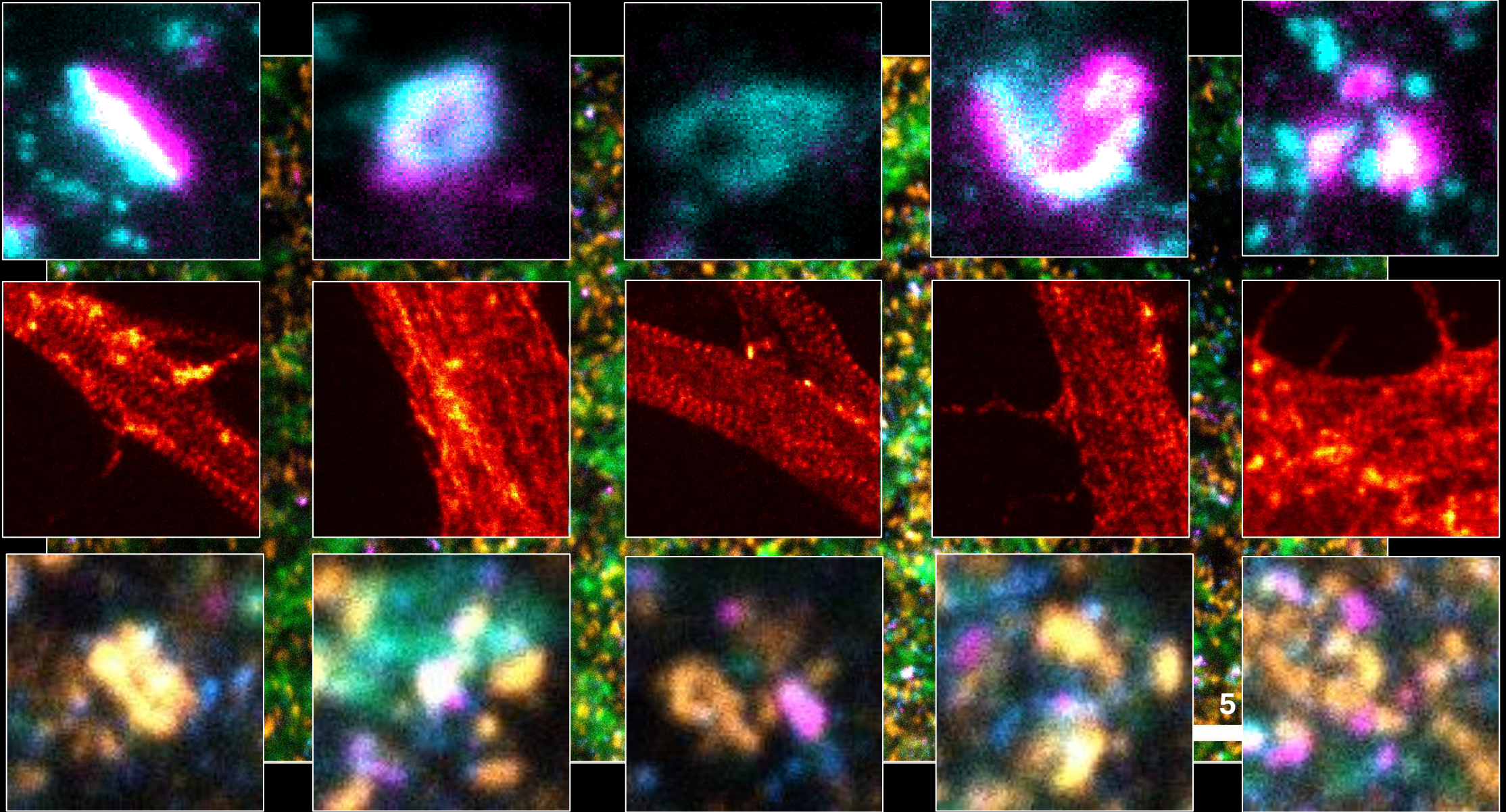
A summer school on advanced optical imaging and photoactivation techniques



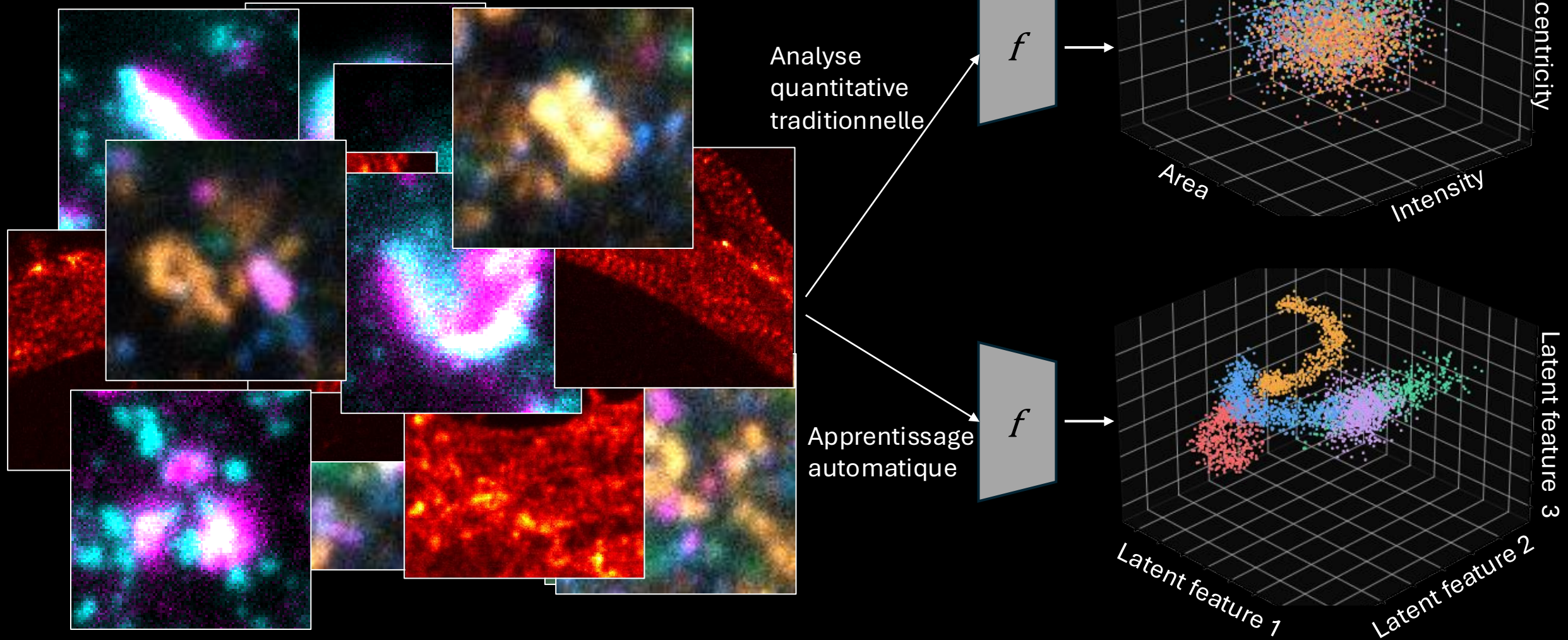


*Machine learning is programming computers to optimize a performance criterion using **example data or past experience***
(Introduction to Machine Learning – Ethem Alpaydin)

Why AI?



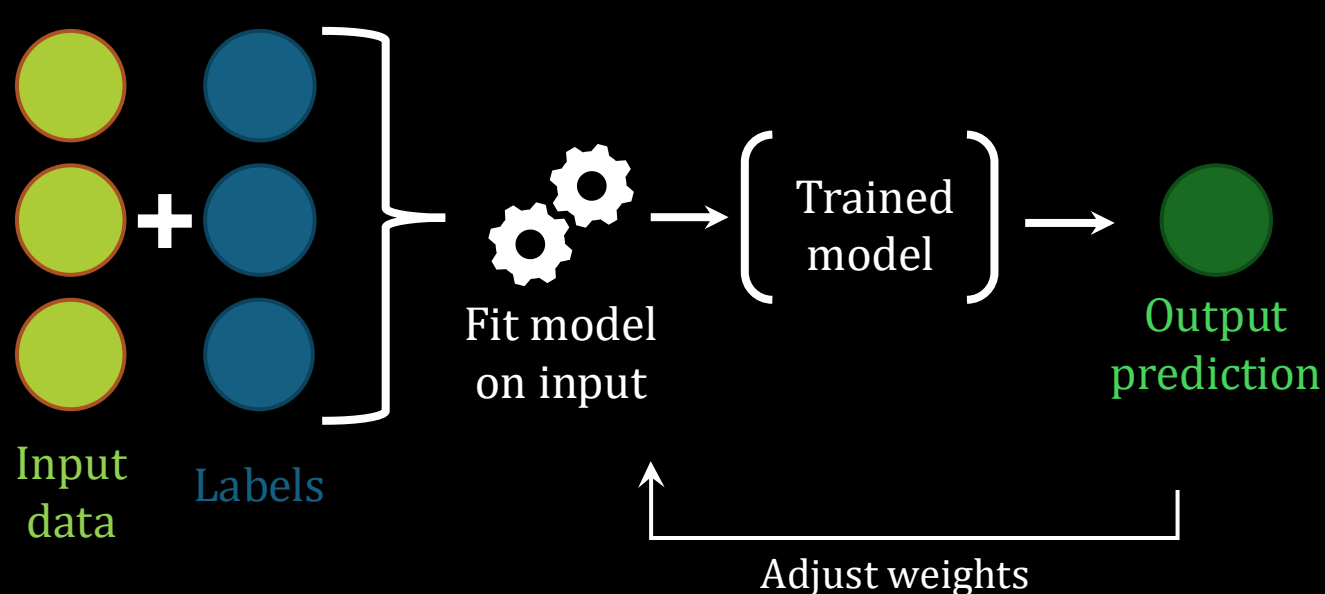
Learn representations from data



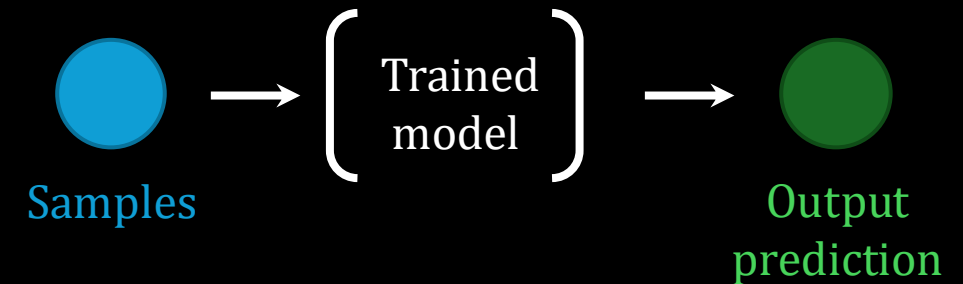
What is machine learning?

- Algorithms that **learn** from **input data** to make **predictions** on **samples**

Training



Inference



Terminology of Machine Learning

- **Features :**

- input variables – the way we represent our data
- e.g. words in an email text (spam recognition), structure on an image (edges, particular objects)

- **Labels/Annotations :**

- target that we try to predict
- e.g. spam or not spam, cat, dog, structure of interest

- **Example :**

- one piece of data
- Labeled or not labeled
- e.g. one image, one e-mail

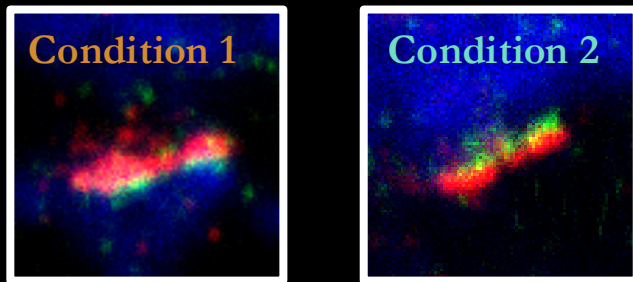
- **Model :**

- Algorithm doing the predictions
- What we want to create by learning from data

Different strategies to consider

Supervised

The data are associated (by a human) to a label



Goal

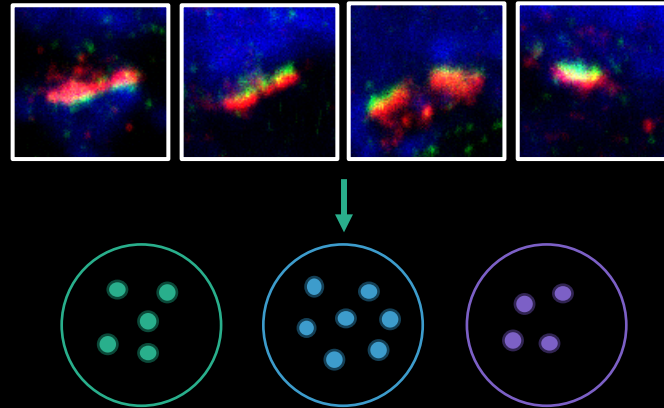
Predict the answer (label) on new images

Challenges

- Obtain annotated datasets
- High “cost” to obtain annotations
 - Human bias
- Under-representation of rare phenotypes

Unsupervised

No labels are provided



Goal

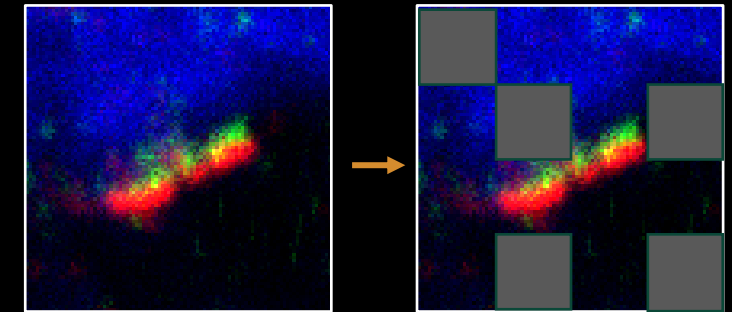
Discover a data structure (groups, patterns)

Challenges

- No ground truth
- Results validation
- Meaning of the discovered groups

Self-Supervised

The data provide the training signal



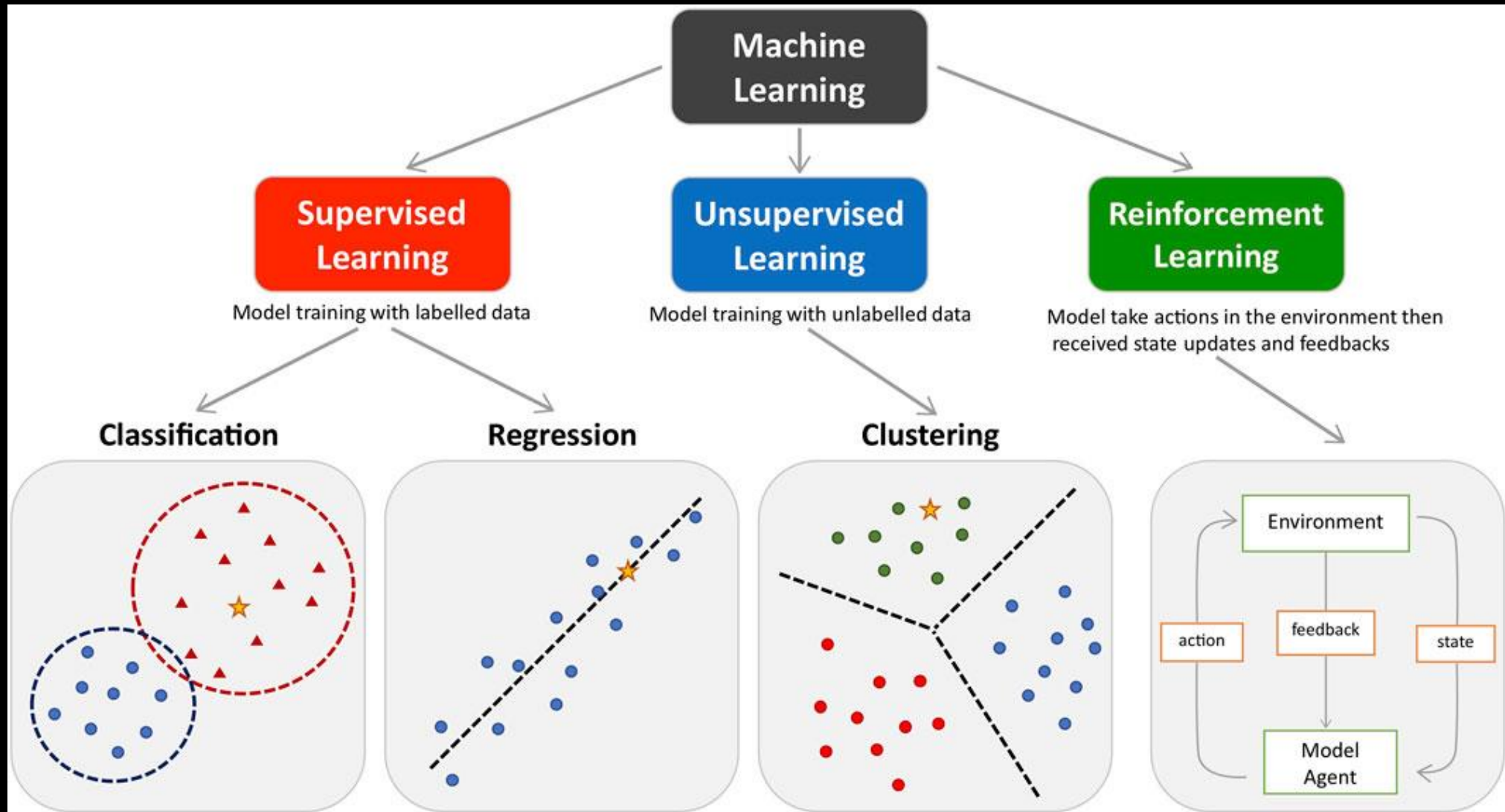
Goal

Learn data representations

Challenges

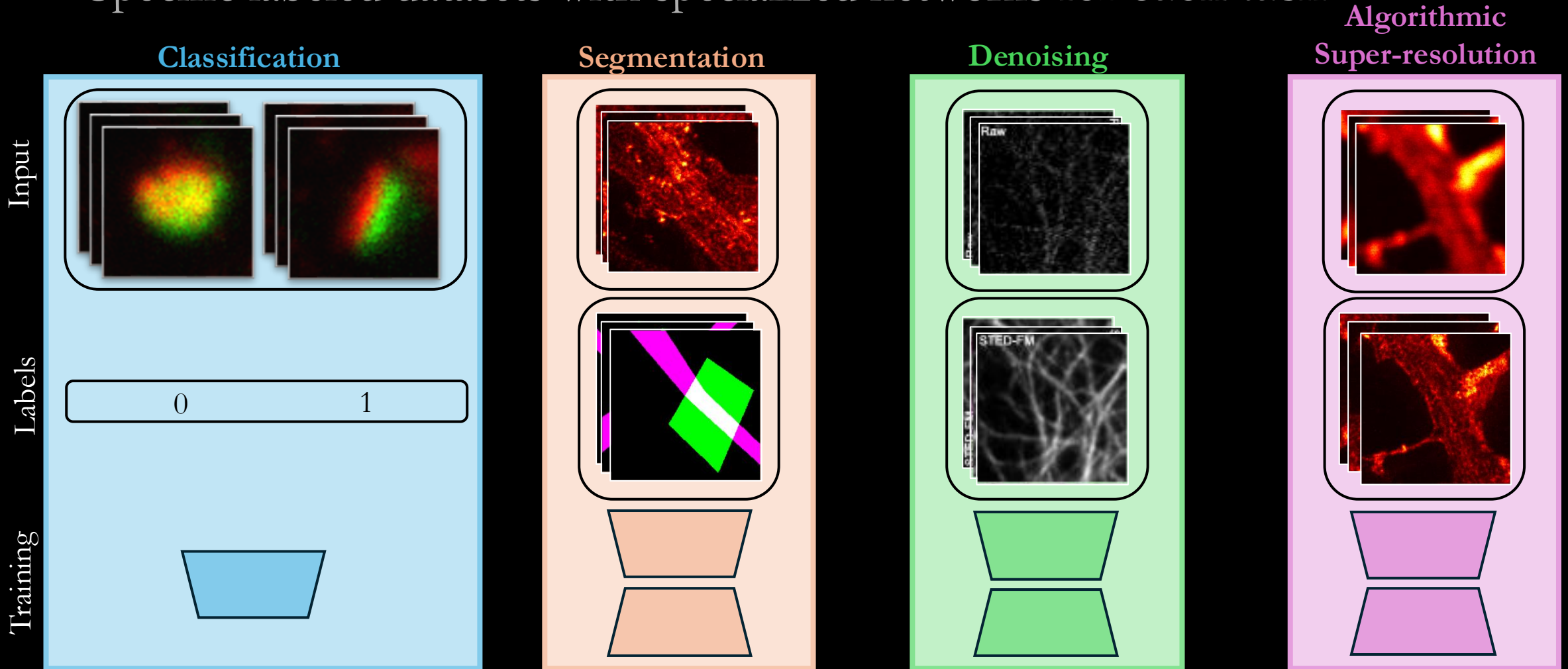
- Very large datasets are required
 - Interpretability
- Representation alignment

Different machine learning strategies



Classical Deep Learning

Specific labeled datasets with specialized networks for each task

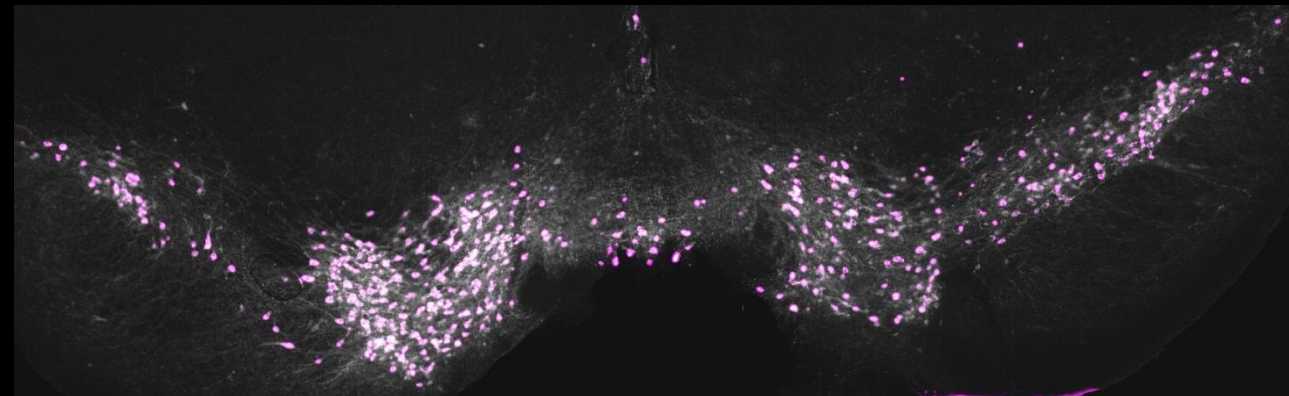
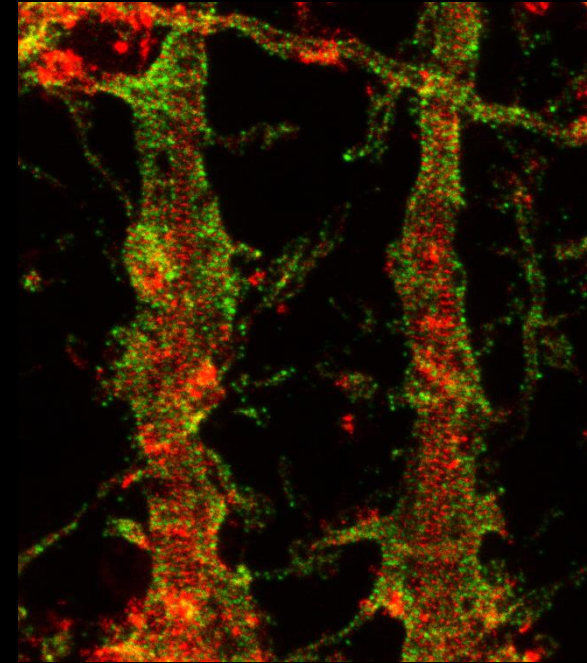
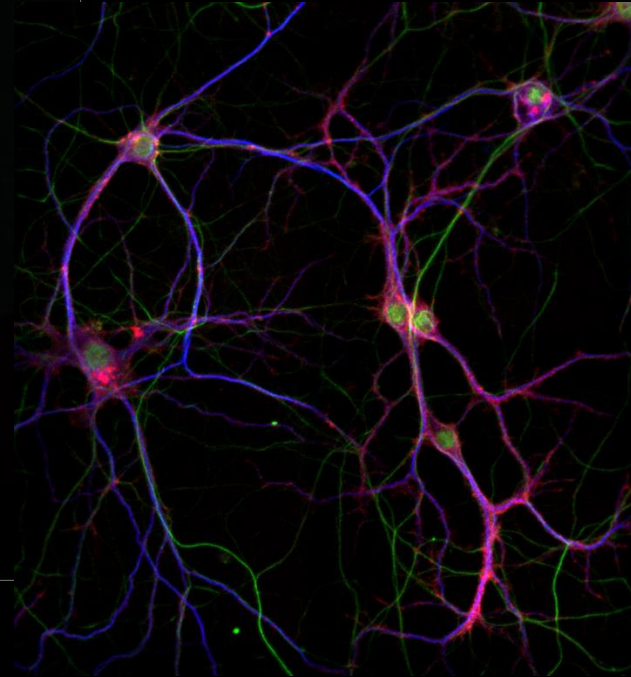
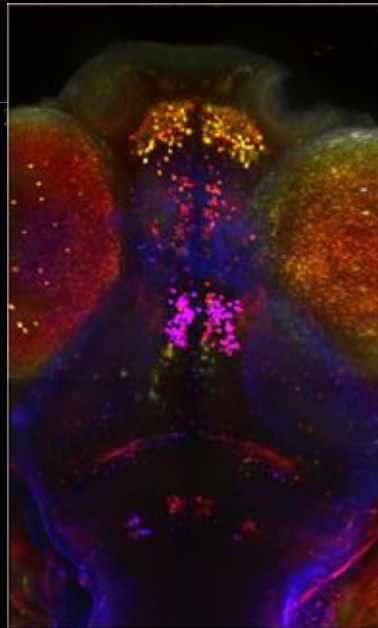
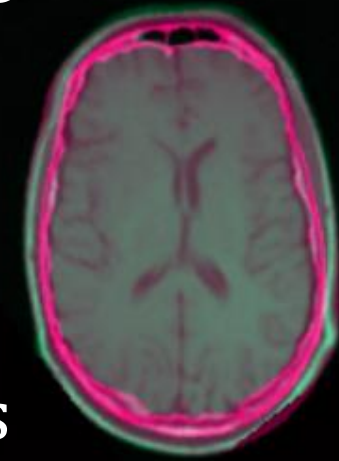
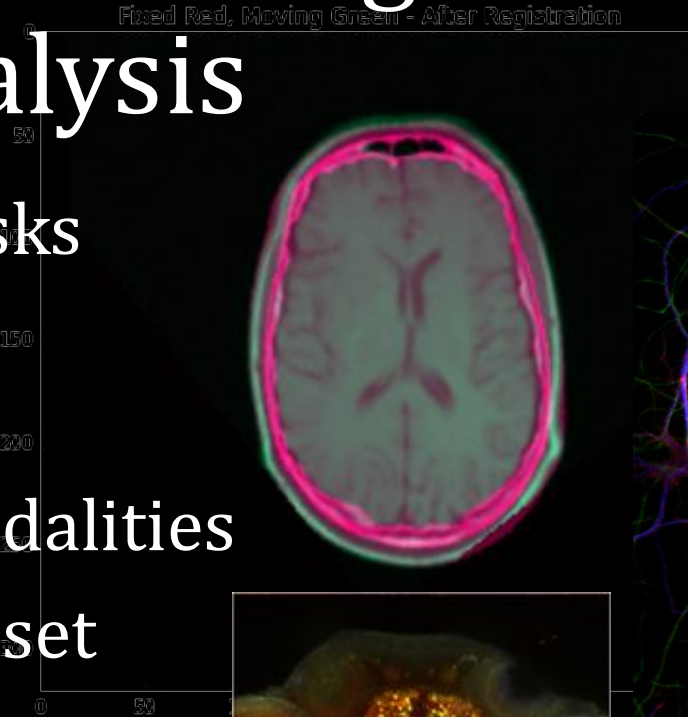


Workflow



Quantitative & High Throughput Biomedical Image Analysis

- Different tasks
- Multiple approaches
- Multiple modalities
- Careful dataset generation



Classical vs ML data analysis approaches

**Classical
Programming**



+



**Machine
Learning**



+





Chihuahua

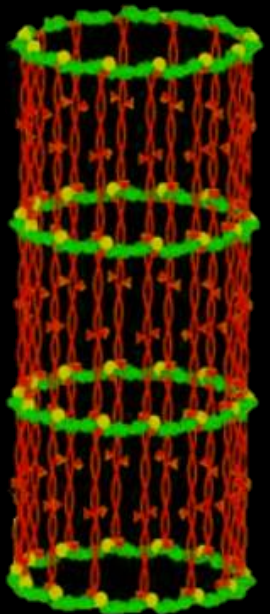
vs



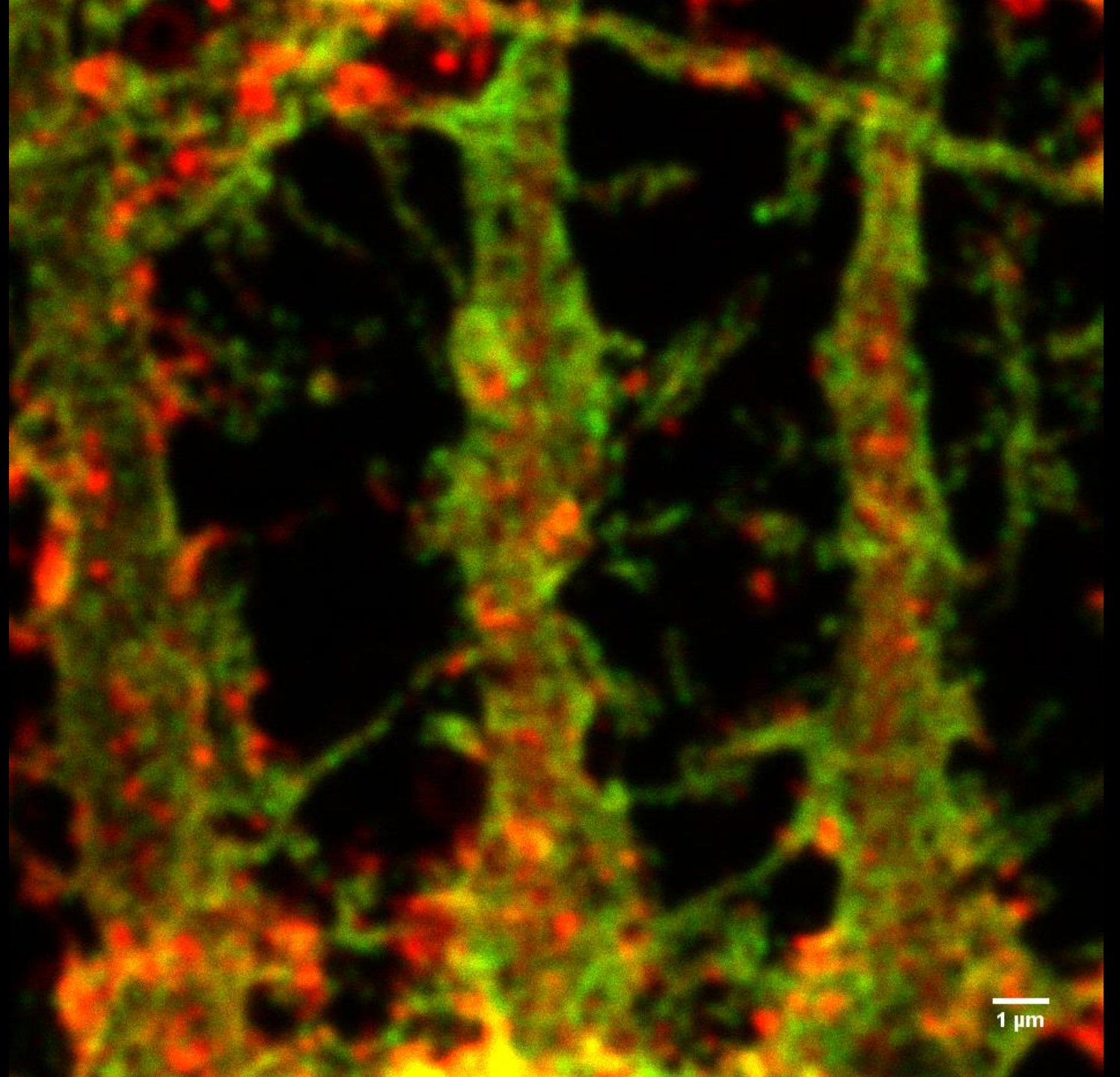
Rat



Test case : detecting
nanoscopic
structures on super-
resolution images

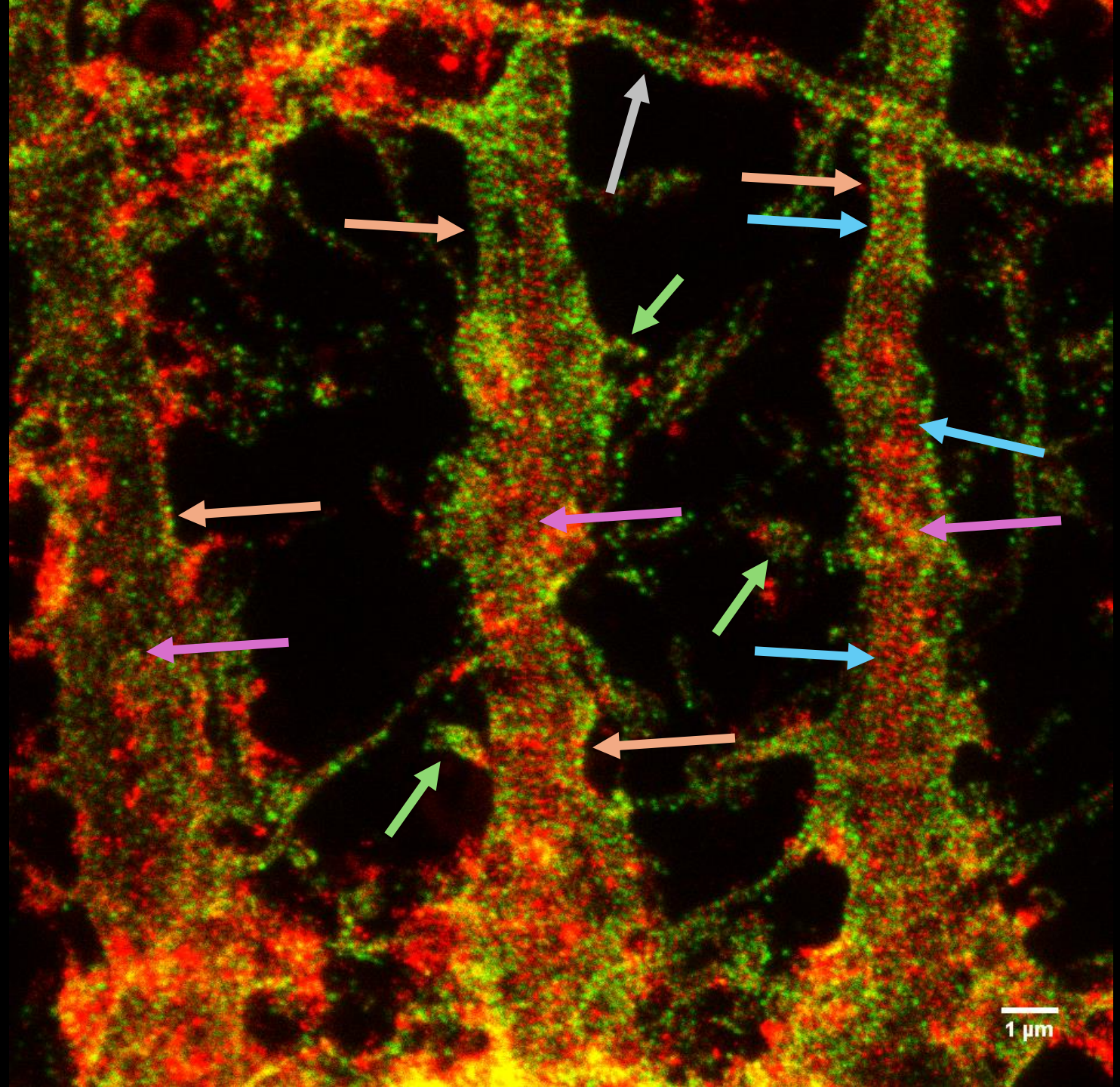


Actin
Spectrin



Features :

- Spine
- Spectrin rings
- Actin rings
- Dendrite
- Axon



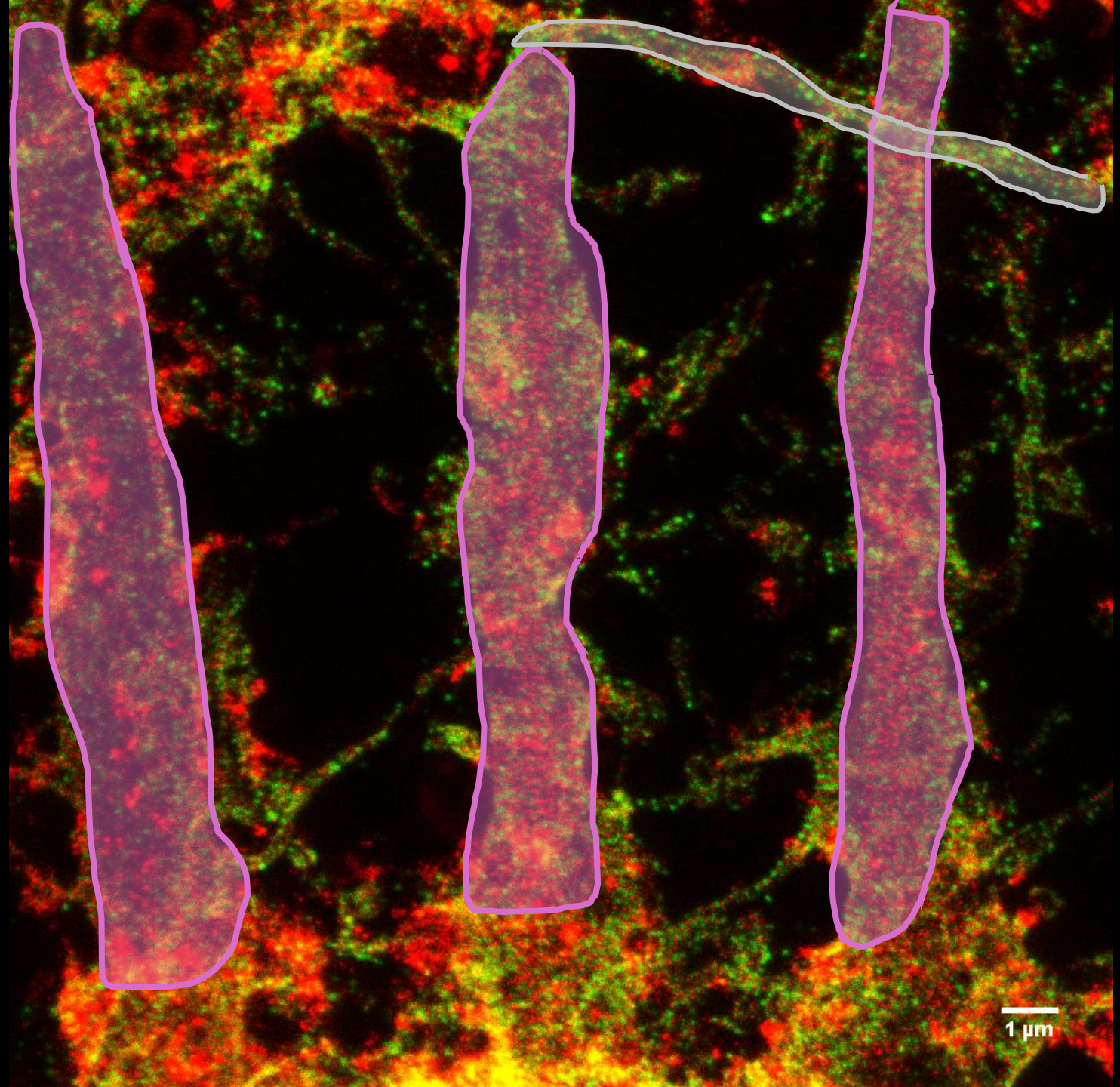
Labels :

A. Image tags

- Spine
- Spectrin rings
- Actin rings
- Dendrite
- Axon

B. Segmentation map

- Dendrite
- Axon



Models :

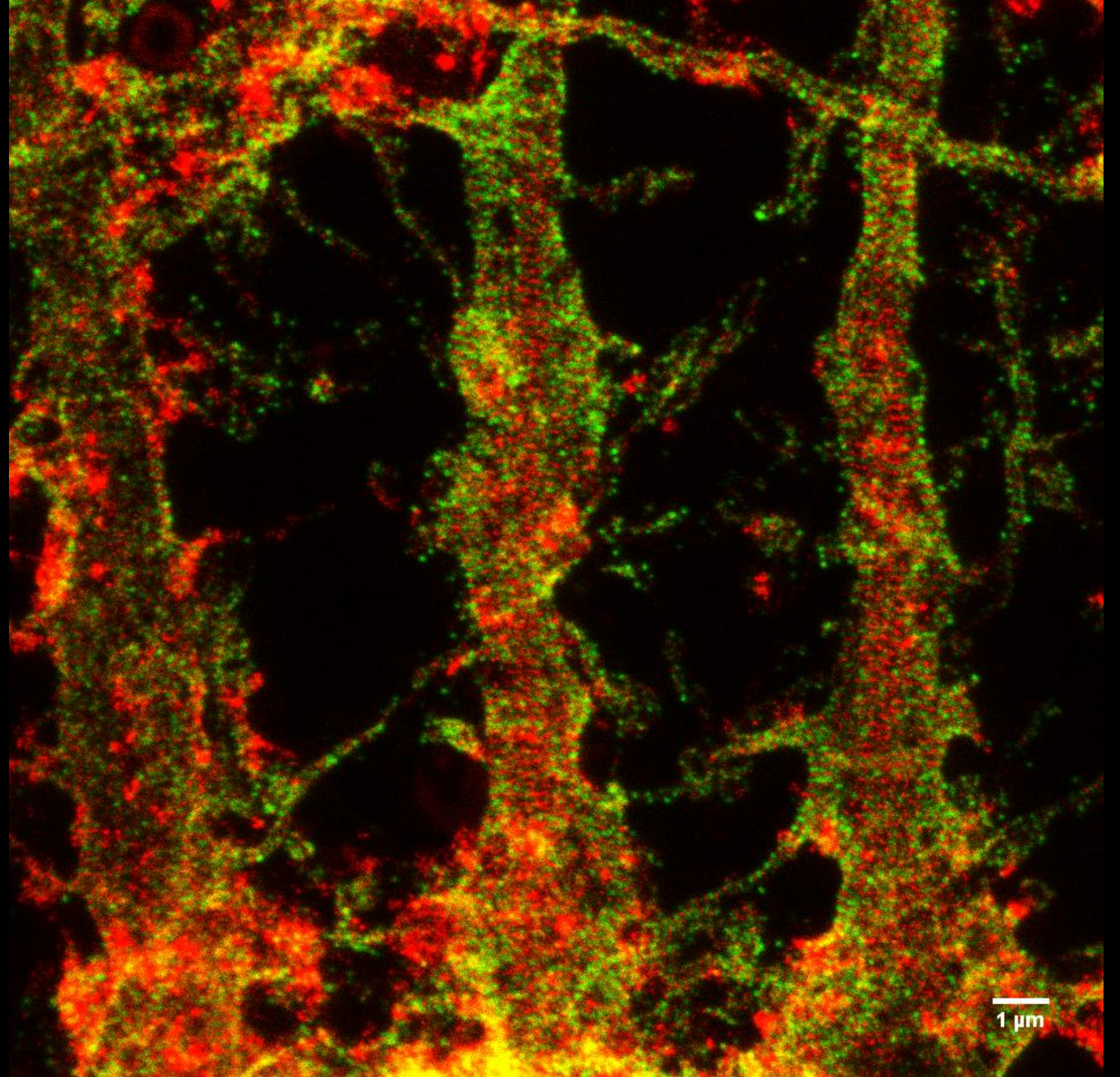
*Defines the relationship
between labels and features*

1. Training :

- The model gradually learn to associate features (F-actin rings and fibers) and labels (expert annotations) by seeing many labeled examples (images)

2. Inference :

- Use the trained model to make usefull predictions (on unlabeled examples)



What to do **before**, during, and after starting a project involving deep learning with your data?

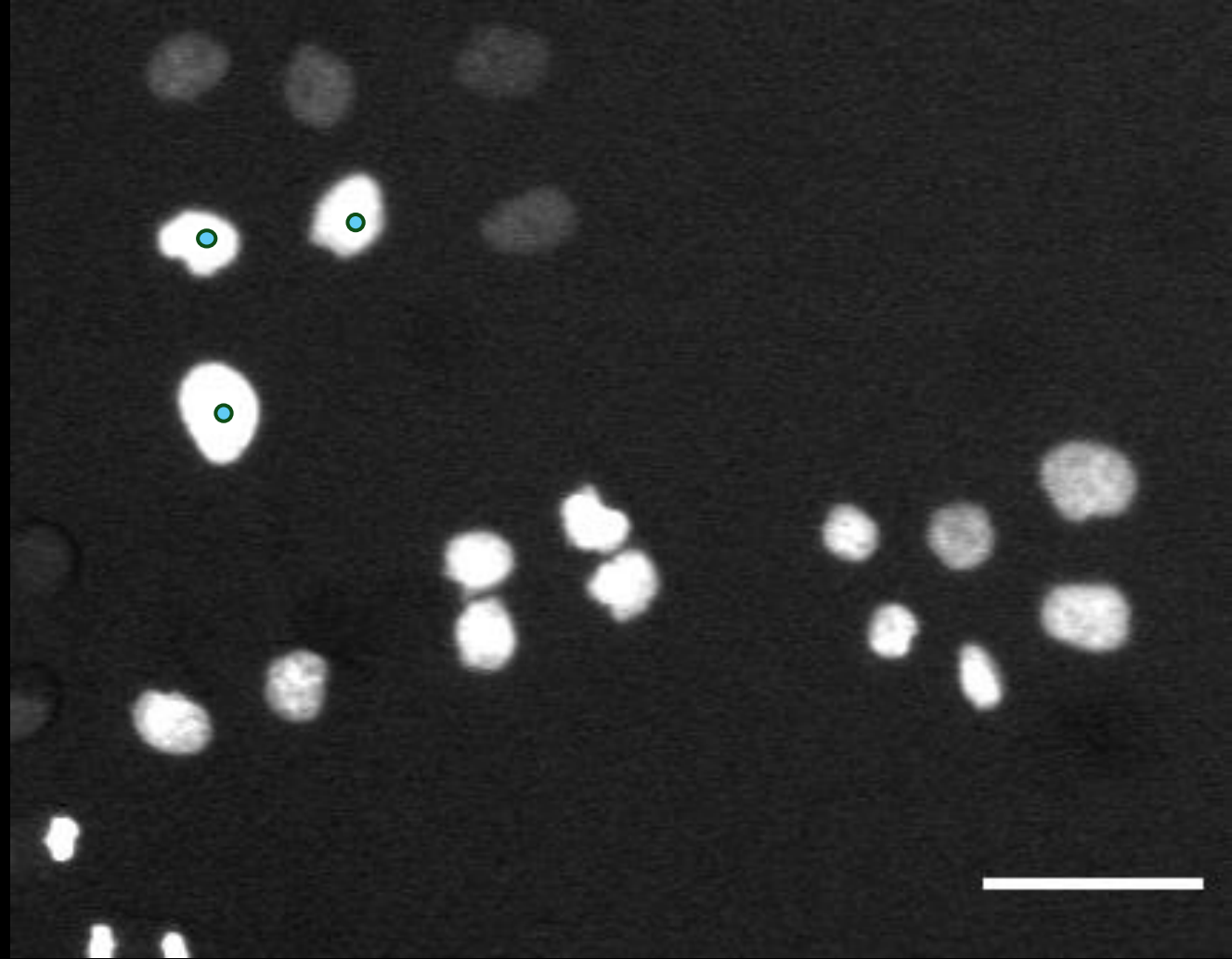
1. Curate your dataset

What to do **before**, during, and after starting a project involving deep learning with your data?

1. Curate your dataset
2. Define the task you want to address

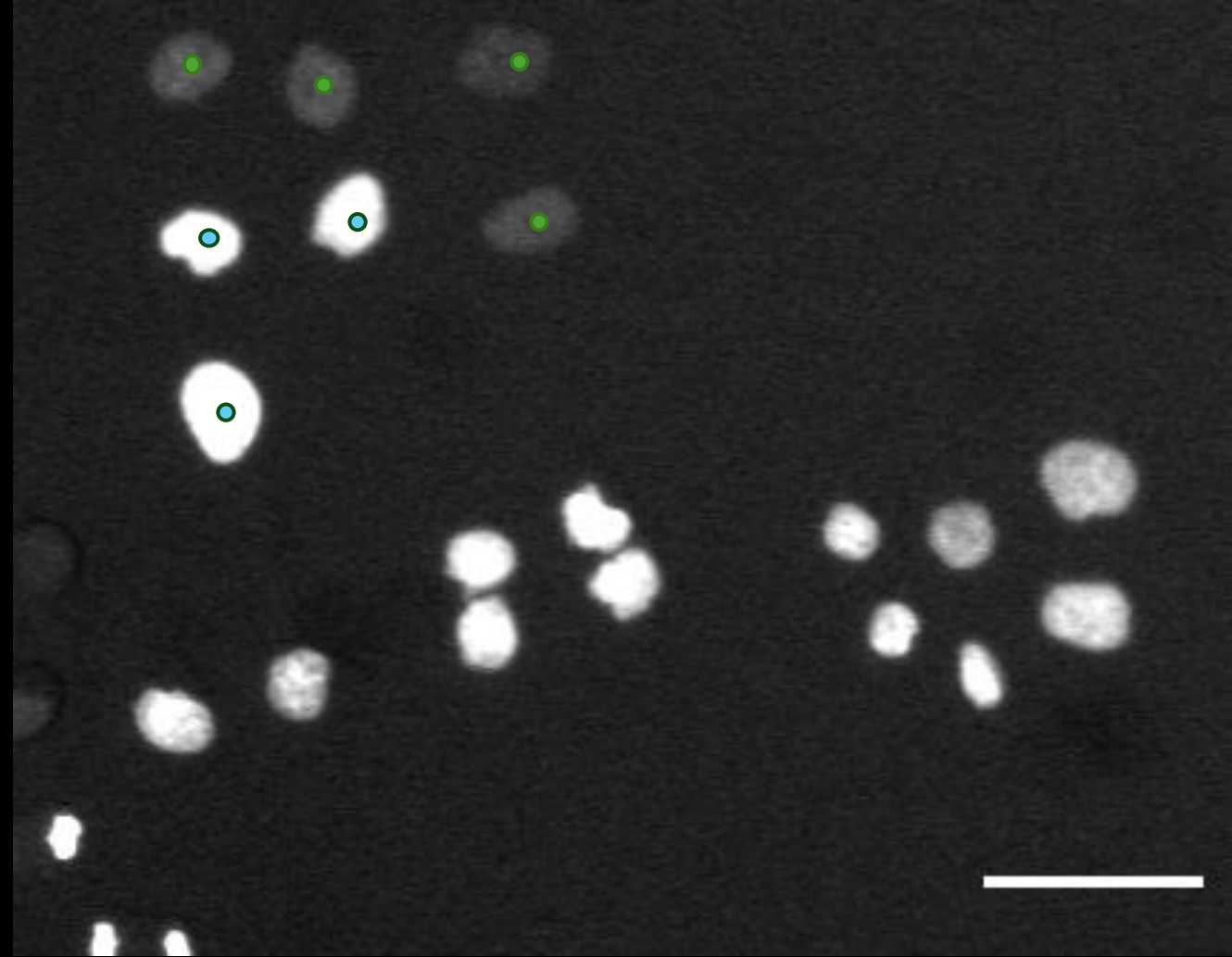
Common tasks

- Detection



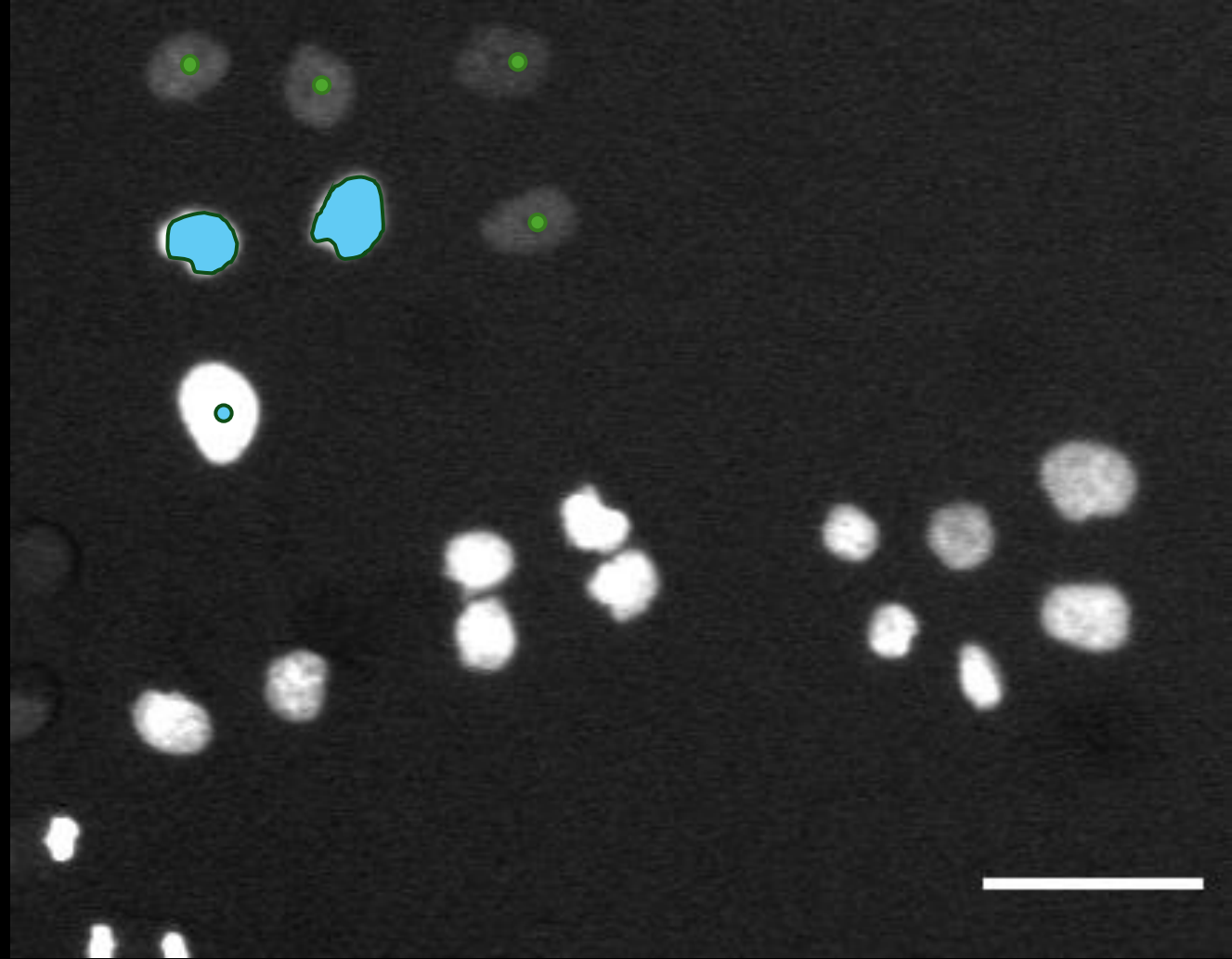
Common tasks

- Detection
- Classification



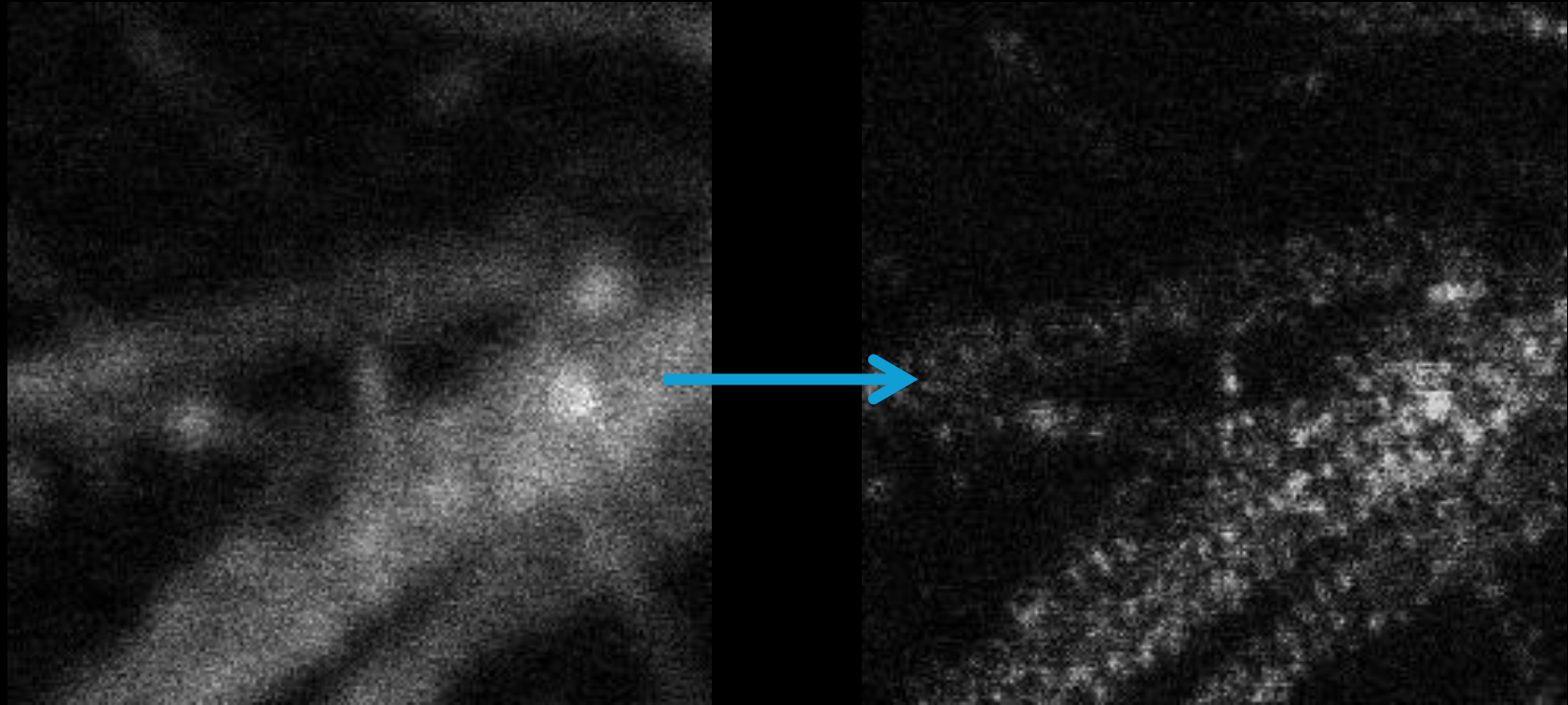
Common tasks

- Detection
- Classification
- Segmentation



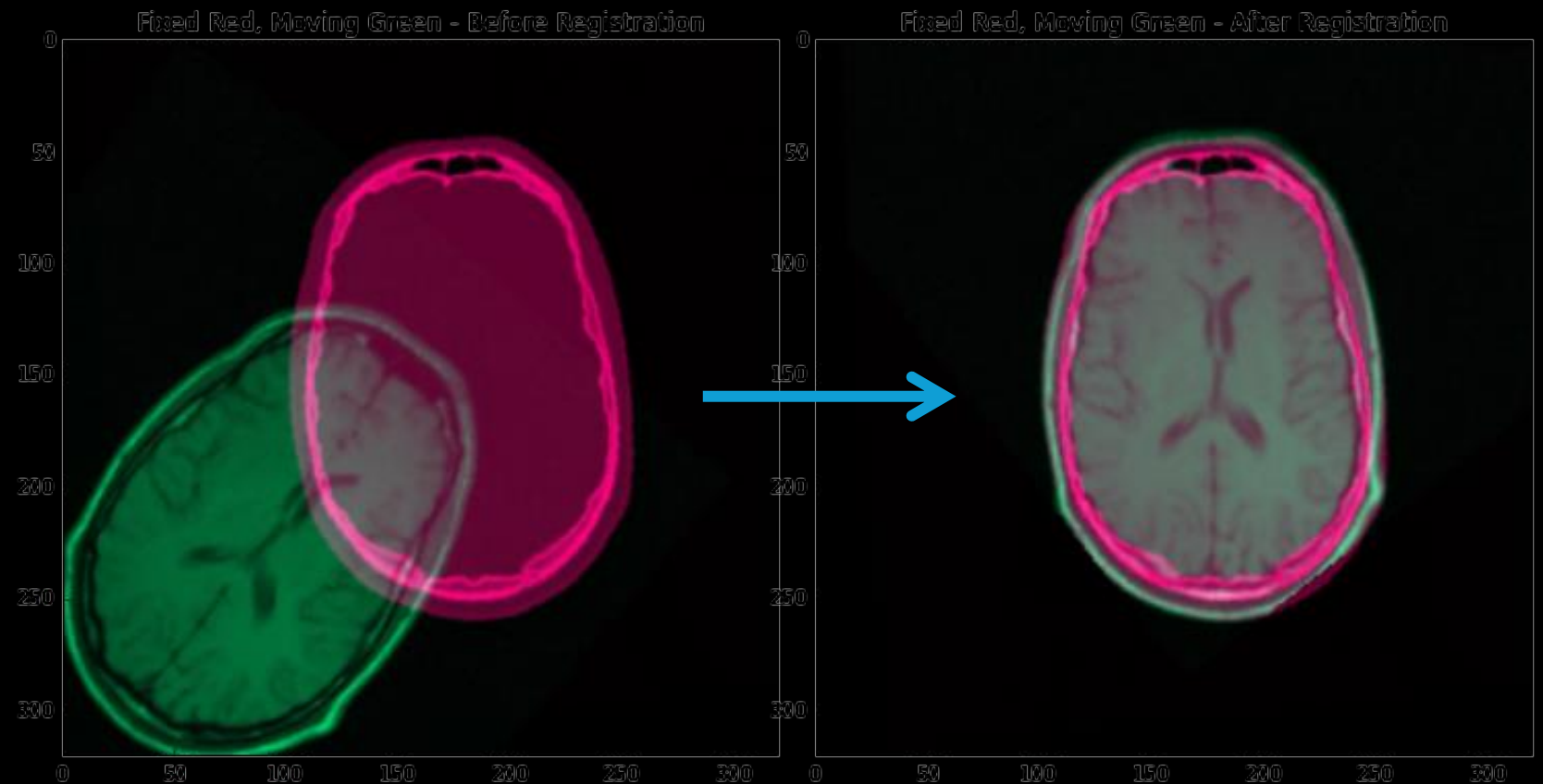
Common tasks

- Detection
- Classification
- Segmentation
- Generation



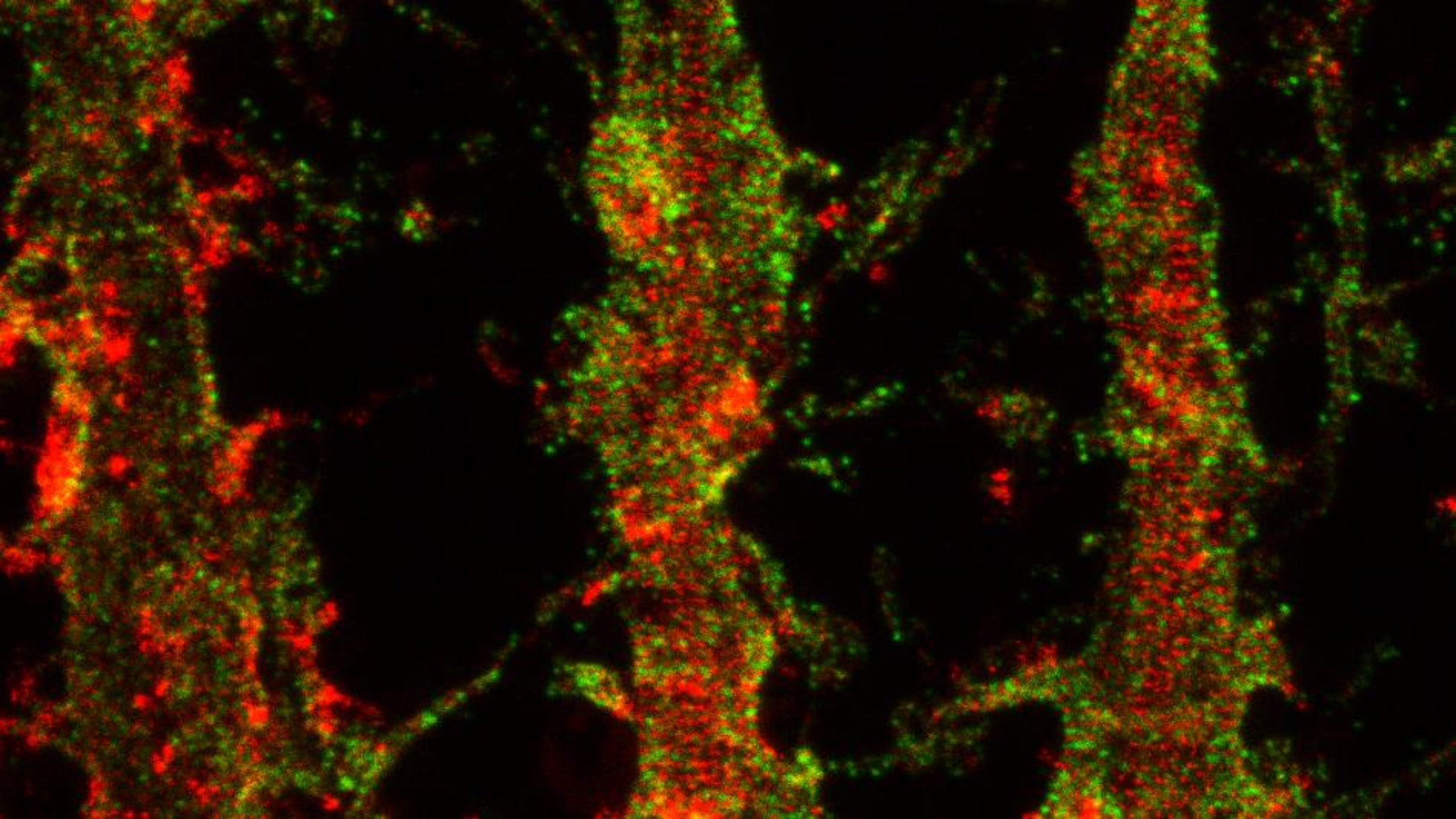
Common tasks

- Detection
- Classification
- Segmentation
- Generation
- Registration



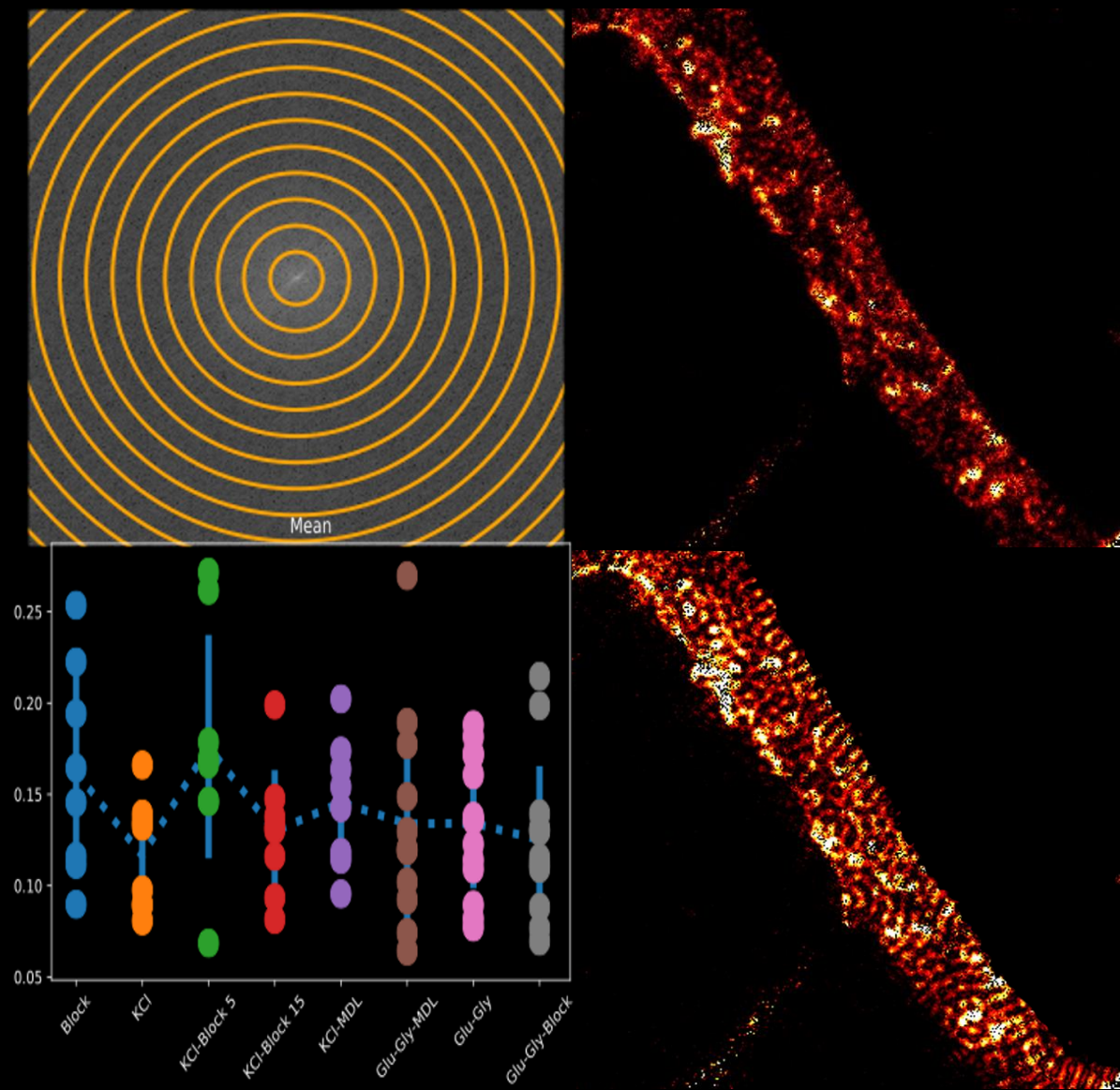
What to do **before**, during, and after starting a project involving deep learning with your data?

1. Curate your dataset
2. Define the task you want to address
3. Try to solve it with a well-established classical approach

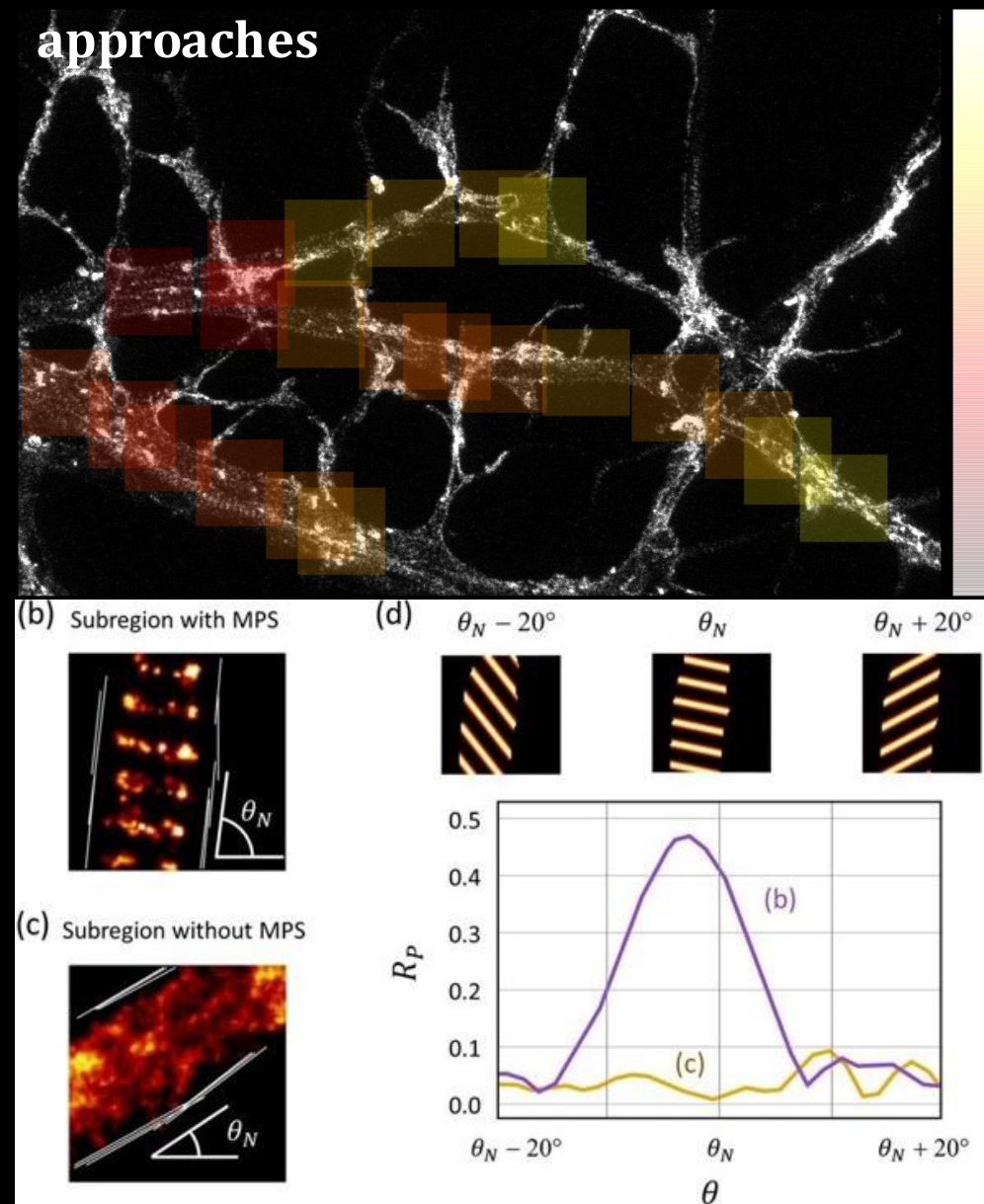


Try other approaches

Fourier analysis

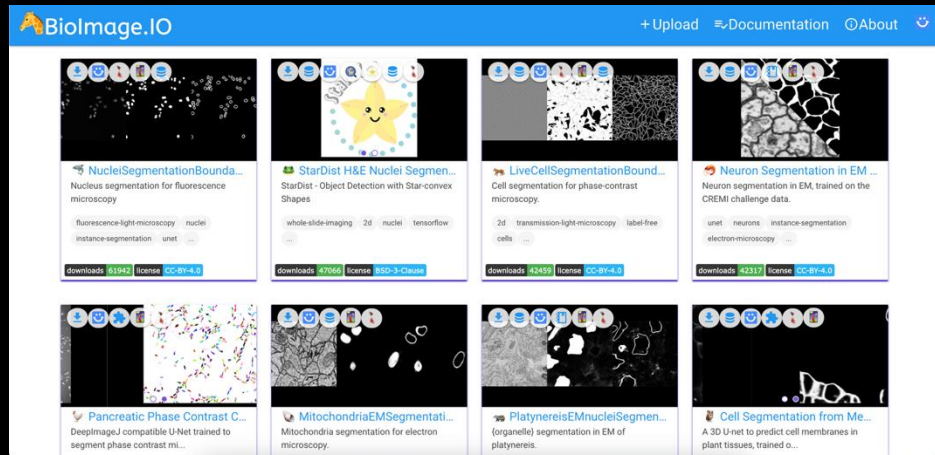


Correlative approaches

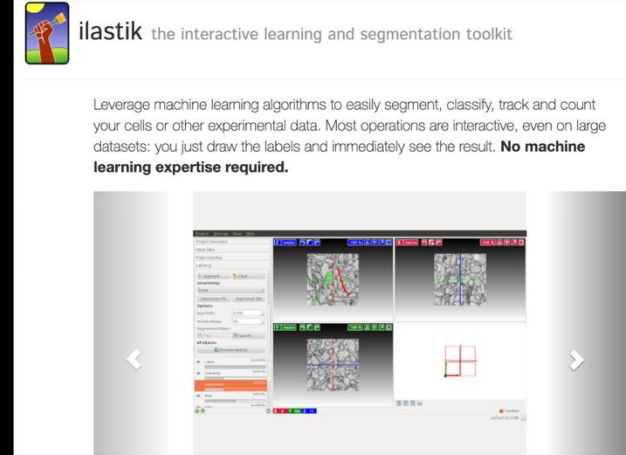


What to do before, **during**, and after starting a project involving deep learning with your data?

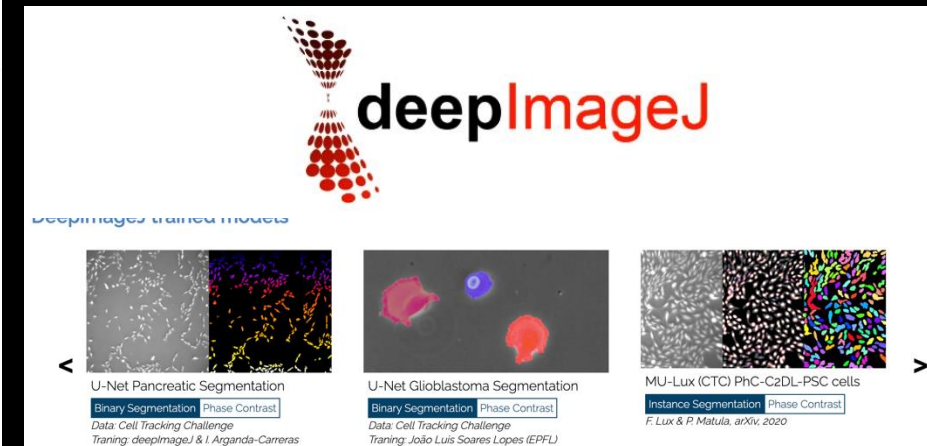
1. Curate your dataset
2. Define the task you want to address
3. Try to solve it with a well-established classical approach
4. Look online for already published user-friendly tools or pre-trained models



Bioimage Model Zoo



Ilastik



Deep ImageJ

What to do before, **during**, and after starting a project involving deep learning with your data?

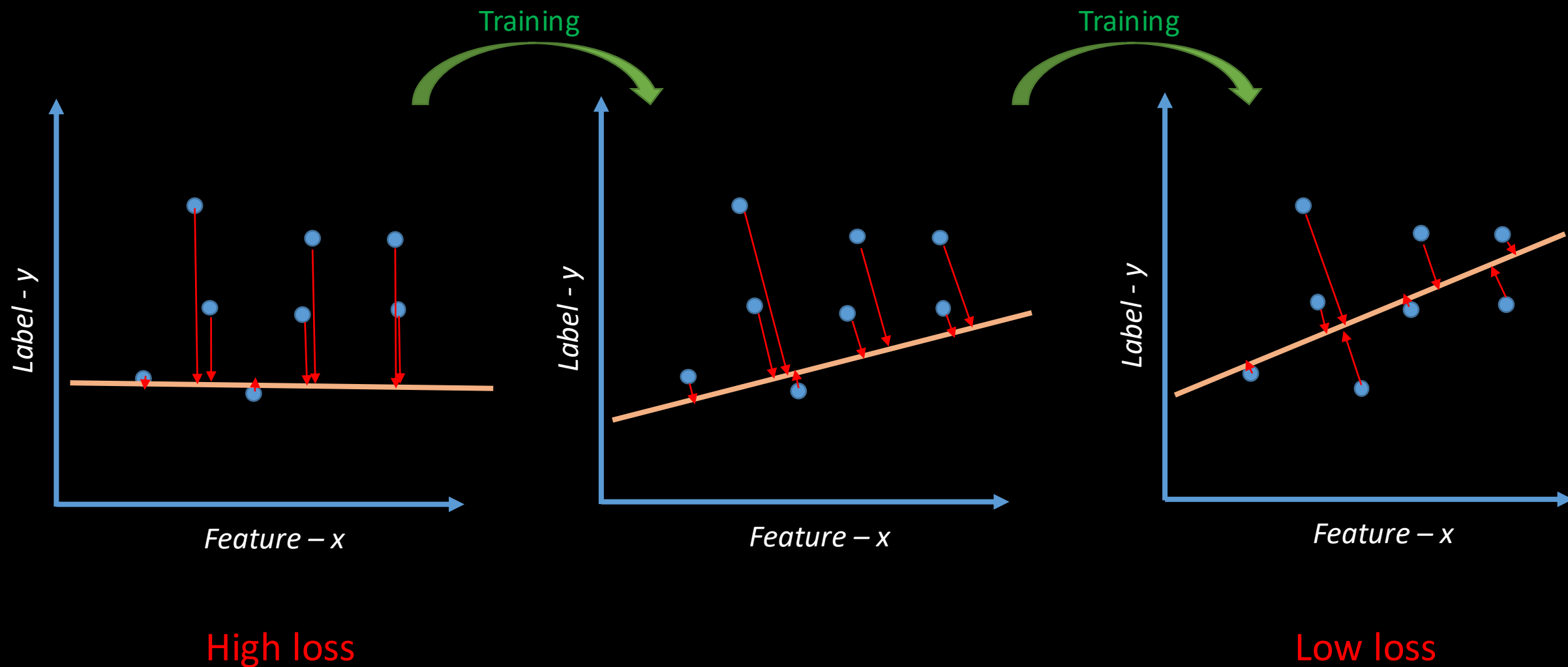
1. Curate your dataset
2. Define the task you want to address
3. Try to solve it with a well-established classical approach
4. Look online for already published user-friendly tools or pre-trained models
5. Refine the task and pipeline by evaluating :
 - The size of your dataset
 - The complexity of the task
 - The annotation requirements
 - Which performance metrics you can use

Training a model

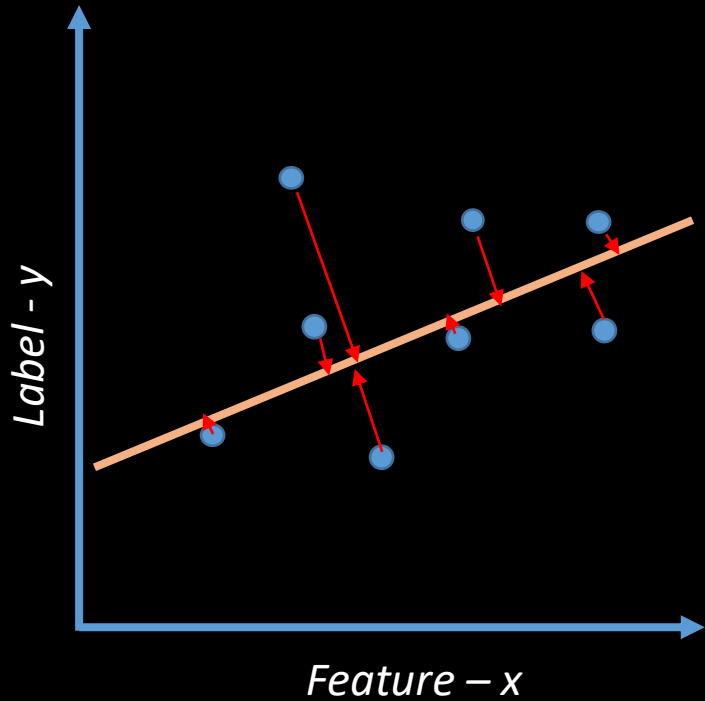
- Examining many examples
- Build a model that will minimize the *loss*
- *Loss* : How bad was a model to predict the features on a single example
 - Loss of 0 = model is perfect
 - High loss = model is bad
- *Goal* : find a set of weights and biases that have low loss on *all examples*.

Loss

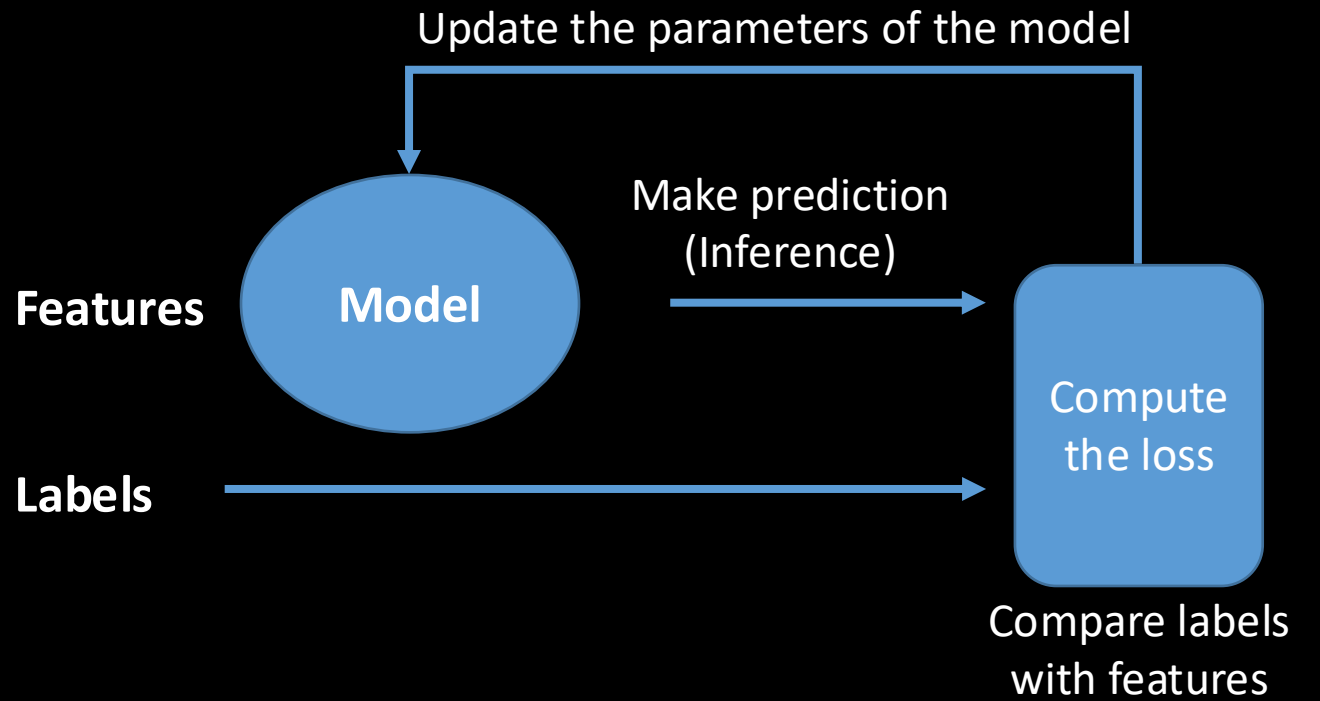
- Predictions
- Examples
- Loss



Loss



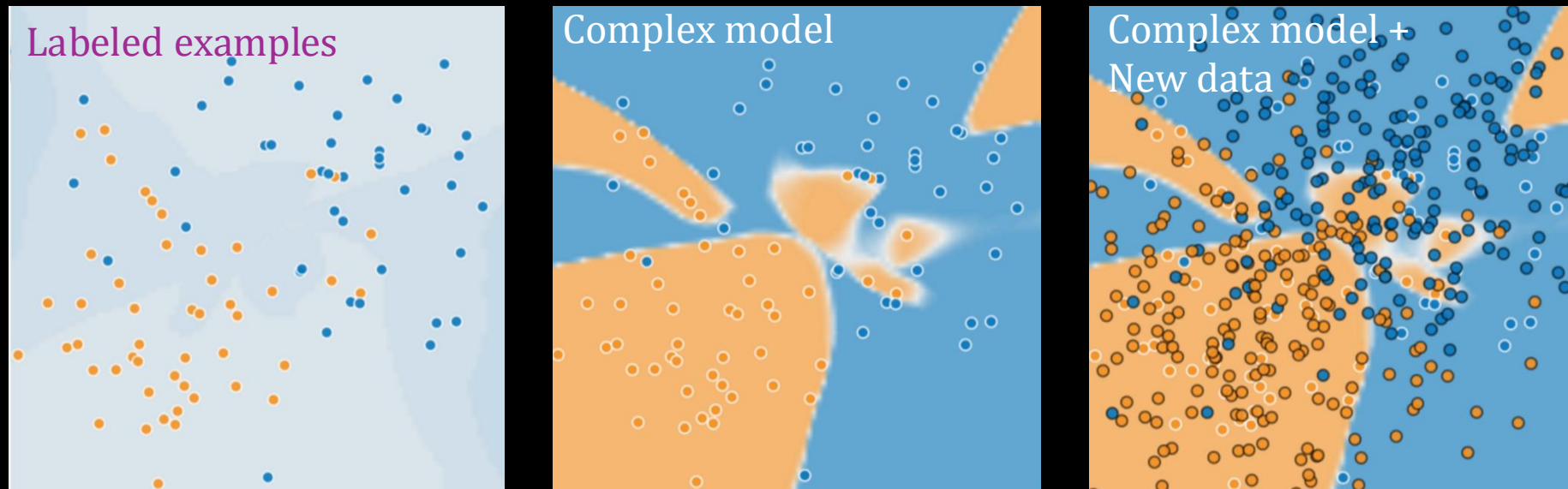
-  Predictions
-  Examples
-  Loss



- Loss function examples :
- *Mean square error*
 - *Cross entropy*

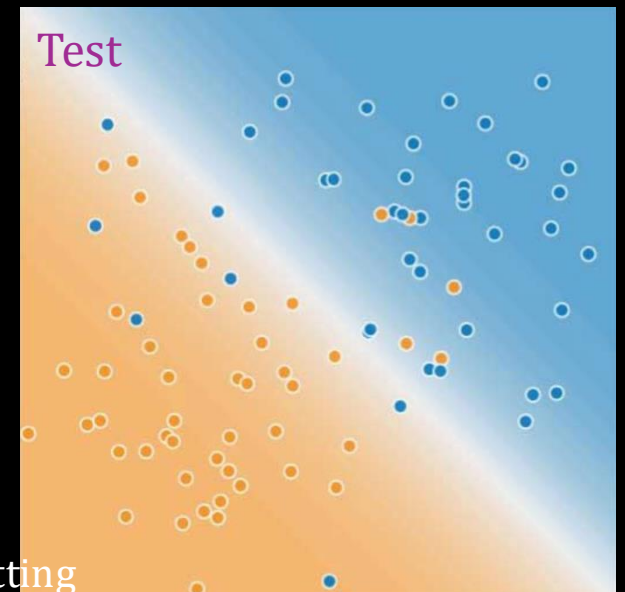
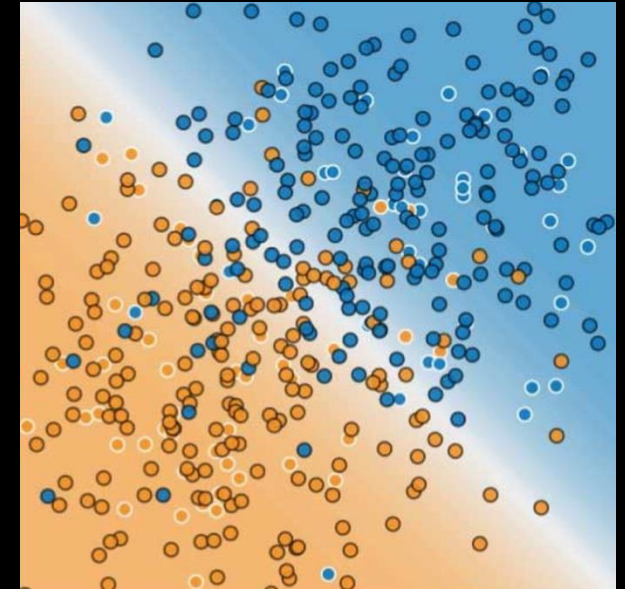
Generalization

- In Machine Learning the goal is to learn a model that will give good prediction on new examples that were never seen
- A too complex model may perform well with the training data set but gives poor prediction on new examples : **overfitting**



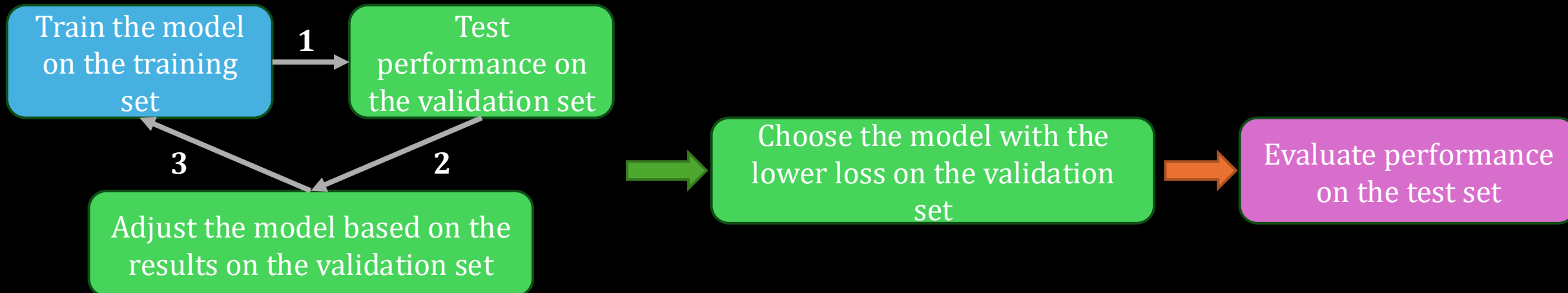
Generalization

- A model that overfits :
 - Low loss on the training dataset
 - High loss on the test dataset (new examples that were never seen before)
- Test set :
 - Large enough
 - But small partition of the dataset to maximize the number of examples used in training
 - Generally 15-25% of the whole dataset
 - Is not used over and over again
 - Representative from the whole dataset
- To prevent overfitting
 - *Try to have a model that is as simple as possible*



Generalization

- Adding a third partition : **Validation set**
- Greatly helps to reduce overfitting
 - Tweaking is done on the validation set
 - The test set is kept completely unknown to the model



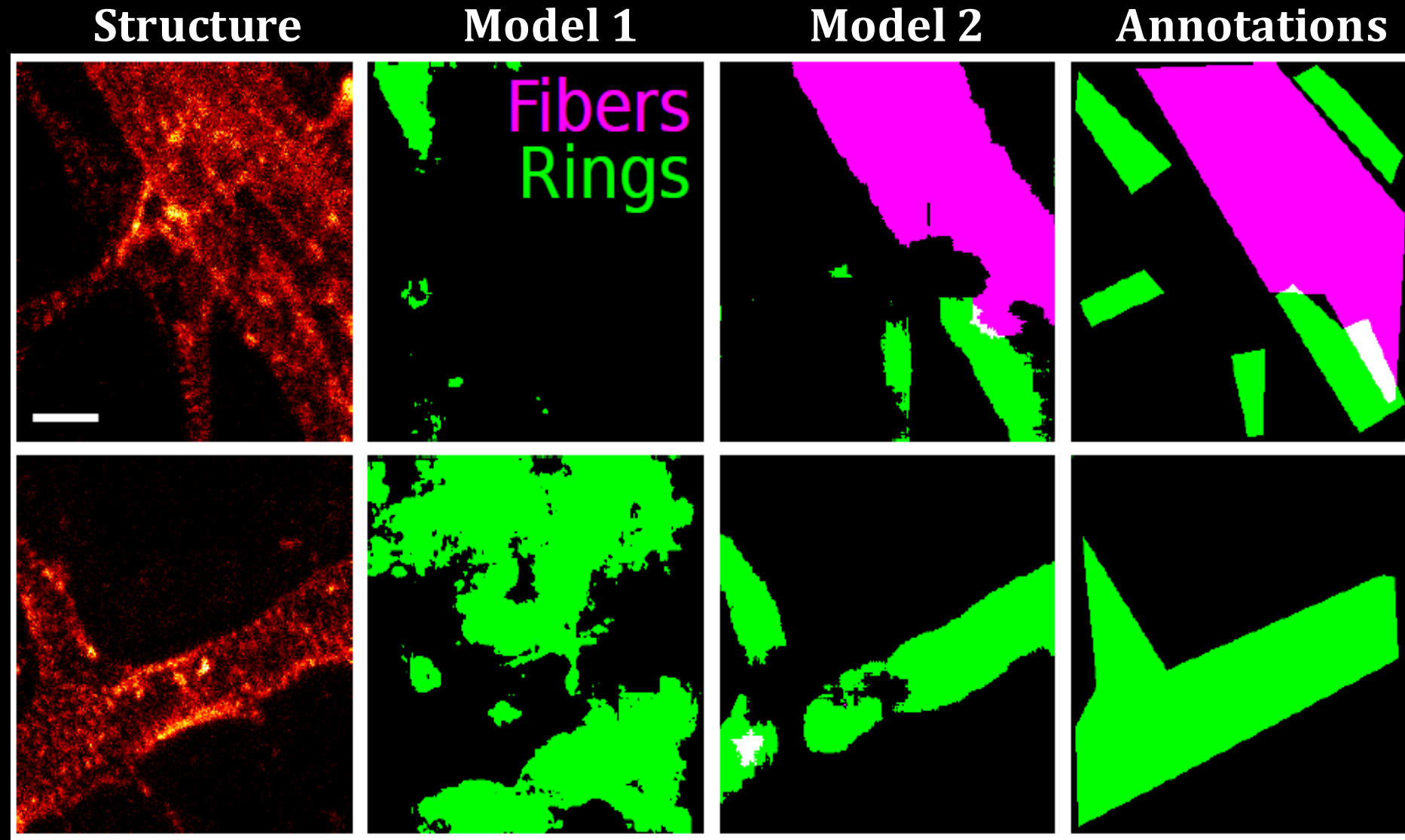
What to do before, **during**, and after starting a project involving deep learning with your data?

1. Curate your dataset
2. Define the task you want to address
3. Try to solve it with a well-established classical approach
4. Look online for already published user-friendly tools or pre-trained models
5. Refine the task and pipeline
6. **Collaborate** with computer scientist or image analysis specialist

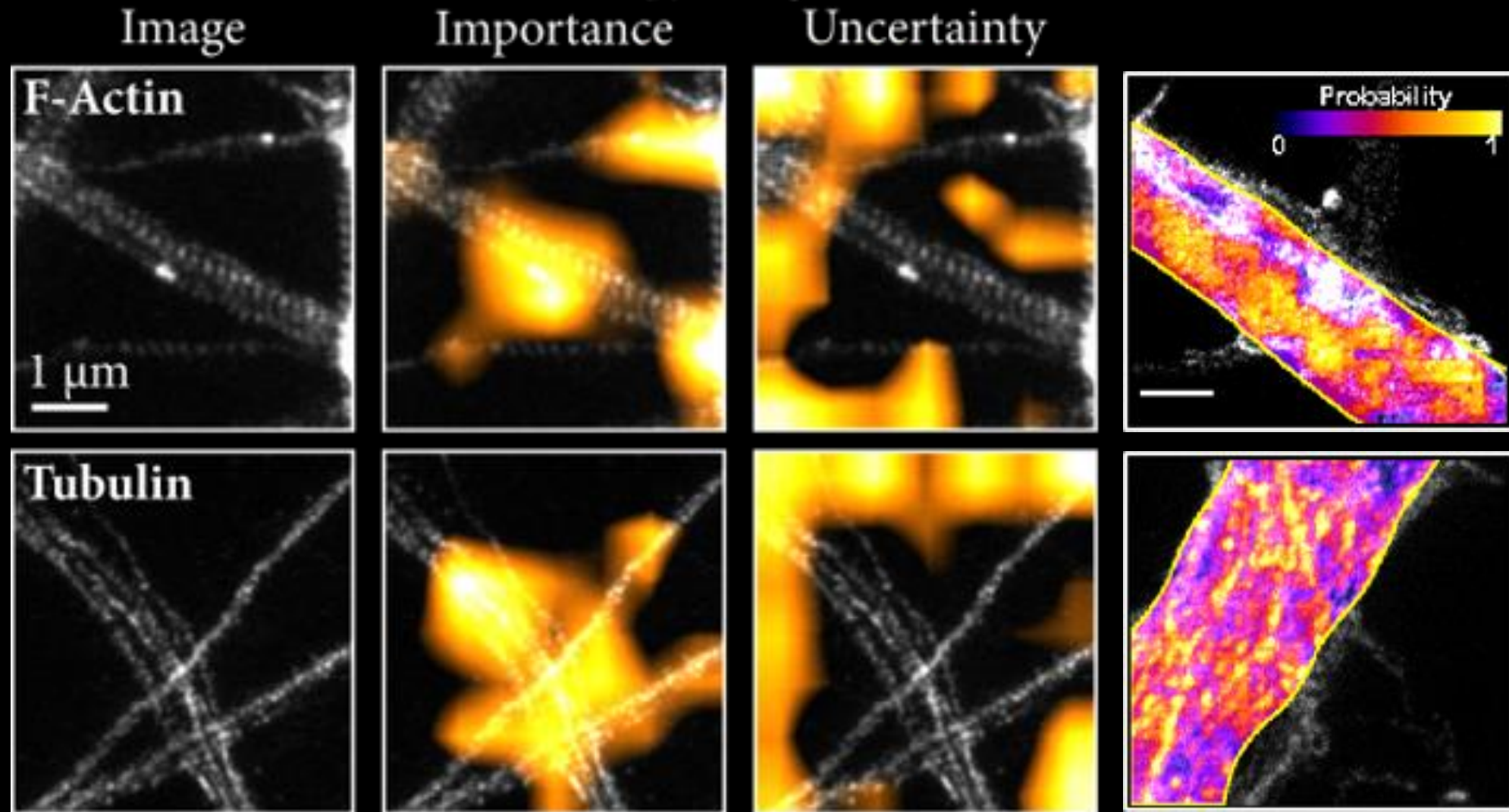
What to do before, during, and after starting a project involving deep learning with your data?

1. Curate your dataset
2. Define the task you want to address
3. Try to solve it with a well-established classical approach
4. Look online for already published user-friendly tools or pre-trained models
5. Refine the task and pipeline
6. **Collaborate** with computer scientist or image analysis specialist
7. **Inspect the results** (look at the images, the segmentation, the classification, etc.)

Inspect results



Inspect the results



Typical pipeline

Data acquisition

- Standardized practices
- Organize the data
- Keep the metadata

Preprocessing

- Signal normalization
- Signal unmixing
- Dataset curation
- Alignment and registration
- Batch normalization
- Illumination correction

Processing

- Classification
- Segmentation
- Resolution Enhancement
- Artificial labeling
- Tracking
- Reconstruction

Data analysis

- Distributions
- Multidimensional analysis
- Feature analysis
- Clustering

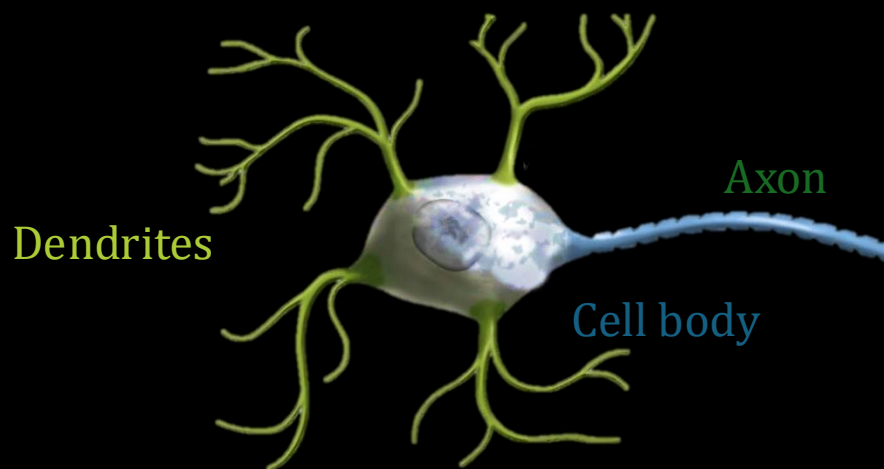
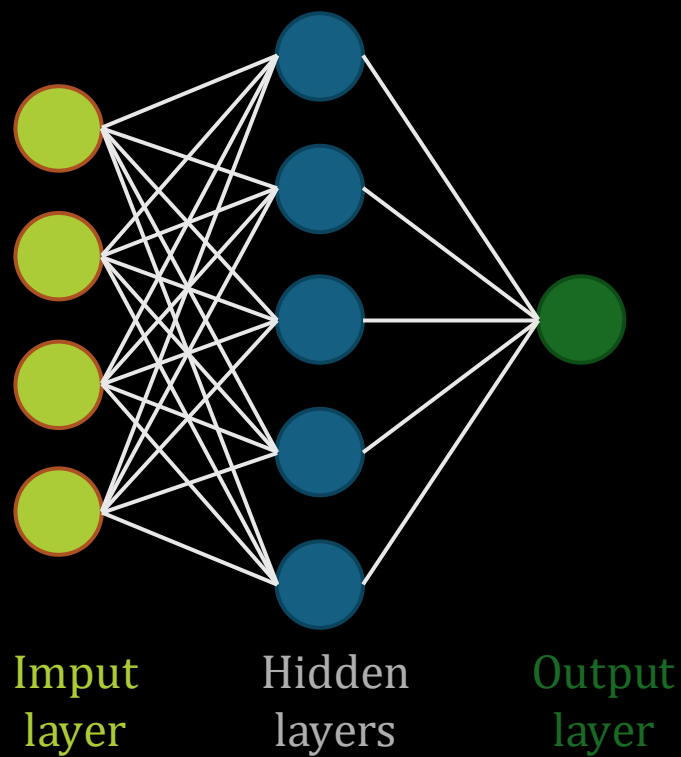
Visualization

- Dimensionality reduction
- Multidimensional representation
- Interactive tools
- Interpretability methods

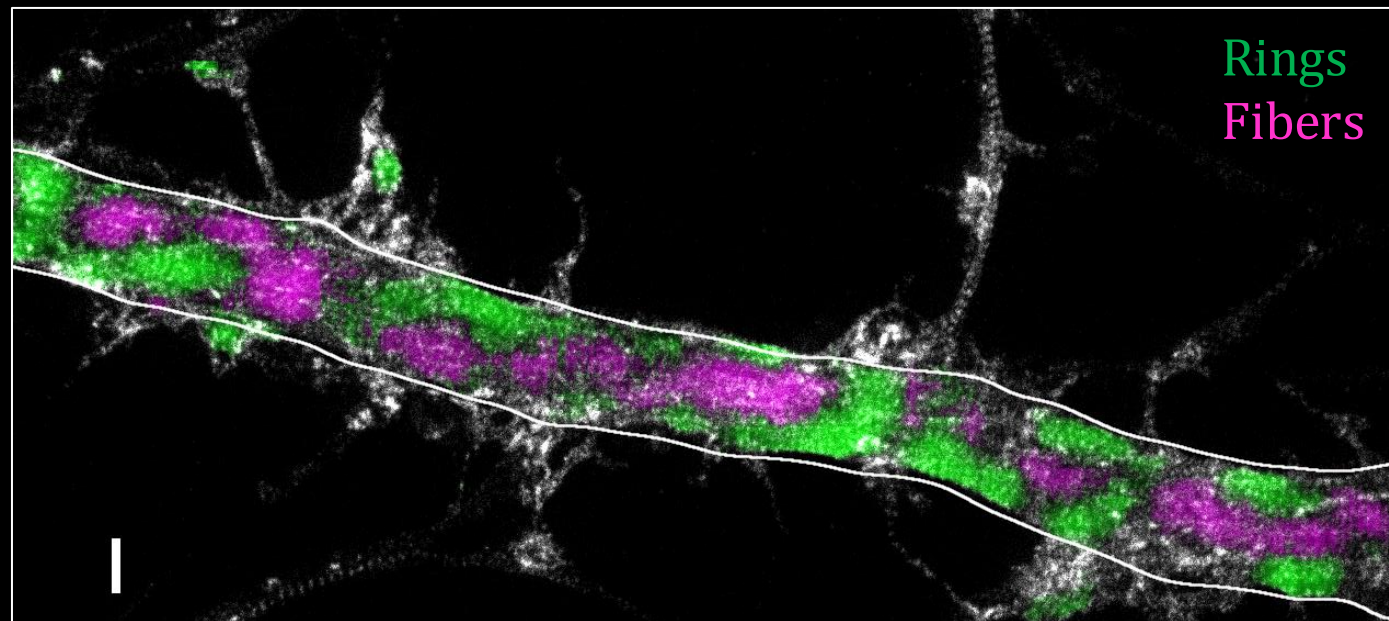
Sharing

- Publish the dataset **with** metadata and annotations
- Open-access code (Github)
- Accessible model (public repository)
- Notebooks

Artificial neural network

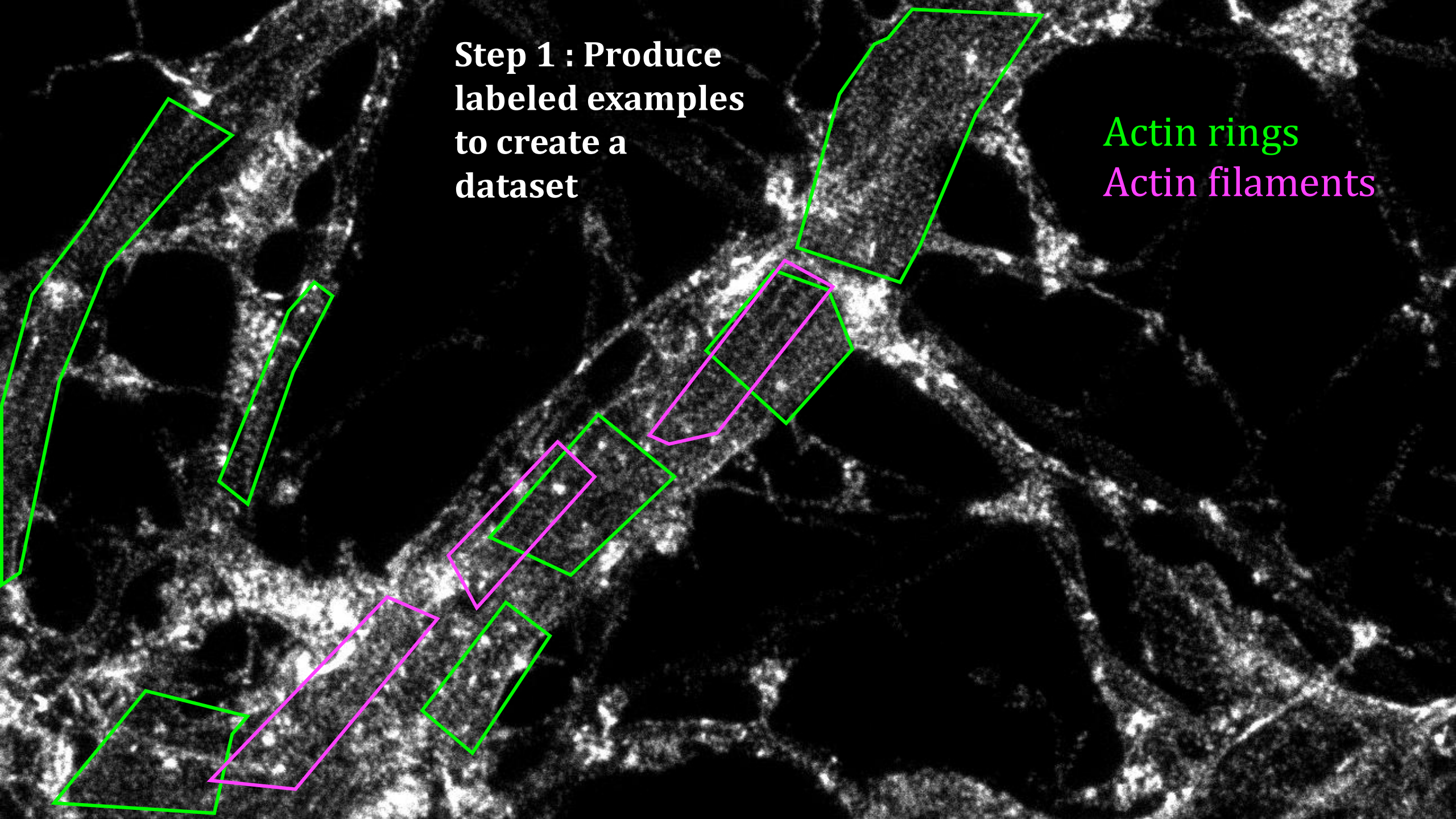


Segmentation map

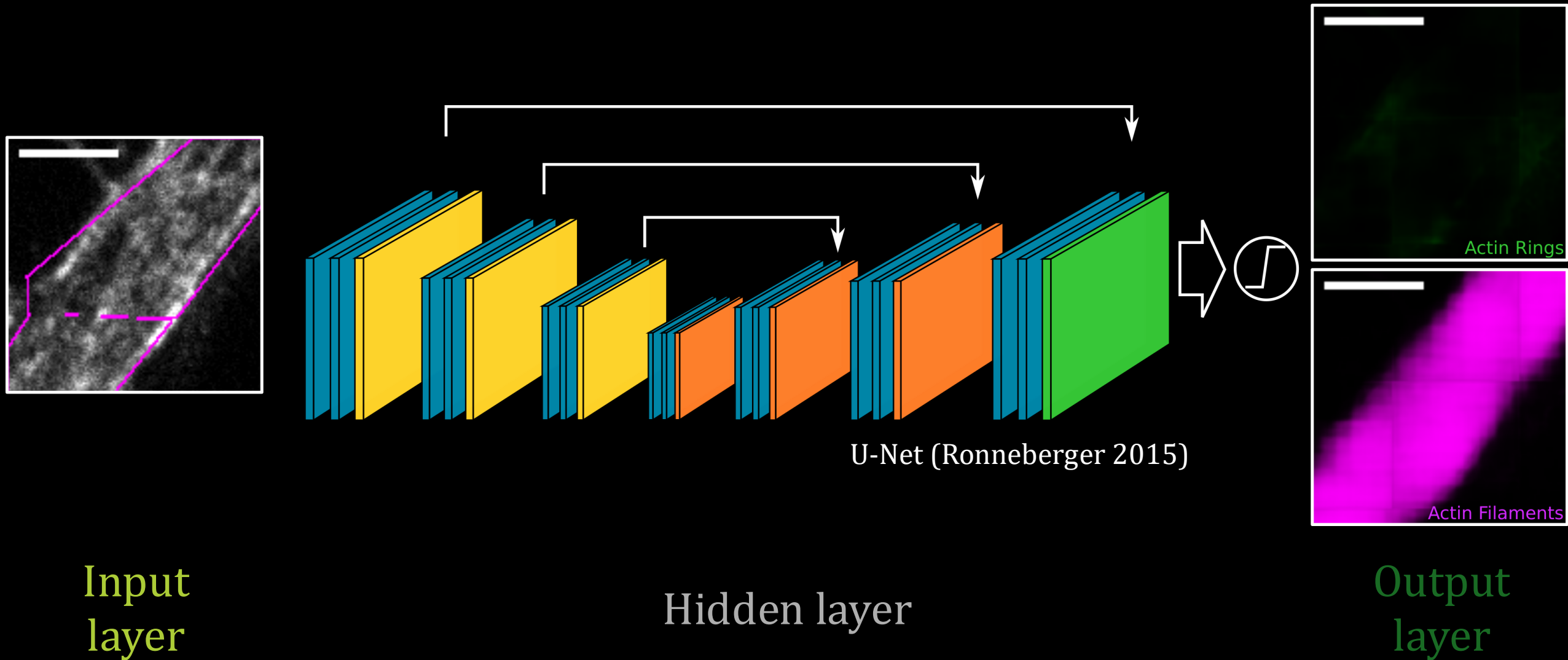


**Step 1 : Produce
labeled examples
to create a
dataset**

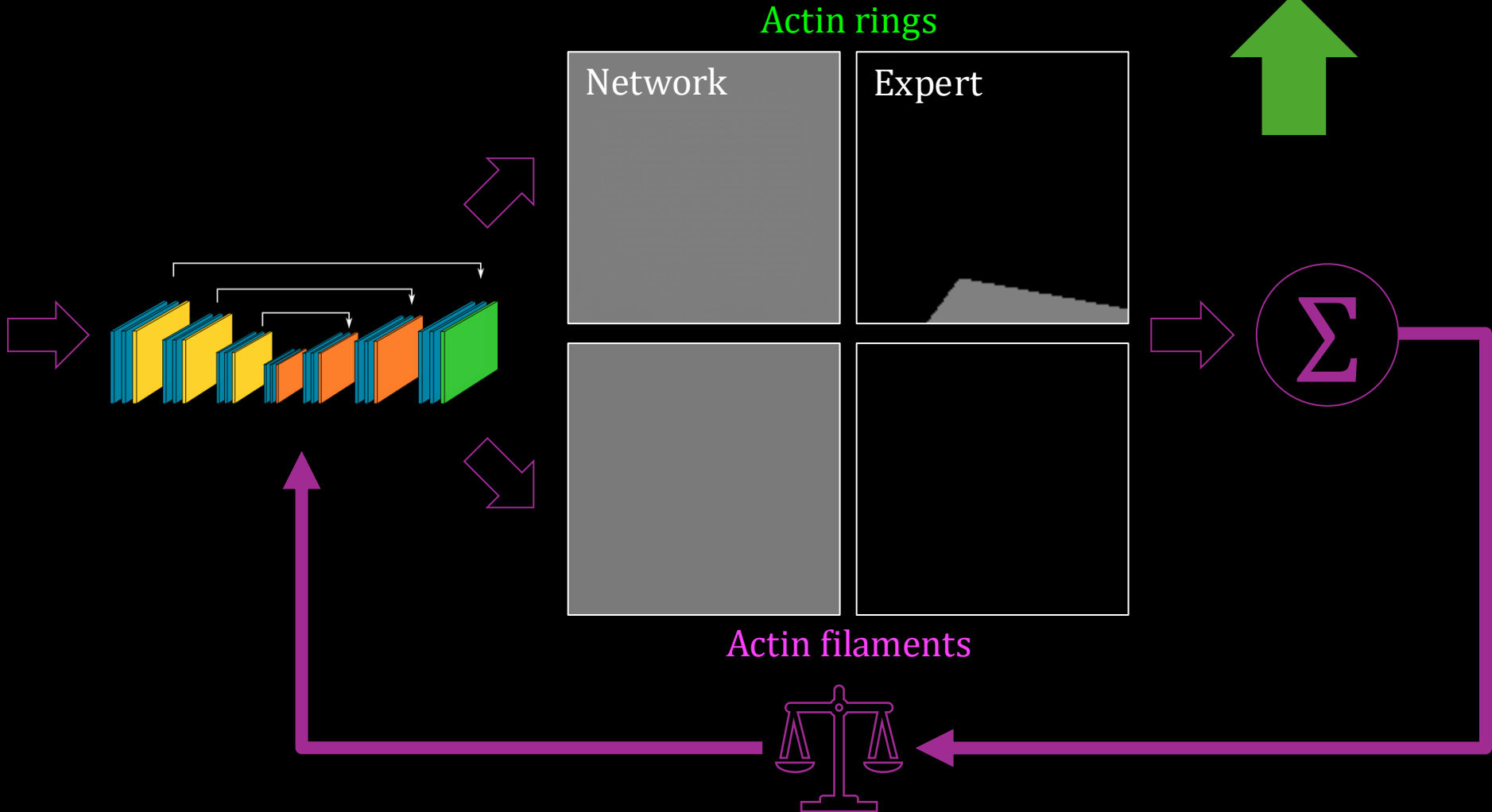
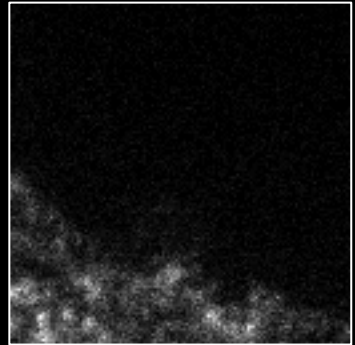
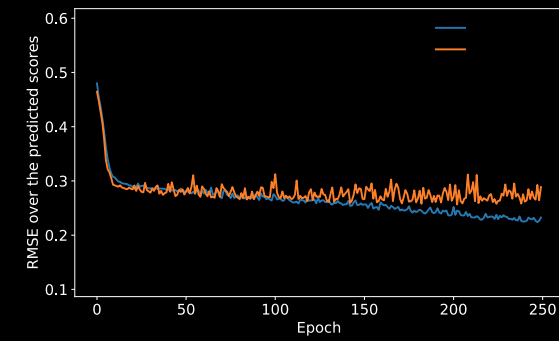
Actin rings
Actin filaments



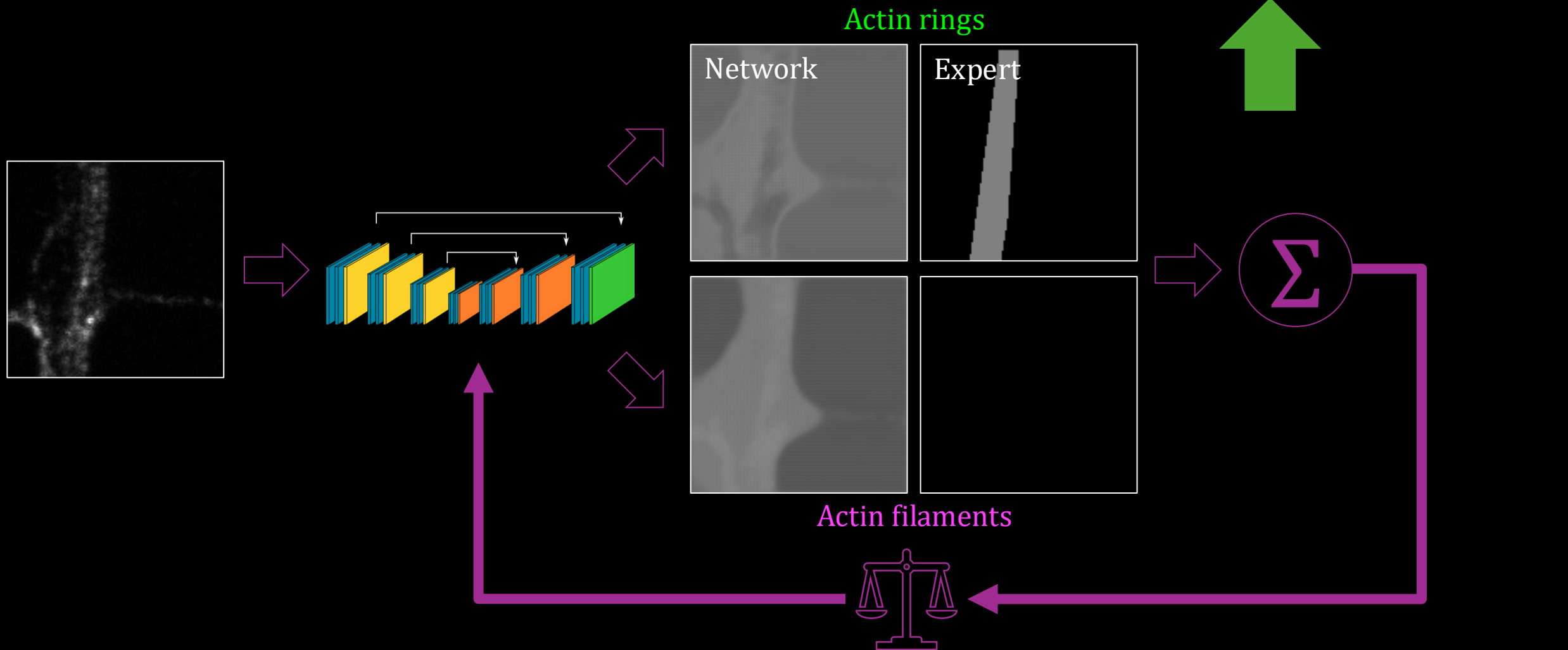
Step 2 : Design a suitable neural network



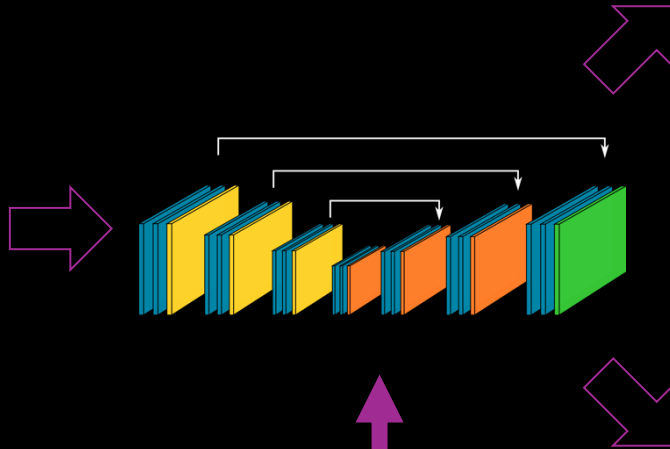
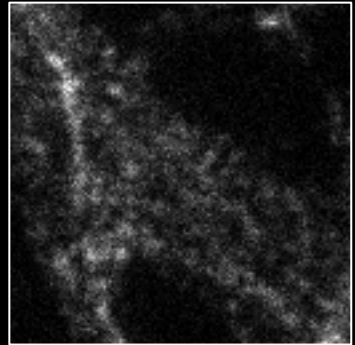
Step 3 : Train the neural network



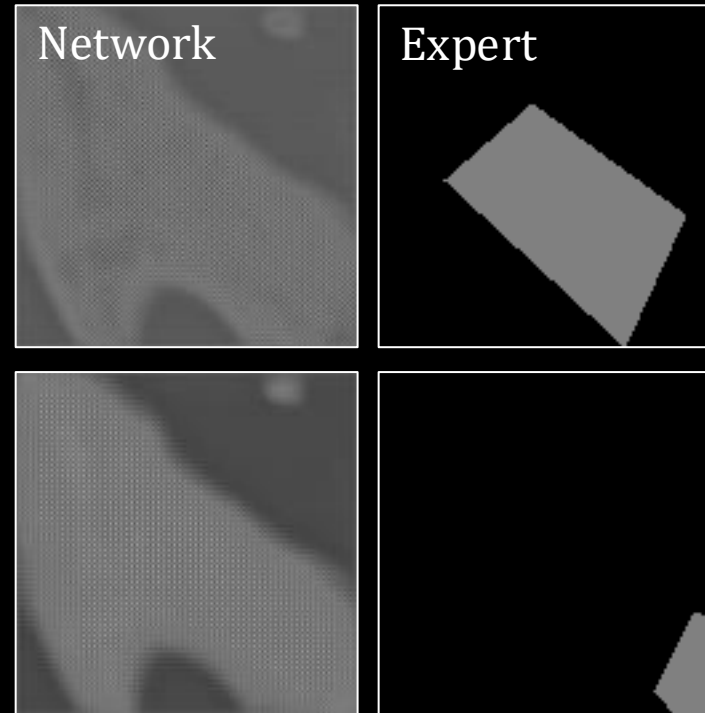
Step 3 : Train the neural network



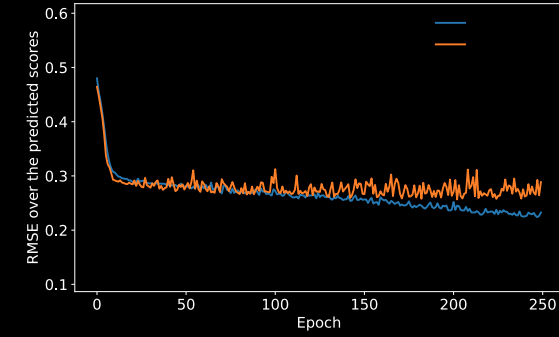
Step 3 : Train the neural network



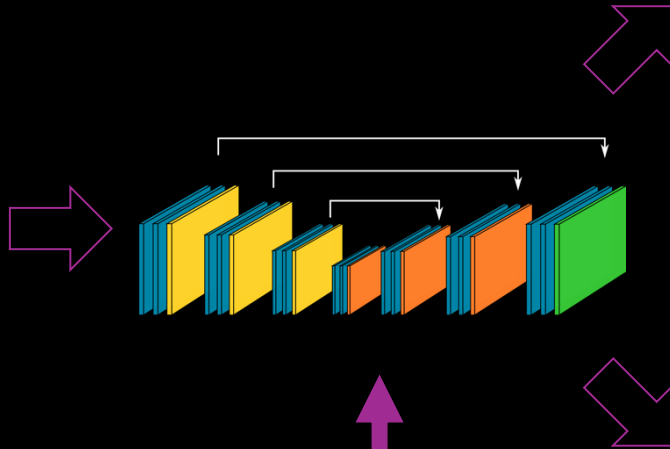
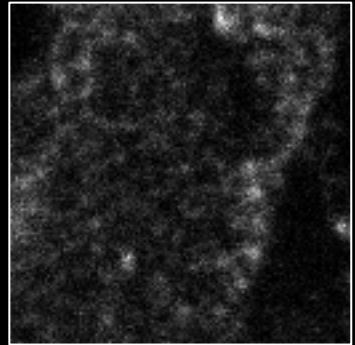
Actin rings



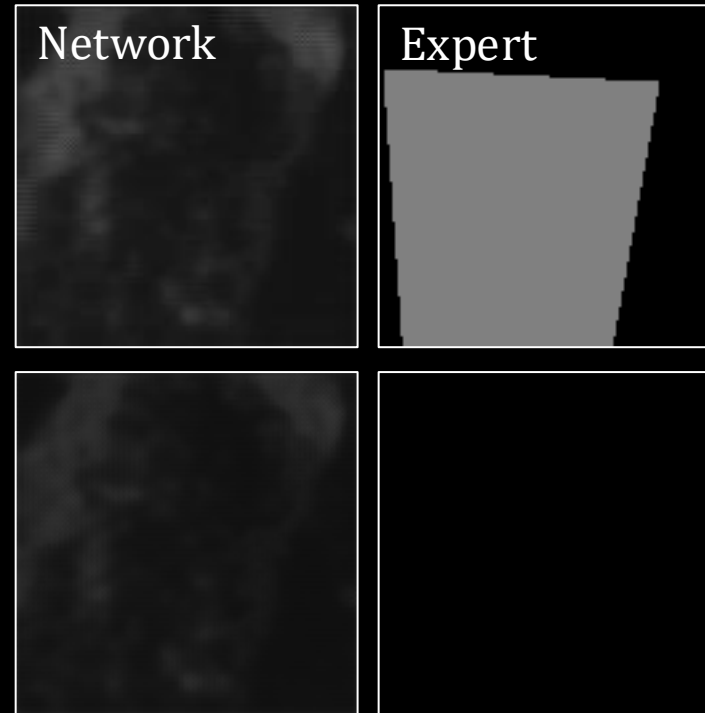
Actin filaments



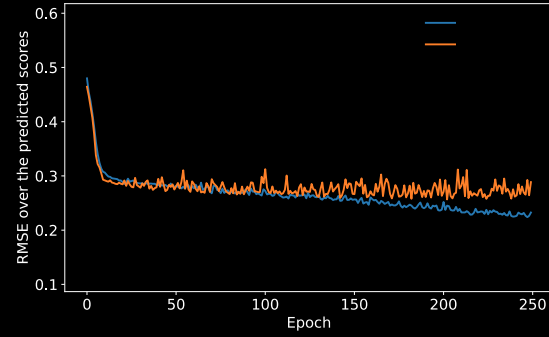
Step 3 : Train the neural network



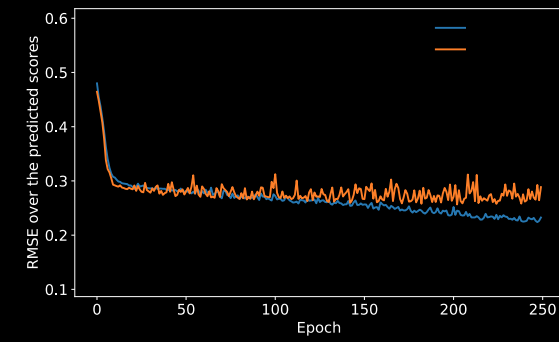
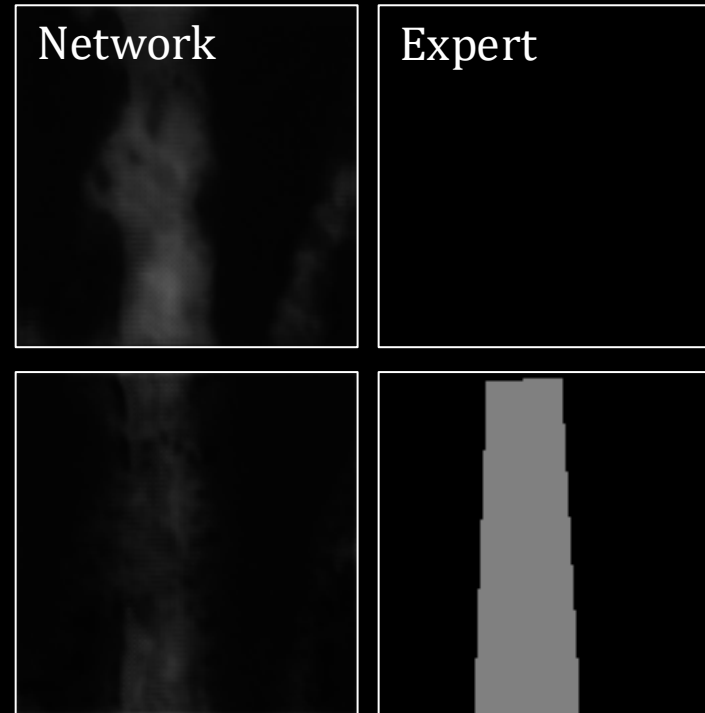
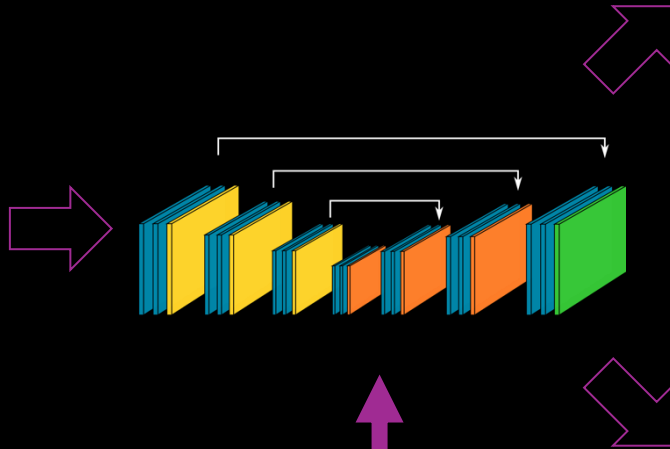
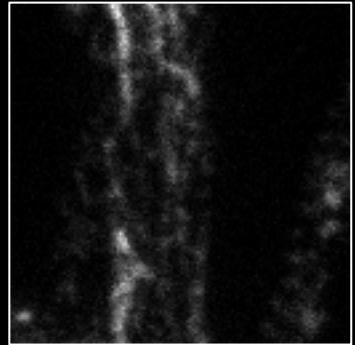
Actin rings



Actin filaments



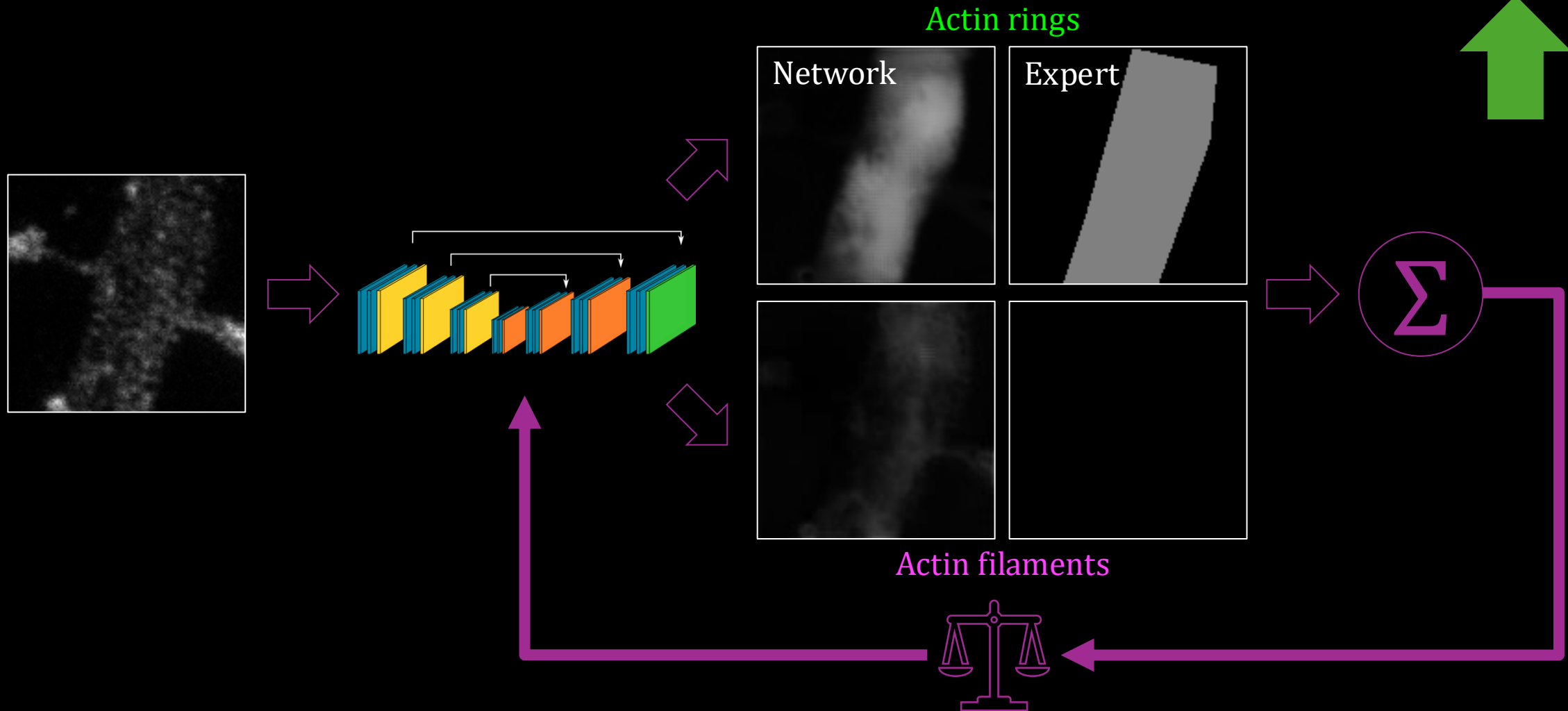
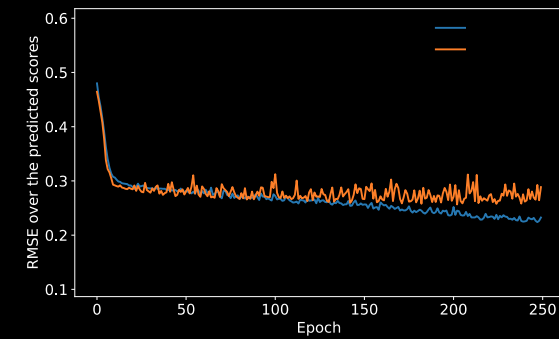
Step 3 : Train the neural network



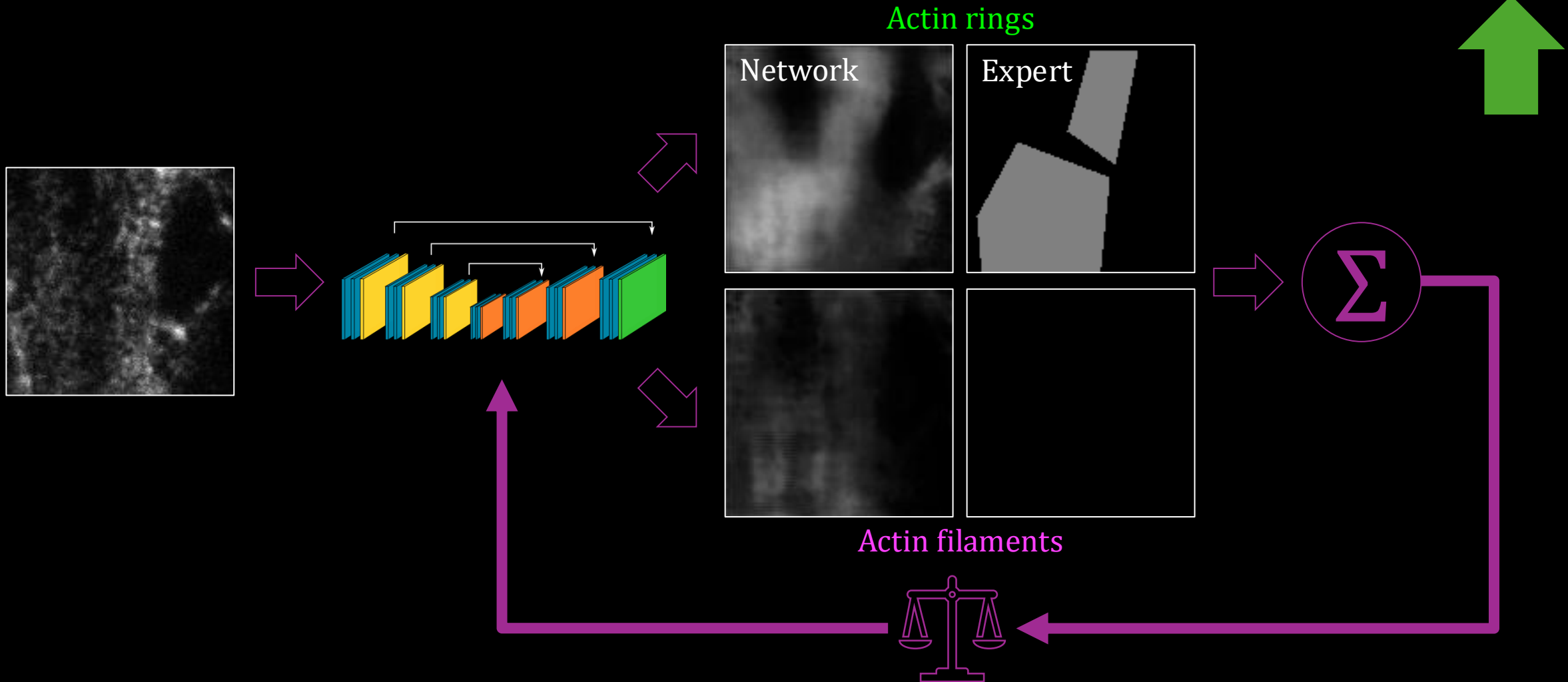
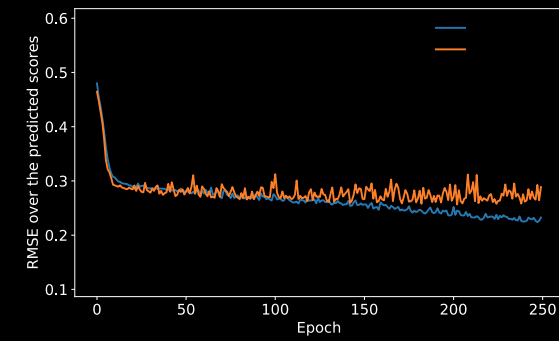
Actin filaments



Step 3 : Train the neural network

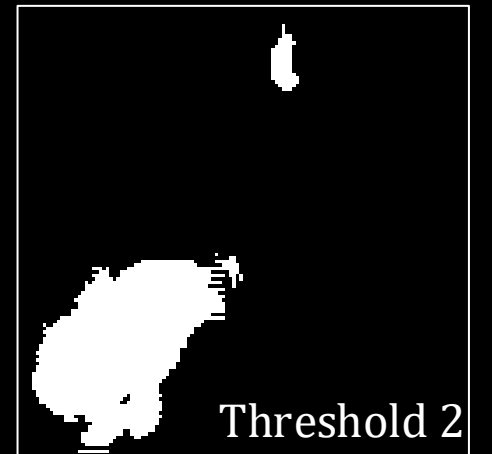
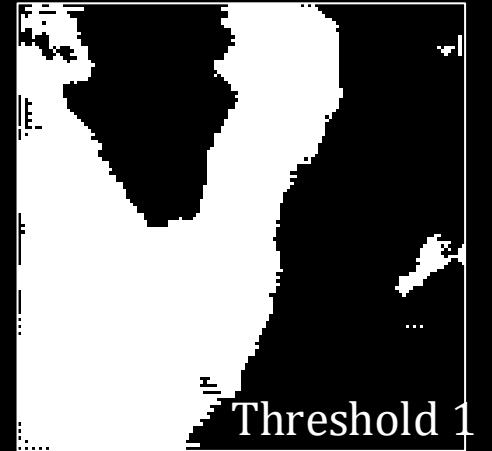
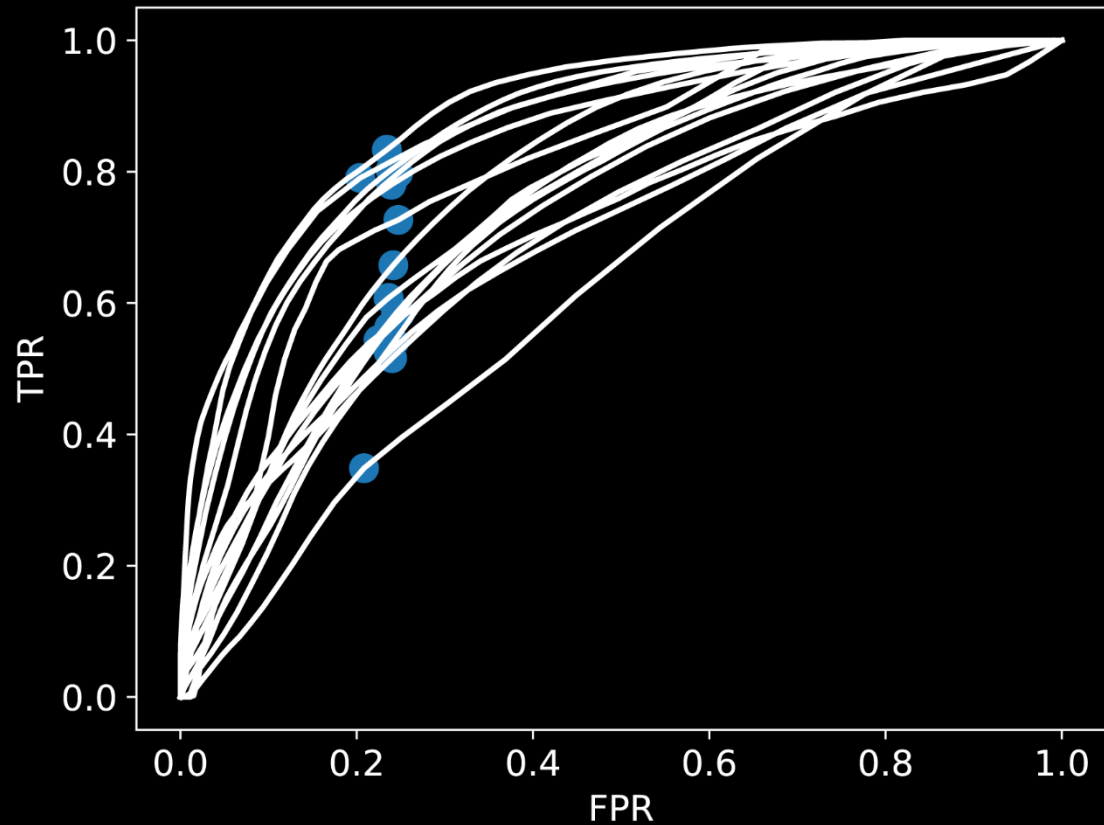


Step 3 : Train the neural network



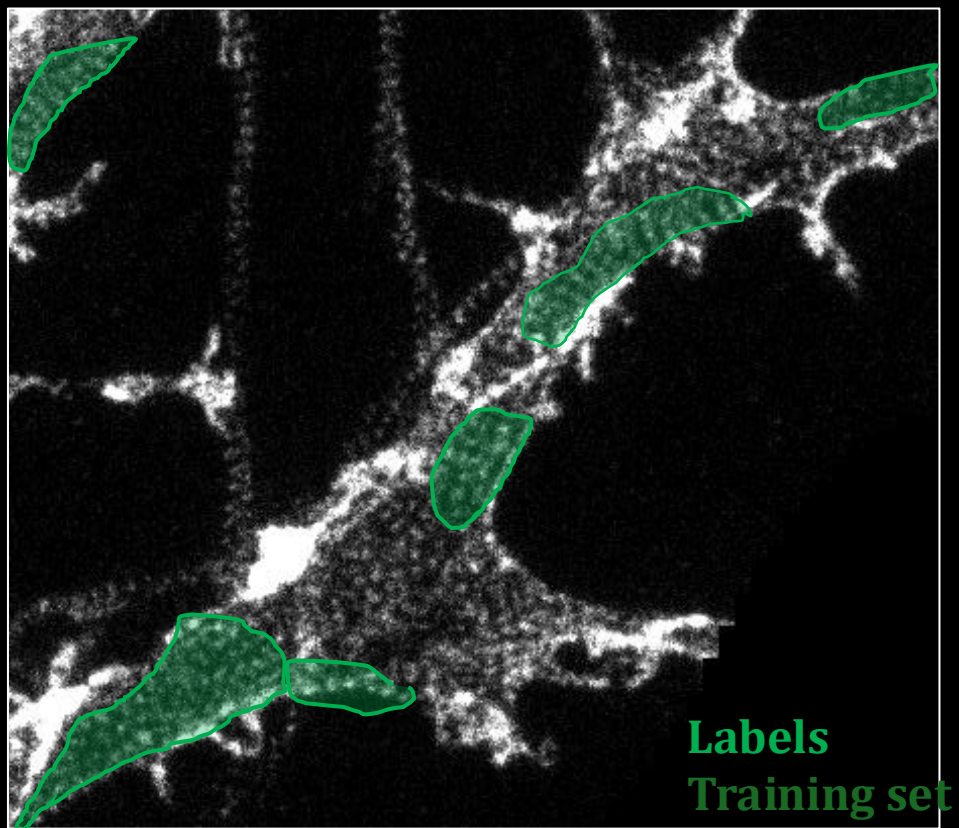
Step 4 : Determine the thresholds

Rings

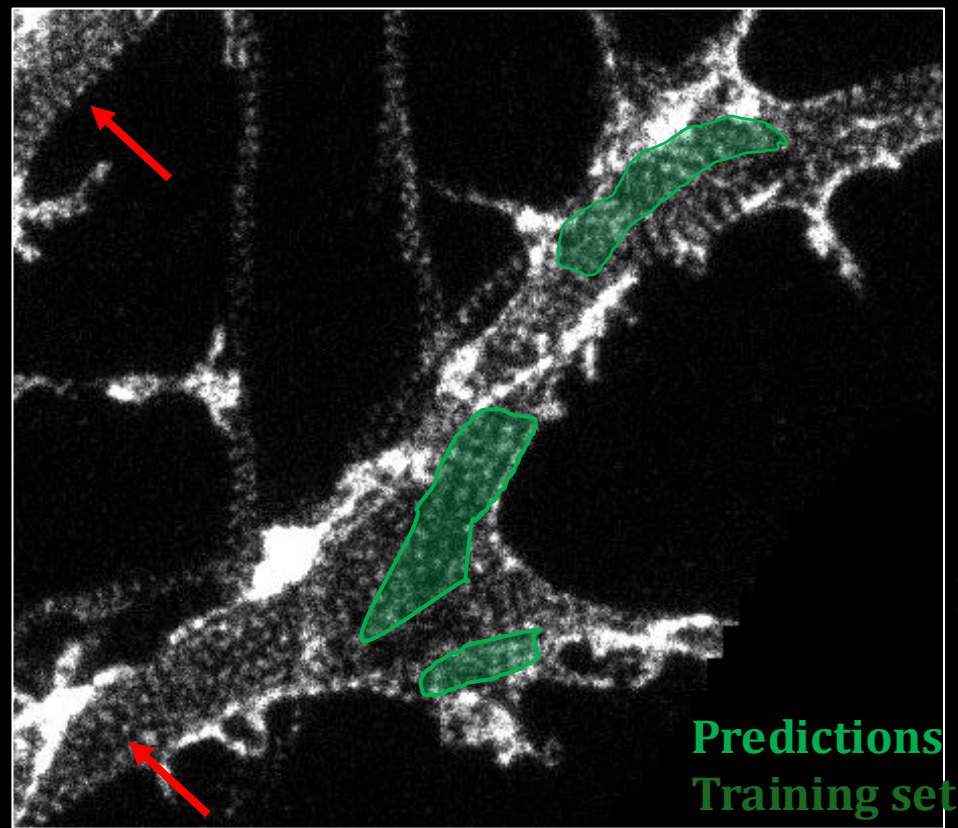


False positive rate of 0.25

Ground truth



Predictions

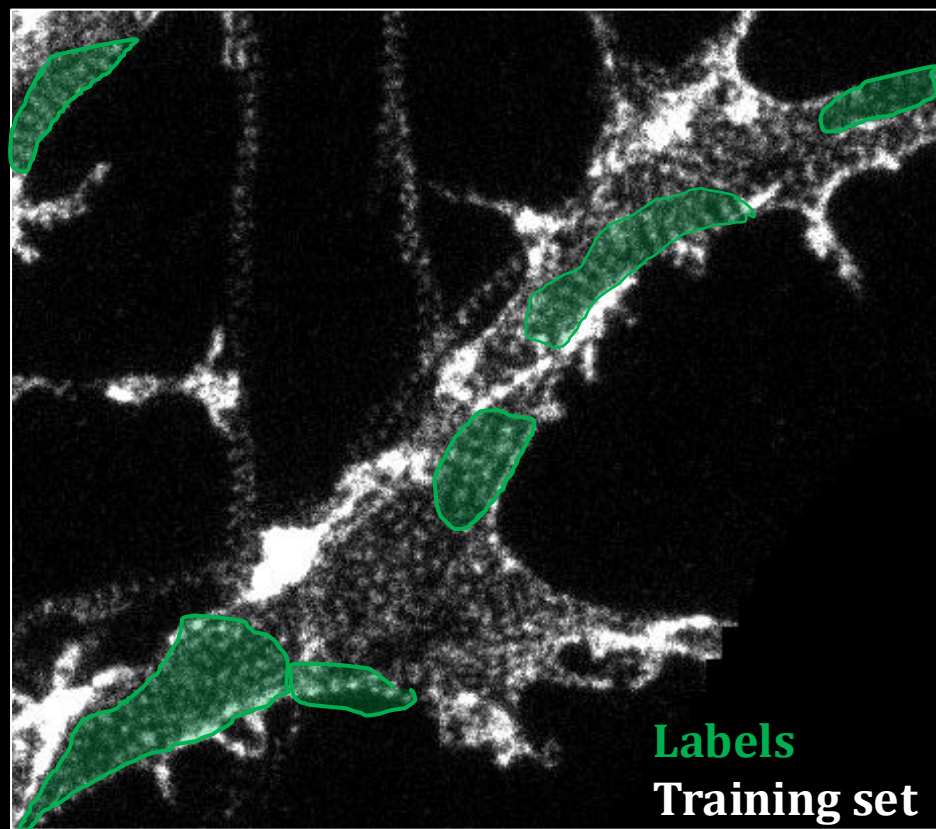


Ground truth and
predictions are very
different :

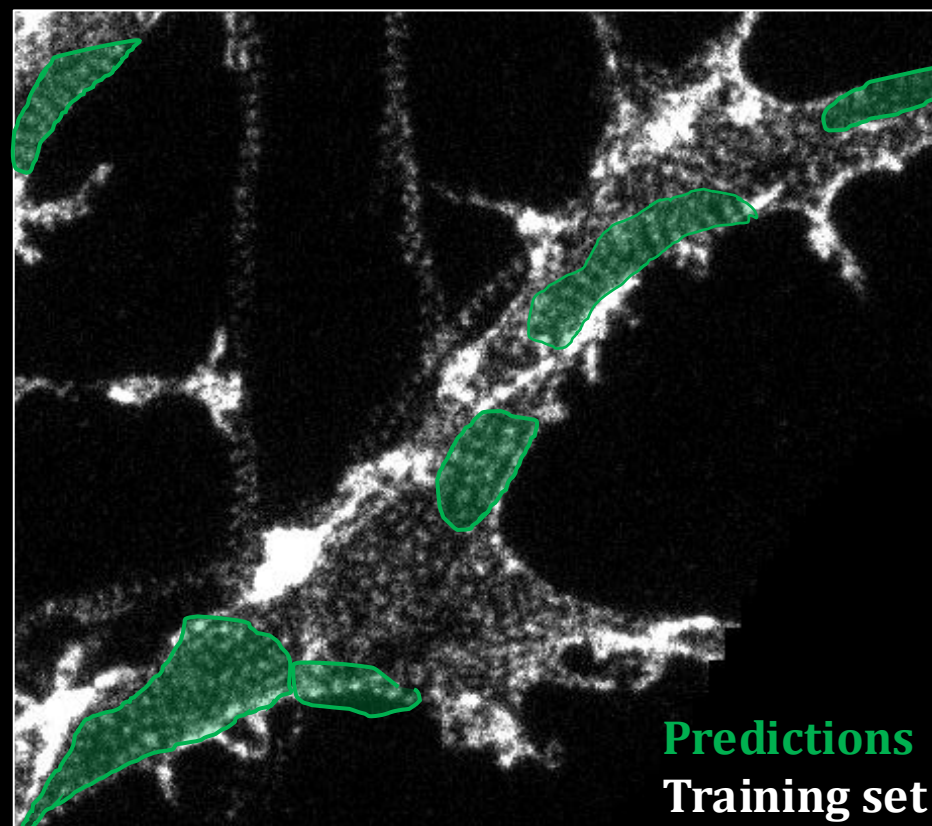
High loss

Model is not optimal

Ground truth



Predictions

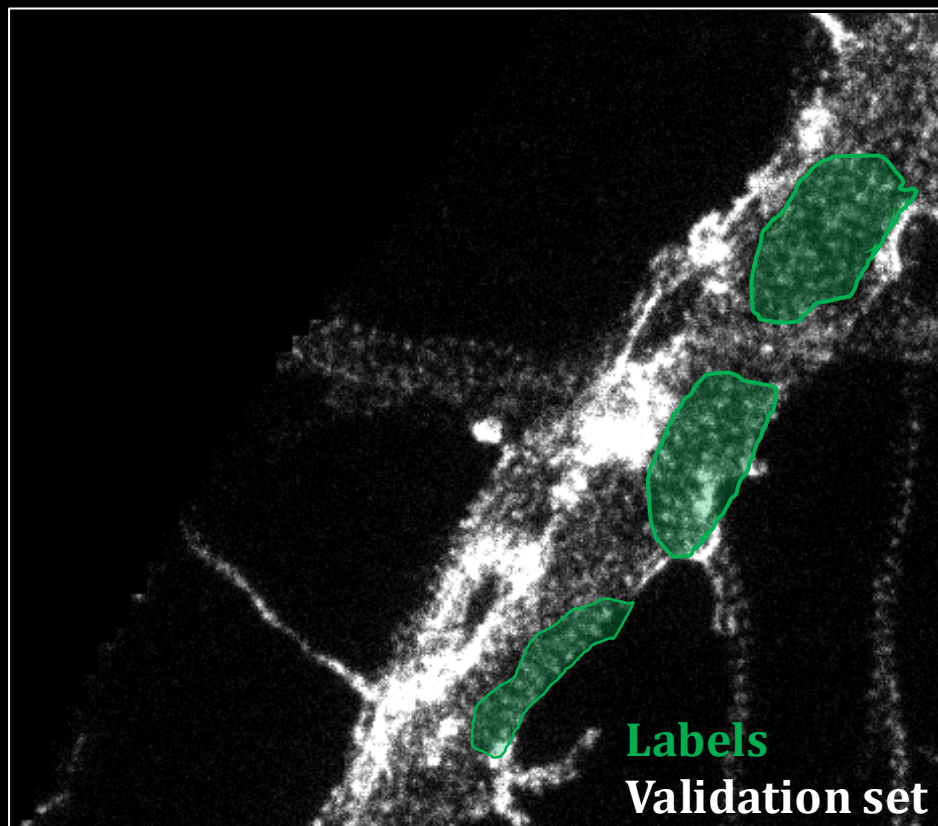


Ground truth and predictions are very almost the same:

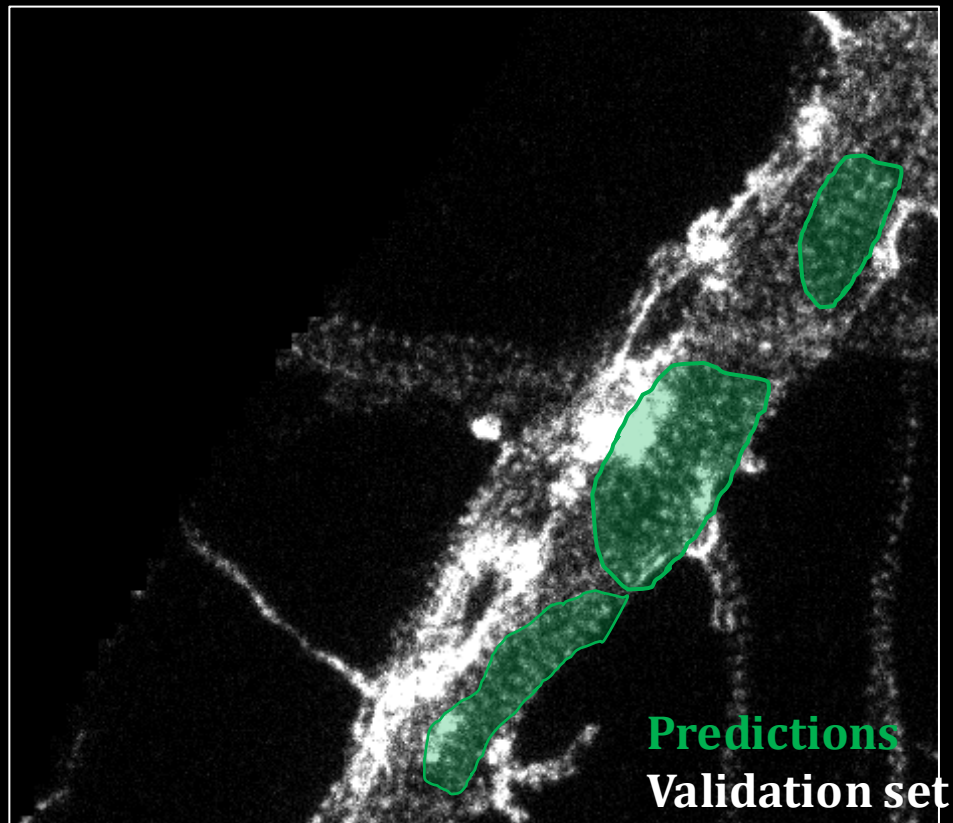
Low loss

Does the model generalize well?

Ground truth



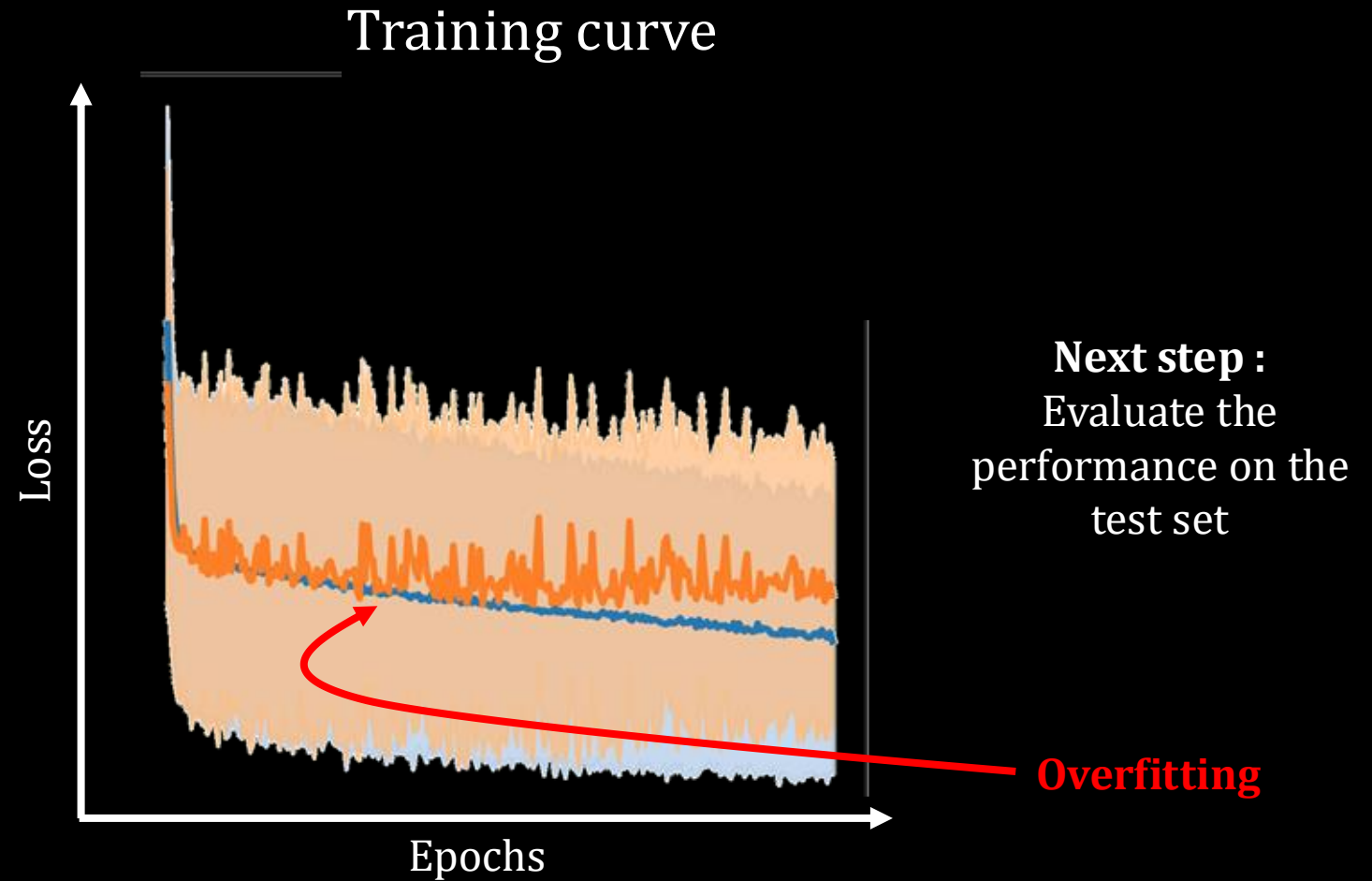
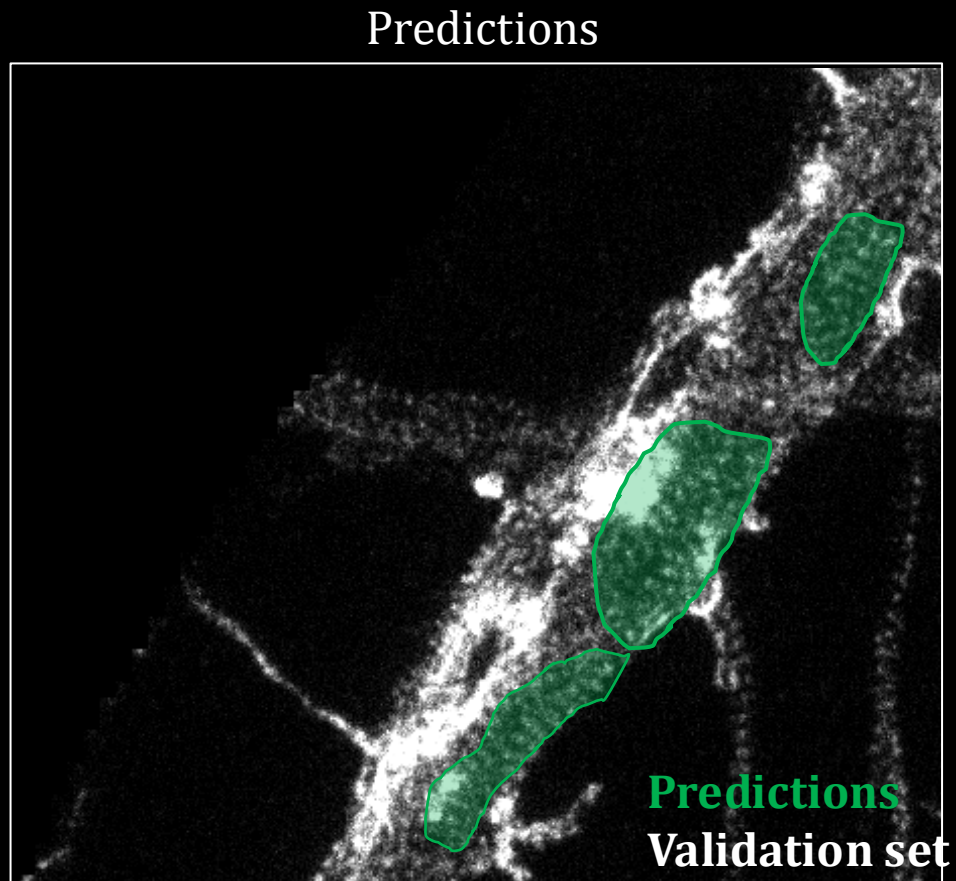
Predictions



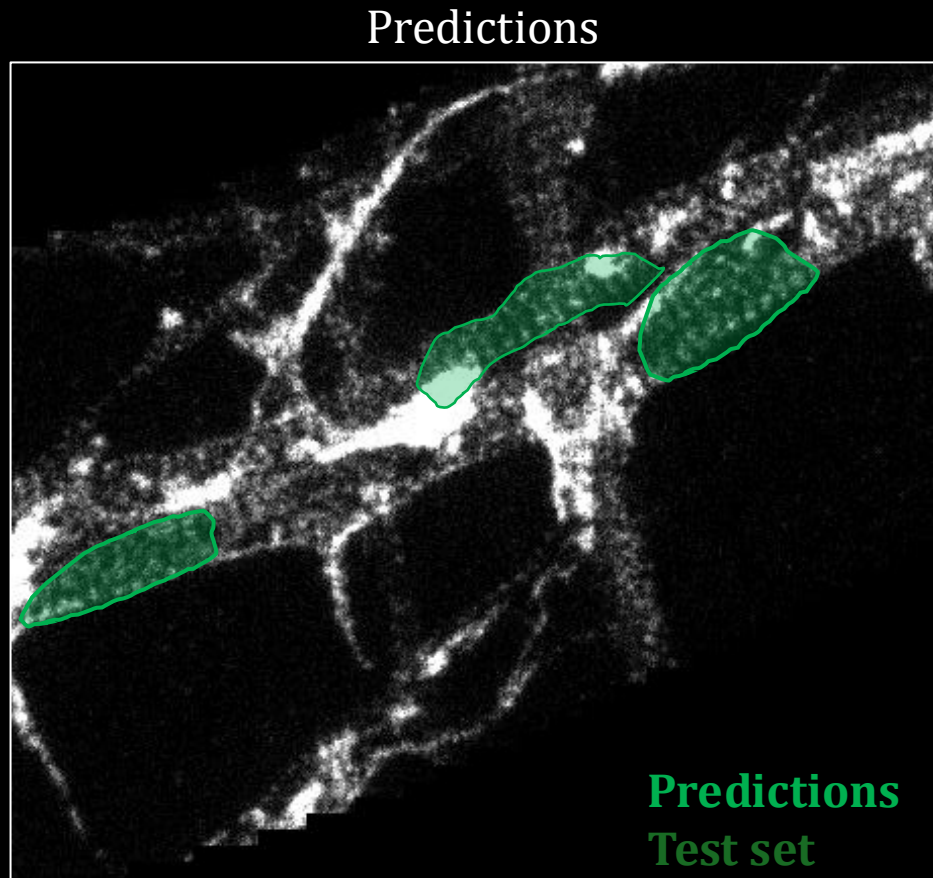
Ground truth and predictions are similar: **Loss was minimized**

Choose the model based on the validation loss

Decide the model to use



Evaluate the performance on the test set



The output is often a probability :

- Use it as it is
- Convert into binary (thresholding)

Classification problems :

- Accuracy
- Precision
- Recall

Pixel-wise classification = segmentation

- Decision threshold
 - Map a logistic regression value (*probability*) into a binary value (*yes/no, class1/class2*)
- **True positive :**
 - Label = Actin rings
 - Prediction = Actin rings
- **False positive :**
 - Label = Background
 - Prediction = Actin rings
- **False negative :**
 - Label = Actin rings
 - Prediction = Background
- **True negative :**
 - Label = Background
 - Prediction = Actin rings

Confusion Matrix

		Predictions	
Labels	Background	True Negative	False Positive
	Actin rings	False Negative	True Positive
		Background	Actin rings

Classification

- Accuracy :

- fraction of right predictions

- $Accuracy = \frac{\text{Number of correct predictions}}{\text{Total predictions}} = \frac{TP+TN}{TP+TN+FP+FN}$

- Precision :

- proportion of positive predictions that were right

- $Precision = \frac{TP}{TP+FP}$

- Recall :

- proportion of positive labels that were identified correctly

- $Recall = \frac{TP}{TP+FN}$

Classification

- Accuracy : fraction of right predictions

- $Accuracy = \frac{\text{Number of correct predictions}}{\text{Total predictions}} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{9+13}{9+13+6+0} = 0.79$

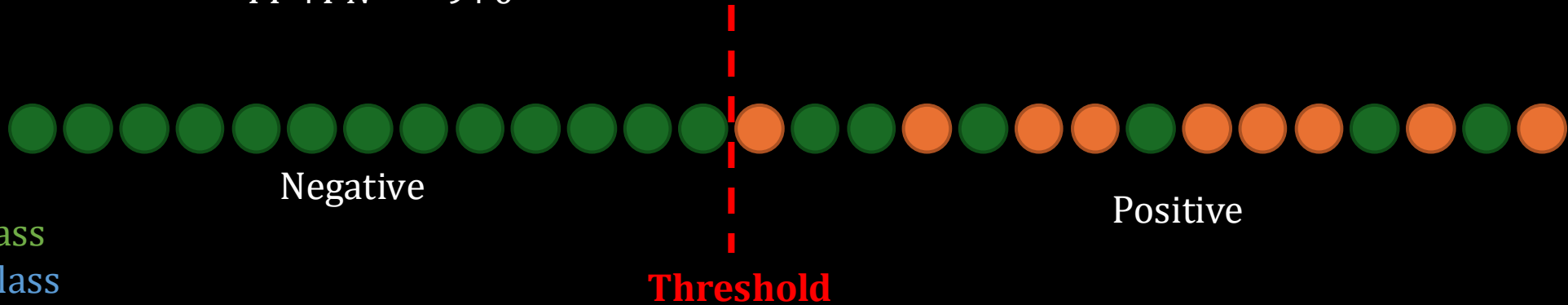
- Precision : proportion of positive predictions that were right

- $Precision = \frac{TP}{TP+FP} = \frac{9}{9+6} = 0.6$

- Recall : proportion of positive labels that were identified correctly

- $Recall = \frac{TP}{TP+FN} = \frac{9}{9+0} = 1.0$

N = 28



Positive class

Negative class

Threshold

Classification

- Accuracy : fraction of right predictions

$$\bullet \text{ Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total predictions}} = \frac{TP+TN}{TP+TN+FP+FN} = \frac{7+16}{7+16+3+2} = 0.82$$

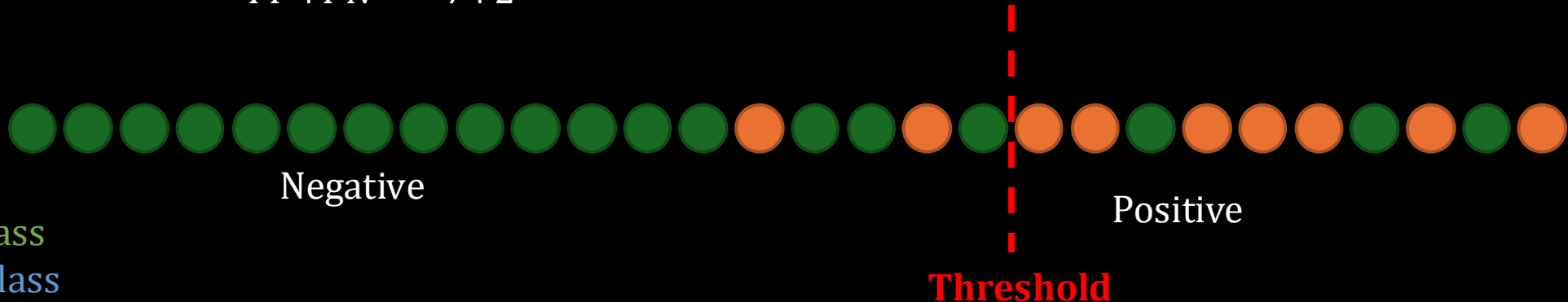
- Precision : proportion of positive predictions that were right

$$\bullet \text{ Precision} = \frac{TP}{TP+FP} = \frac{7}{7+3} = 0.7$$

- Recall : proportion of positive labels that were identified correctly

$$\bullet \text{ Recall} = \frac{TP}{TP+FN} = \frac{7}{7+2} = 0.78$$

N = 28

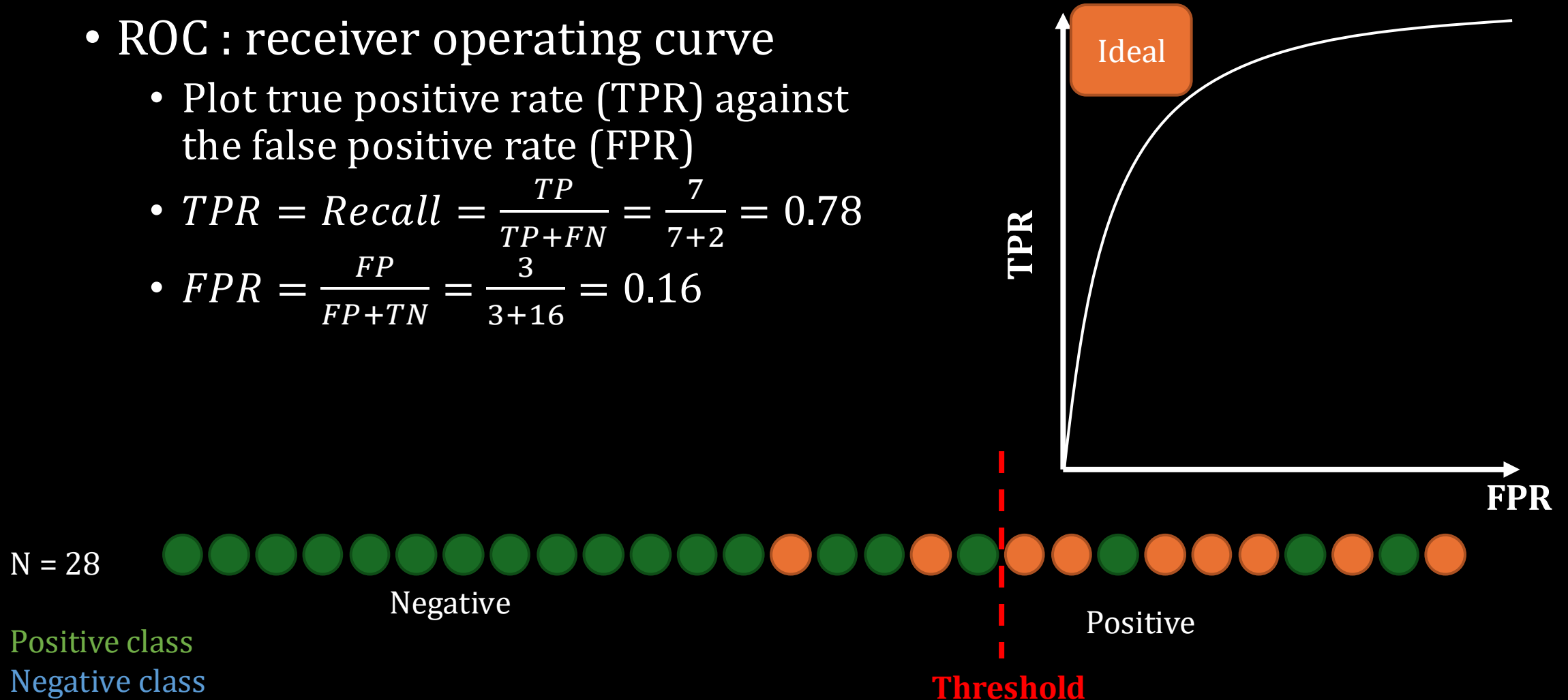


Positive class

Negative class

Classification – ROC curve

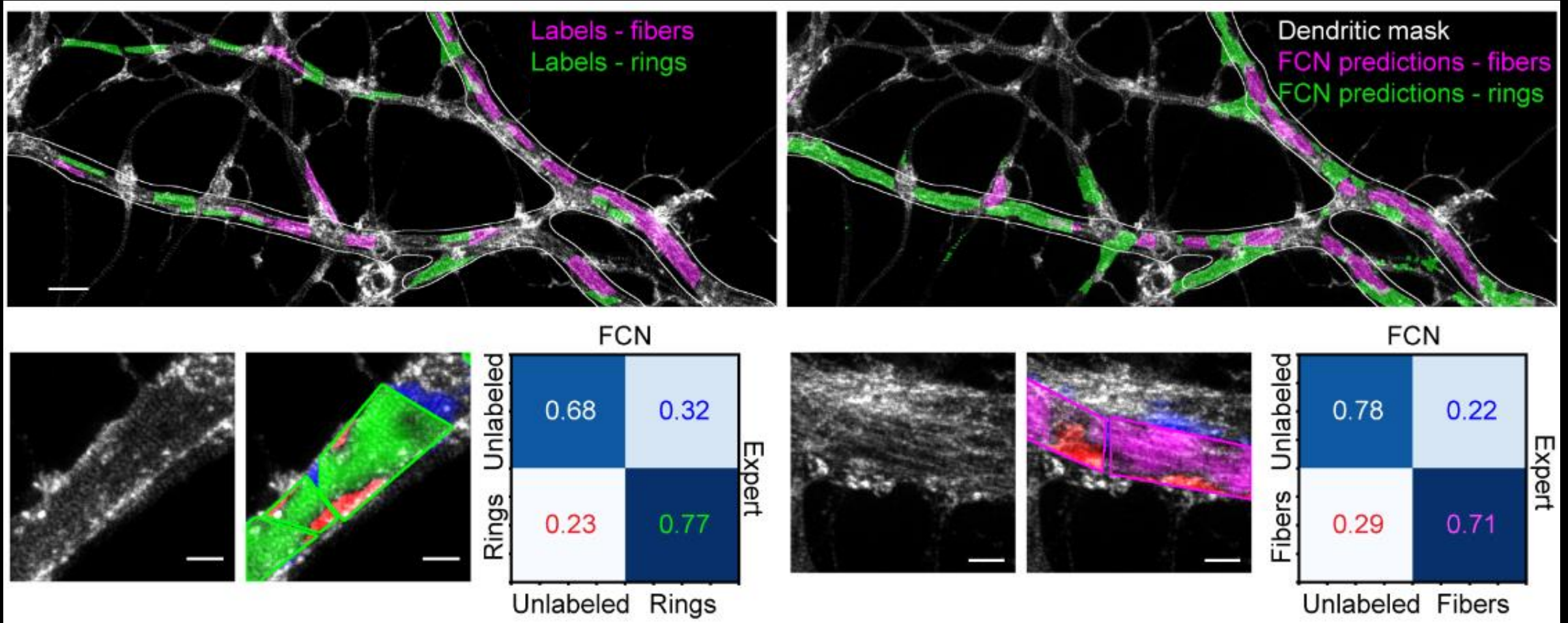
- ROC : receiver operating curve
 - Plot true positive rate (TPR) against the false positive rate (FPR)
 - $TPR = Recall = \frac{TP}{TP+FN} = \frac{7}{7+2} = 0.78$
 - $FPR = \frac{FP}{FP+TN} = \frac{3}{3+16} = 0.16$



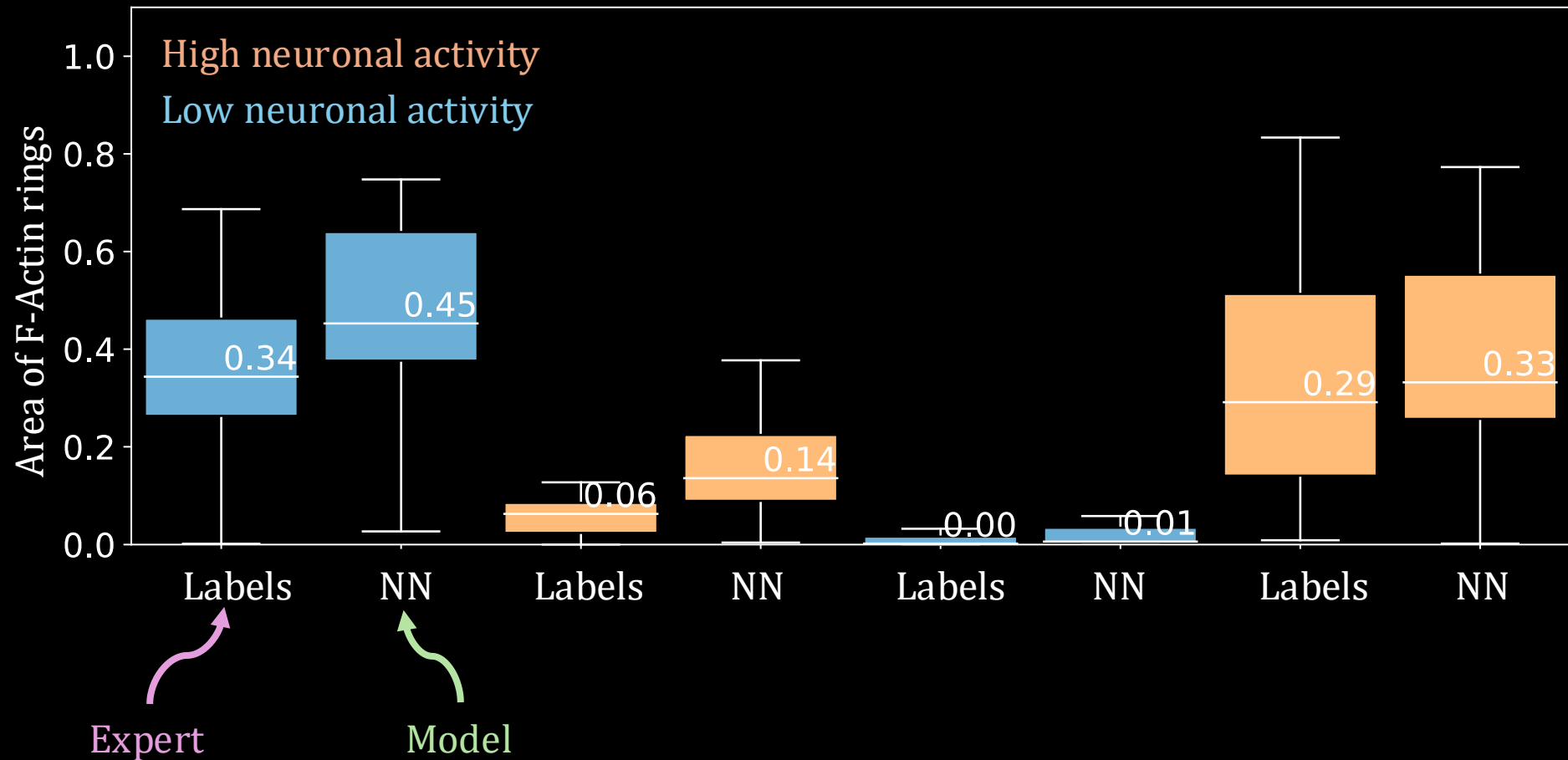
Pixel-wise classification : semantic segmentation

← **Feature 1 : Rings**
Feature 2 : Fibers →

Step 5 : Evaluate performance on the test set



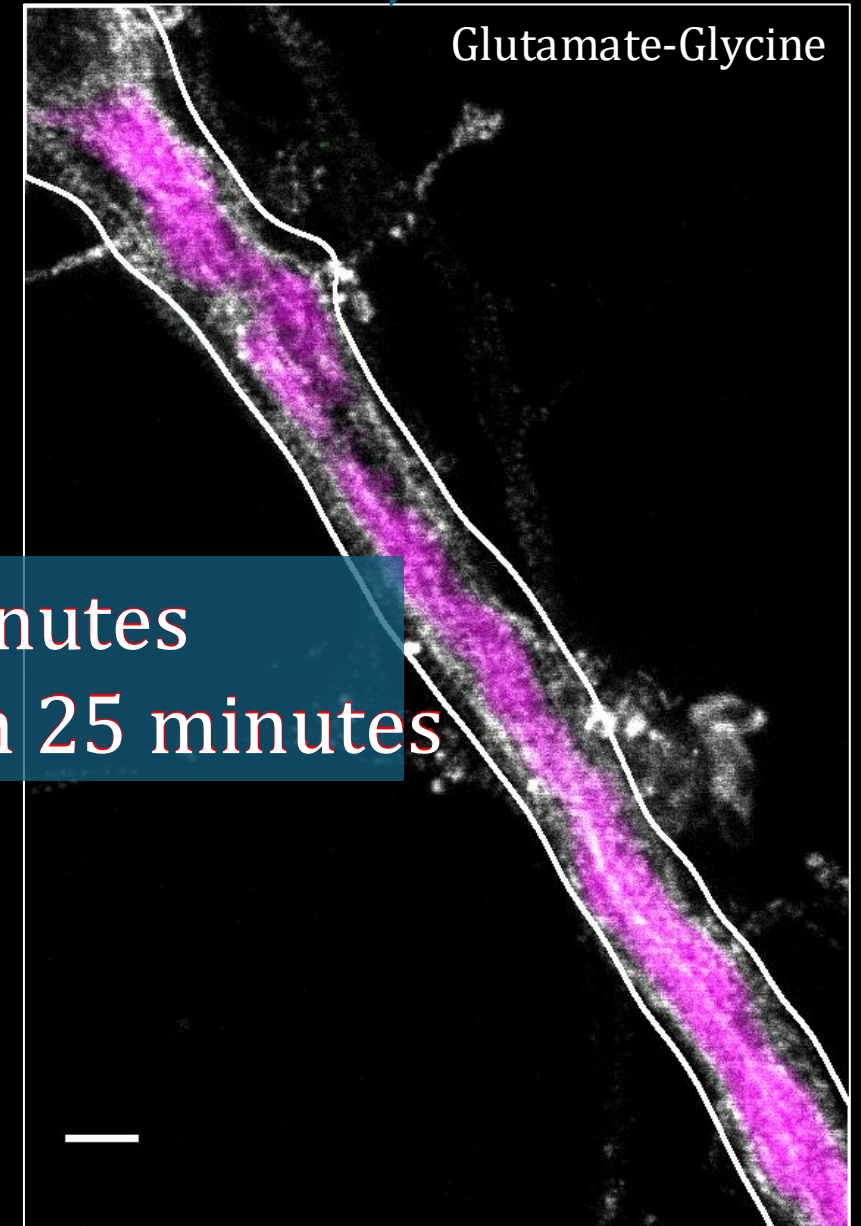
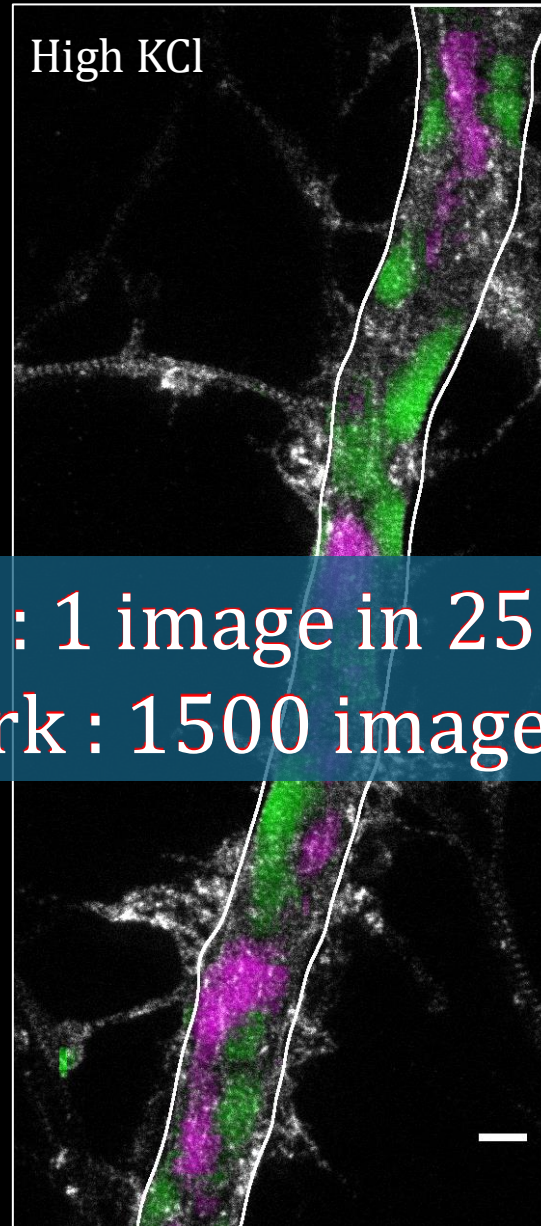
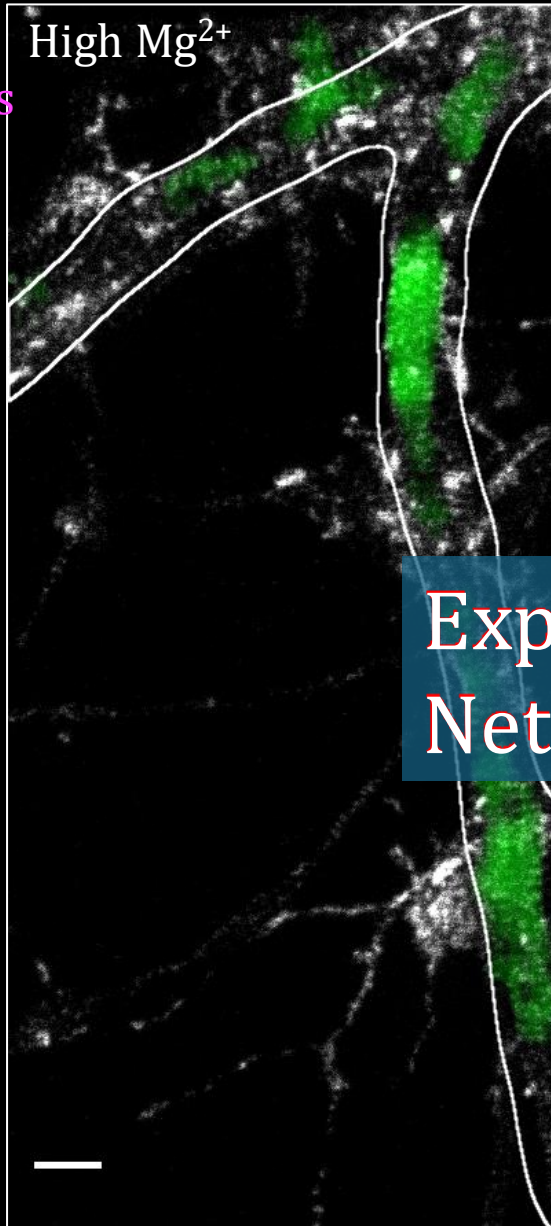
Step 5 : Evaluate performance on the test set



Step 6 : Inference on real data

Increasing neuronal activity

Actin rings
Actin fibers



Expert : 1 image in 25 minutes
Network : 1500 images in 25 minutes

Automating the analysis of two-photon microscopy images of the dopaminergic neuronal circuit

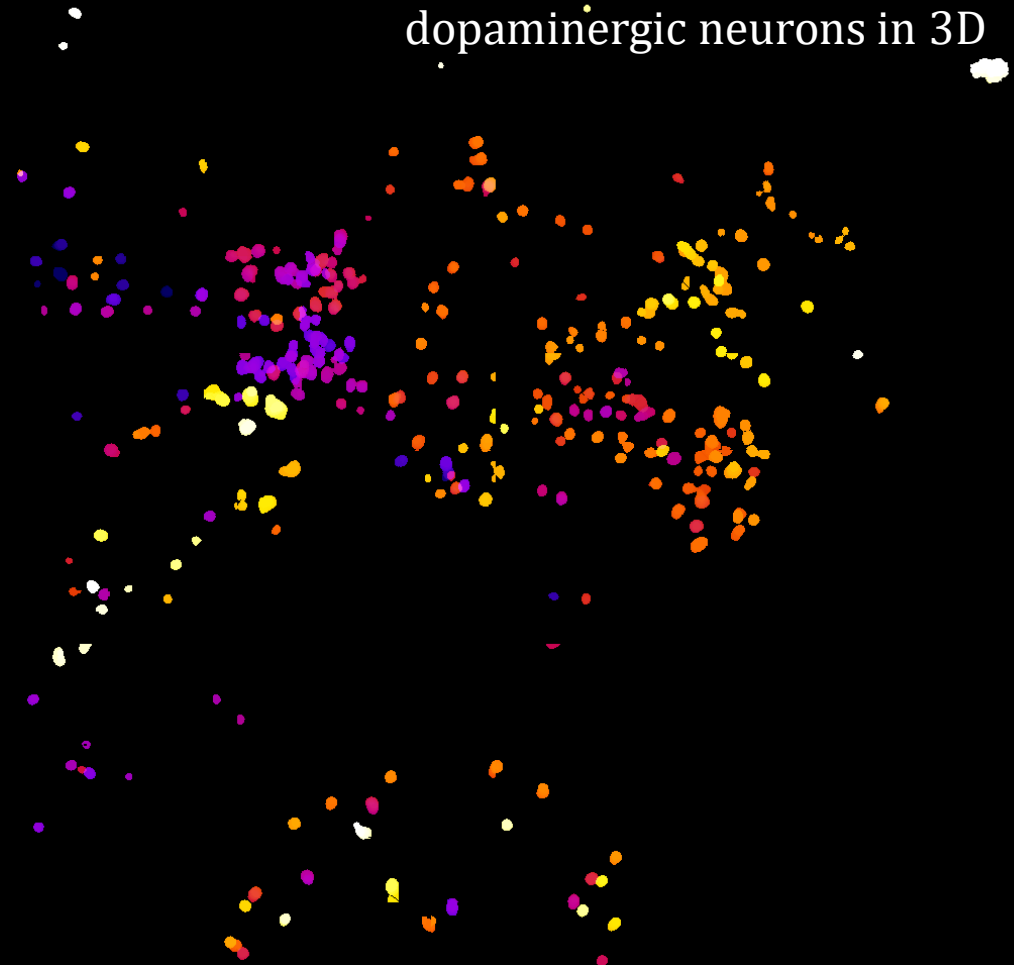
Original z-stack of
DAT-GFP neurons
in the zebrafish brain



Deep learning

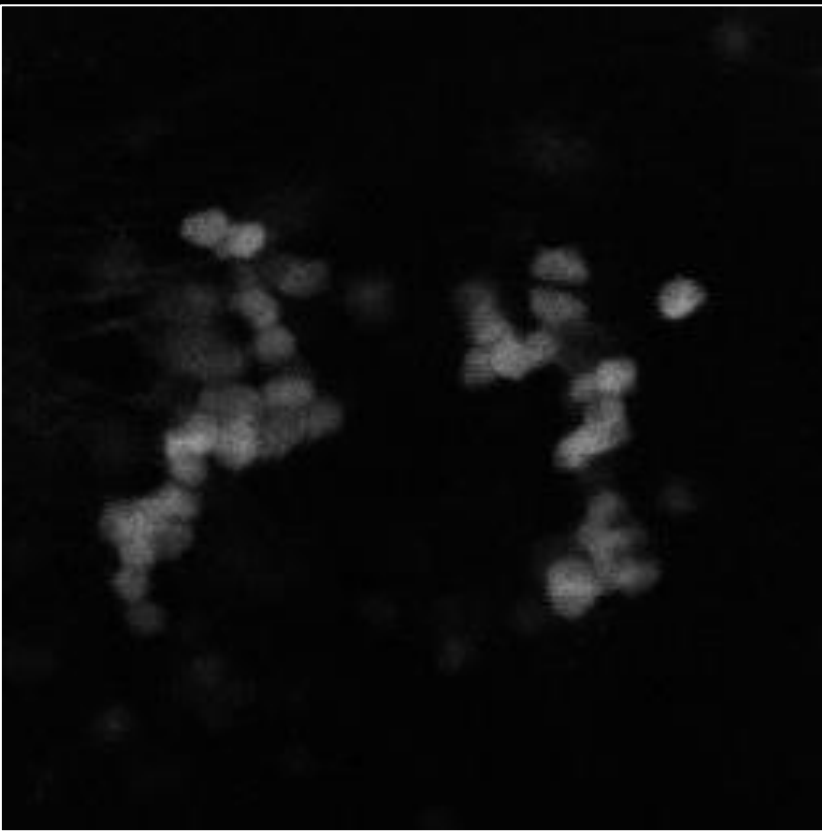


Segmentation map of
dopaminergic neurons in 3D

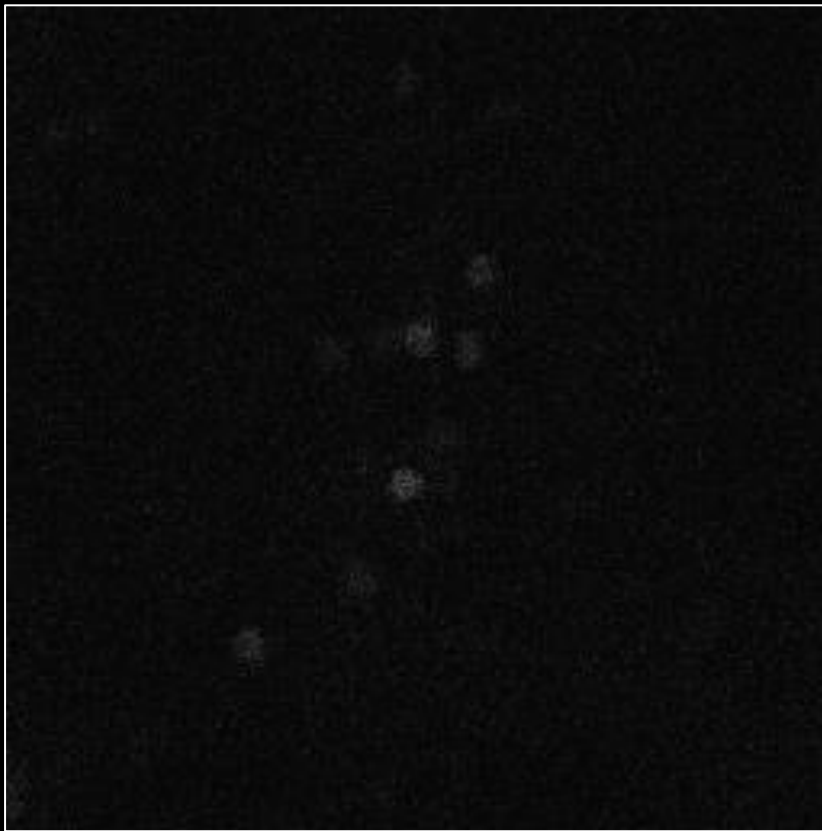


Detection challenges

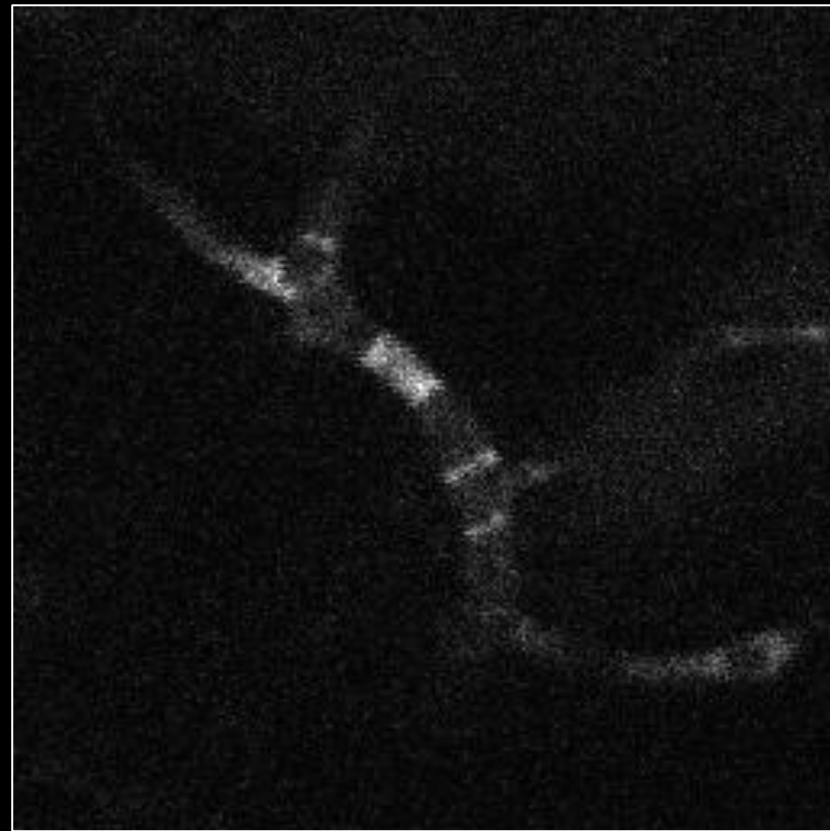
Dense regions



Intensity and contrast variations

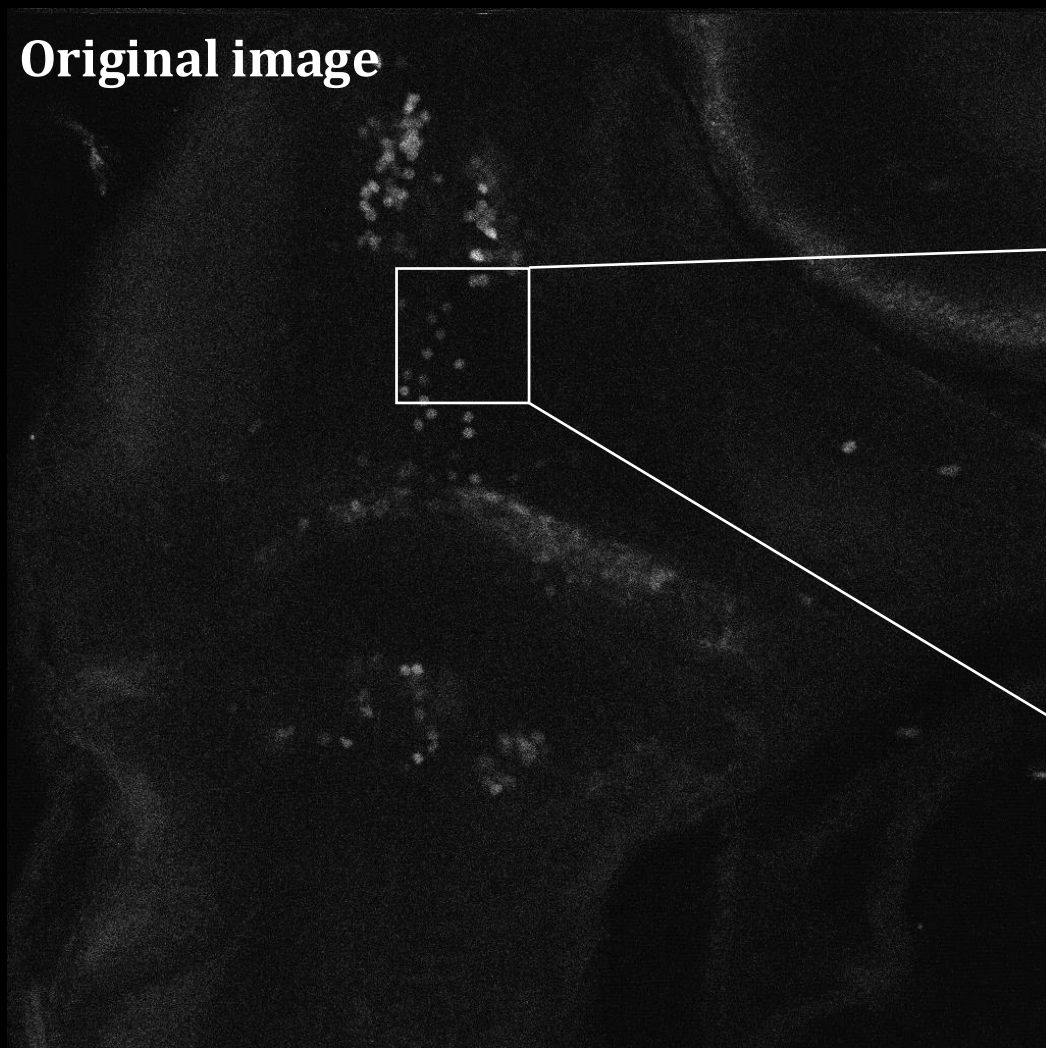


Undesirable foreground

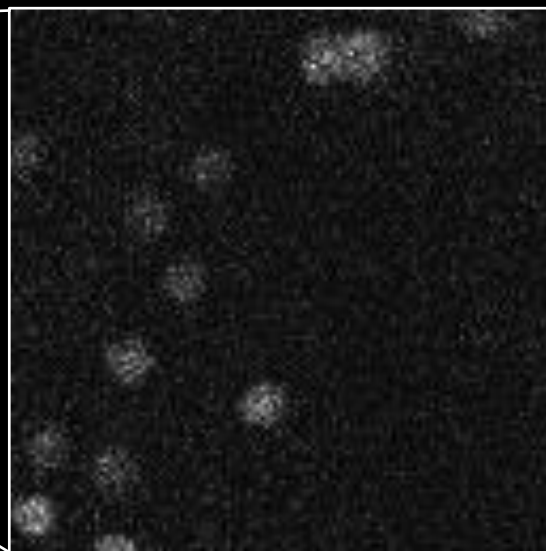


Dataset

Original image

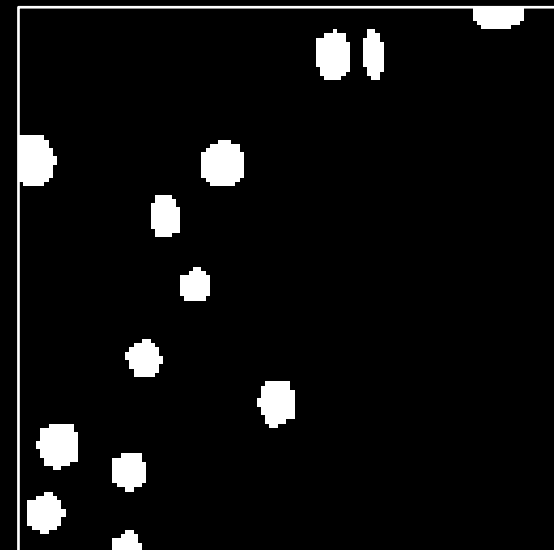


Crop
128 x 128 pixel

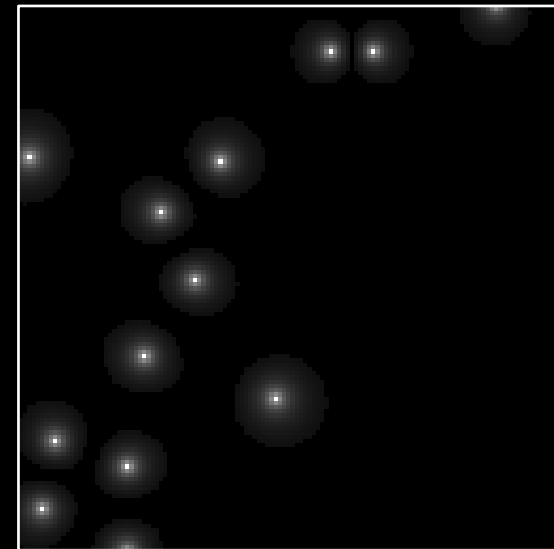


60% Training
25% Validation
15% Testing

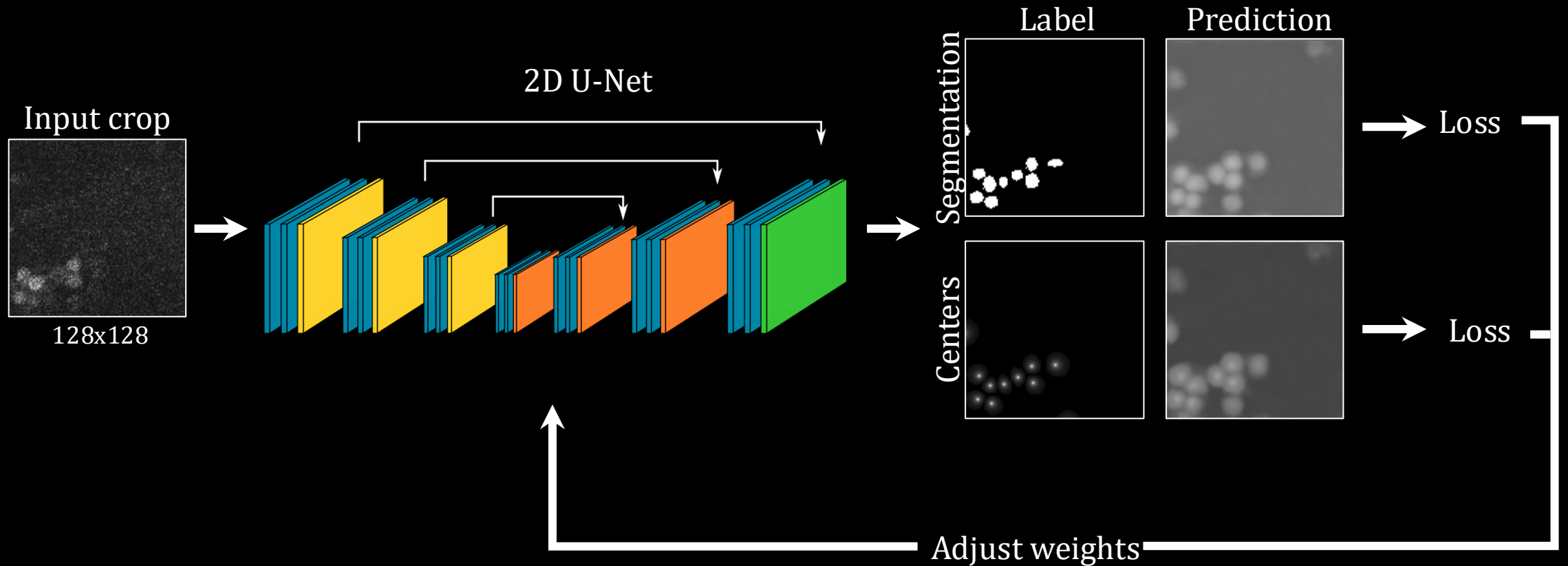
Manual labels

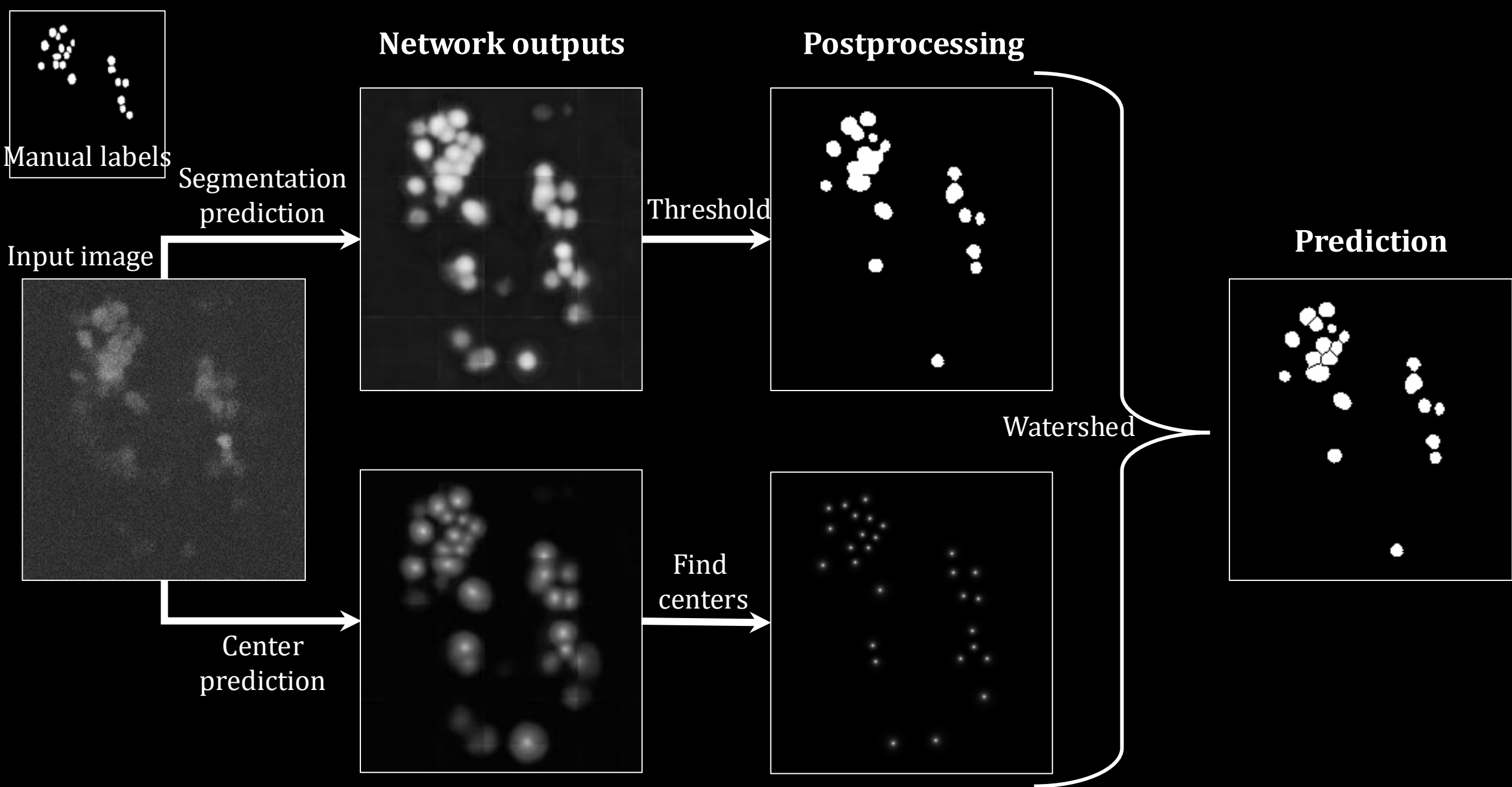


Cell centers

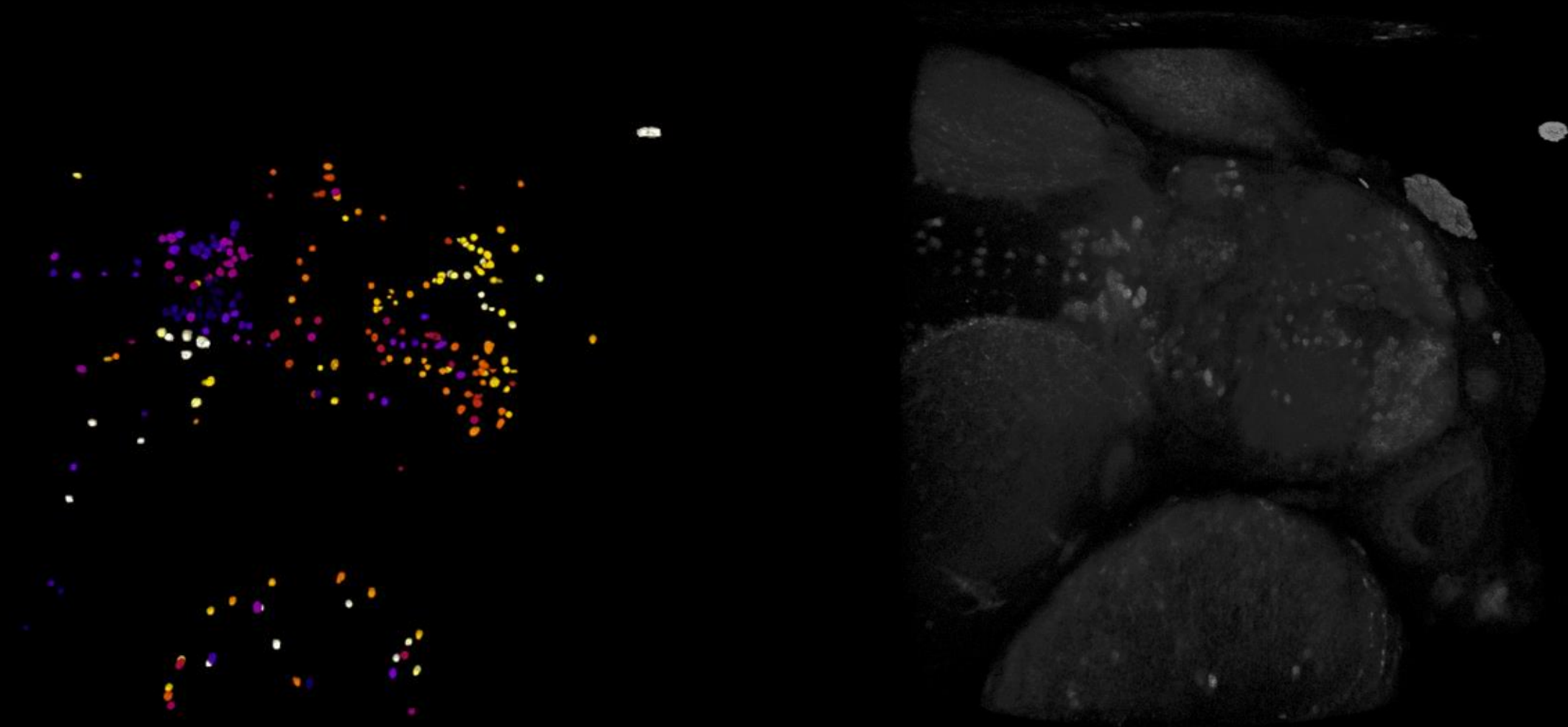


Fully convolutional network : U-Net





Automated 3D segmentation

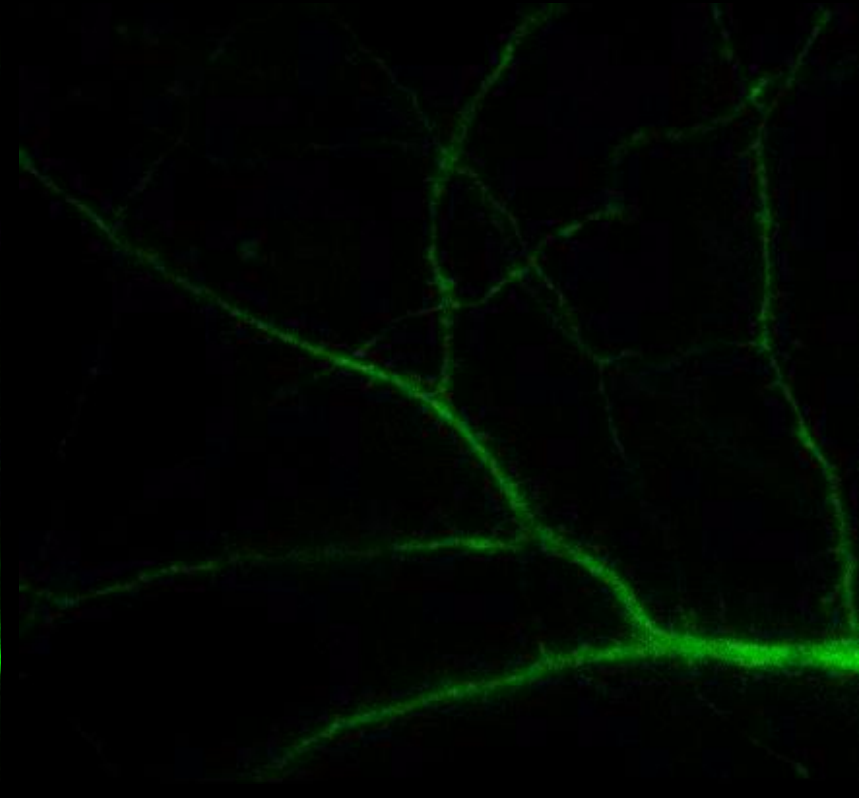
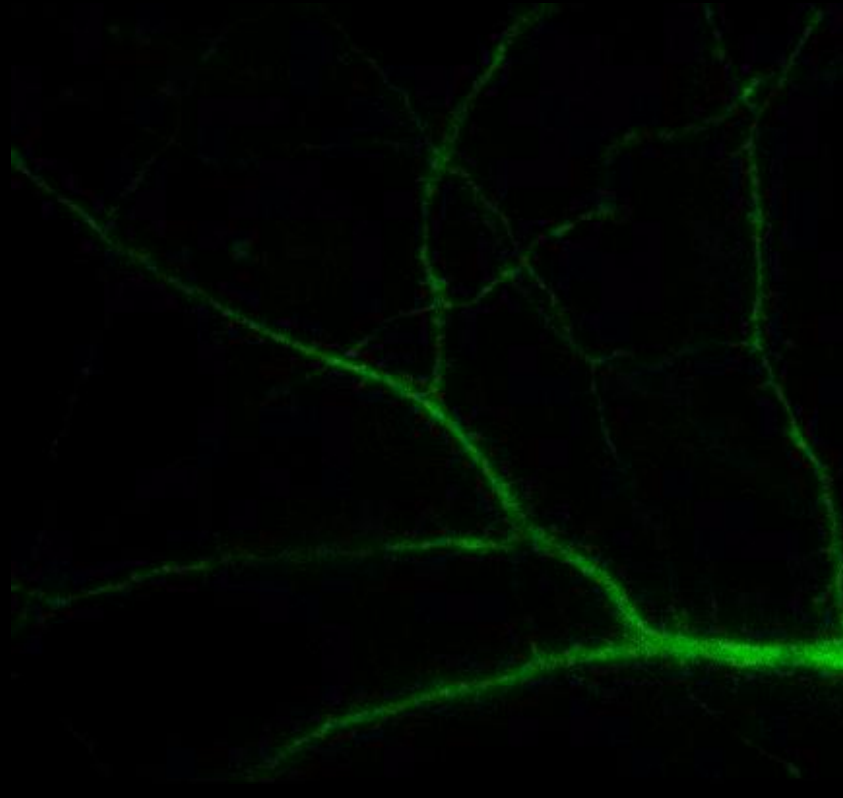
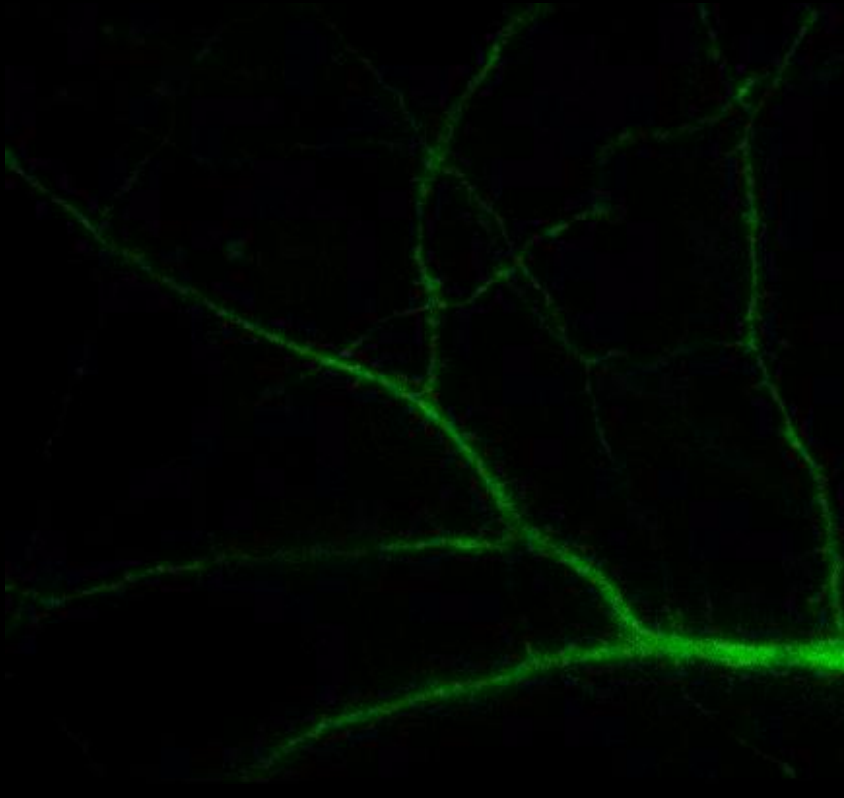
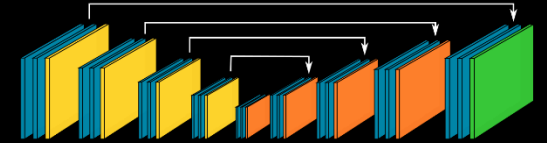


Detection of neuronal activity with a 3D neural network

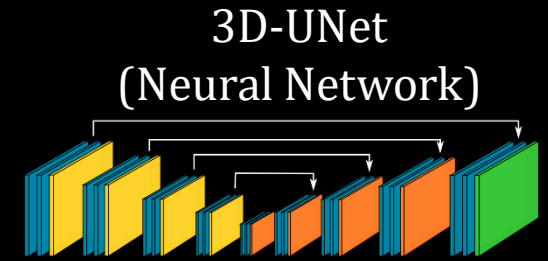
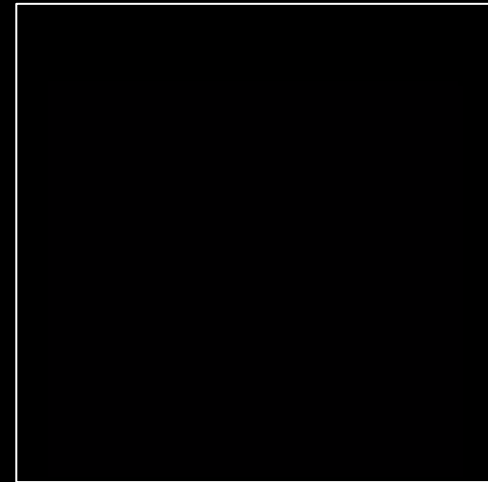
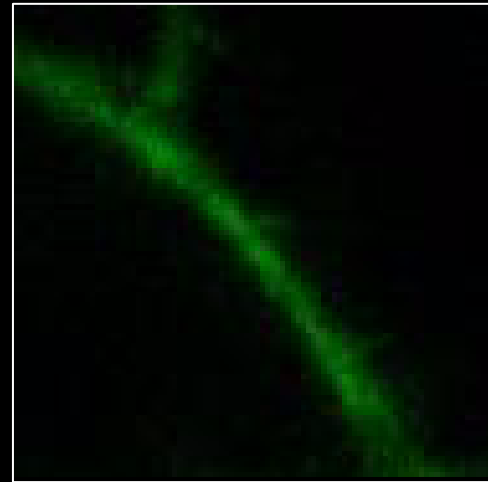
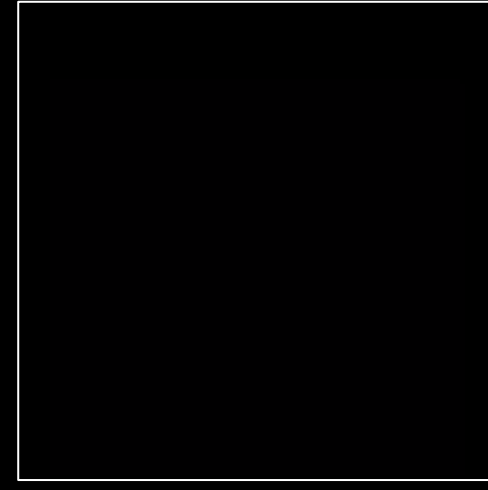
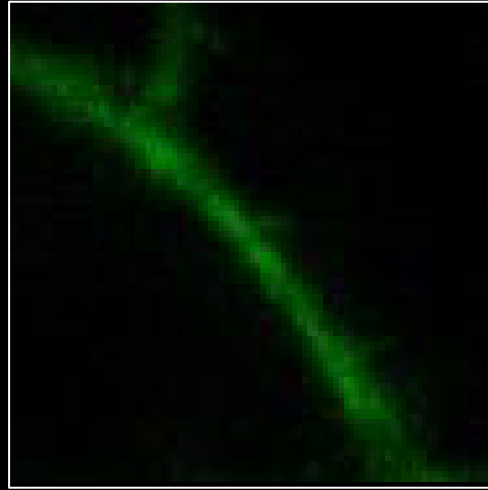
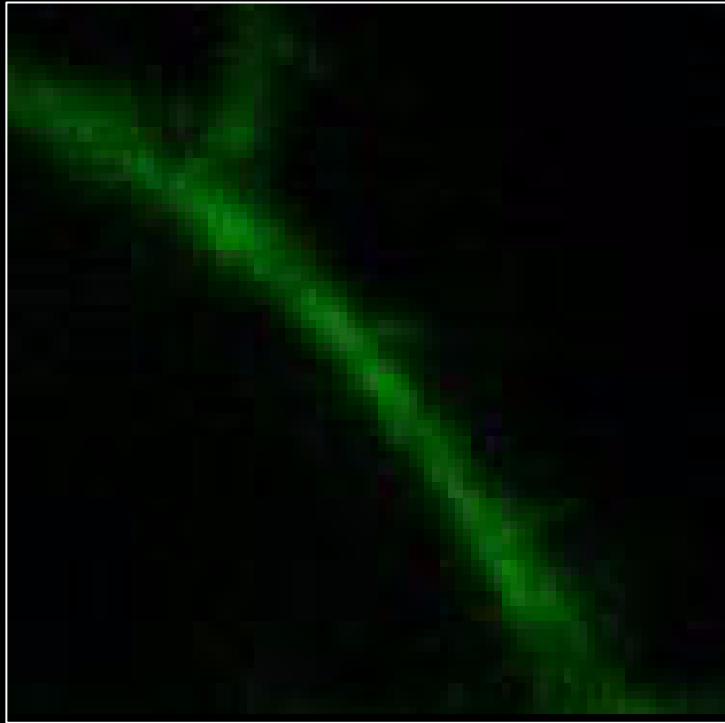
Widefield imaging – GCaMP6f
spontaneous miniature Ca^{2+} events

Foreground detection
(classical Matlab analysis)

3D-UNet
(Neural Network)



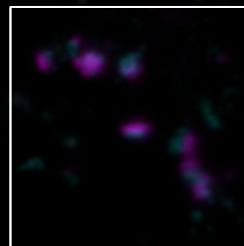
Increased segmentation precision with a 3D neural network



Foreground detection
(classical Matlab analysis)

Challenges

Metric evaluation



Bakas *et al.*, **Sci Data** (2017); Bakas *et al.*, **arXiv** (2019); Chen *et al.*, **ECCV** (2018); Menze *et al.*, **IEEE Trans Med Imaging** (2019); Taha and Hanbury, **BMC Med Imaging** (2019); Zhou *et al.*, **Array** (2019);

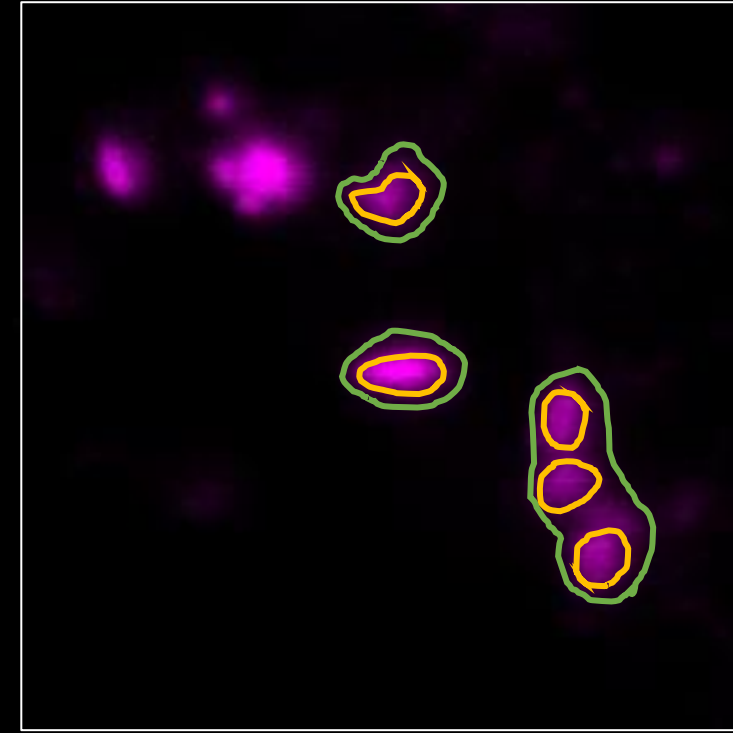
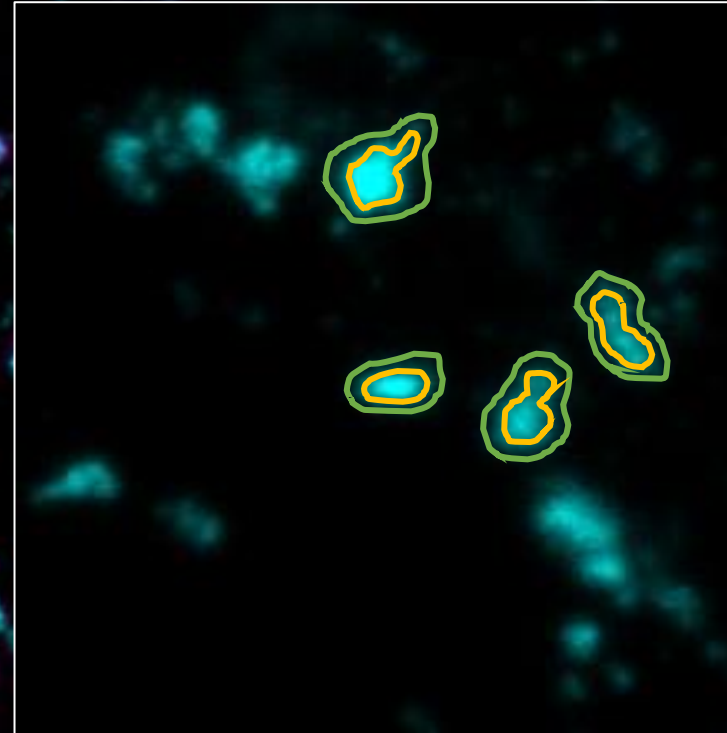
5 μm



Challenges

Metric evaluation

Bakas *et al.*, **Sci Data** (2017); Bakas *et al.*, **arXiv** (2019); Chen *et al.*, **ECCV** (2018); Menze *et al.*, **IEEE Trans Med Imaging** (2019); Taha and Hanbury, **BMC Med Imaging** (2019); Zhou *et al.*, **Array** (2019);

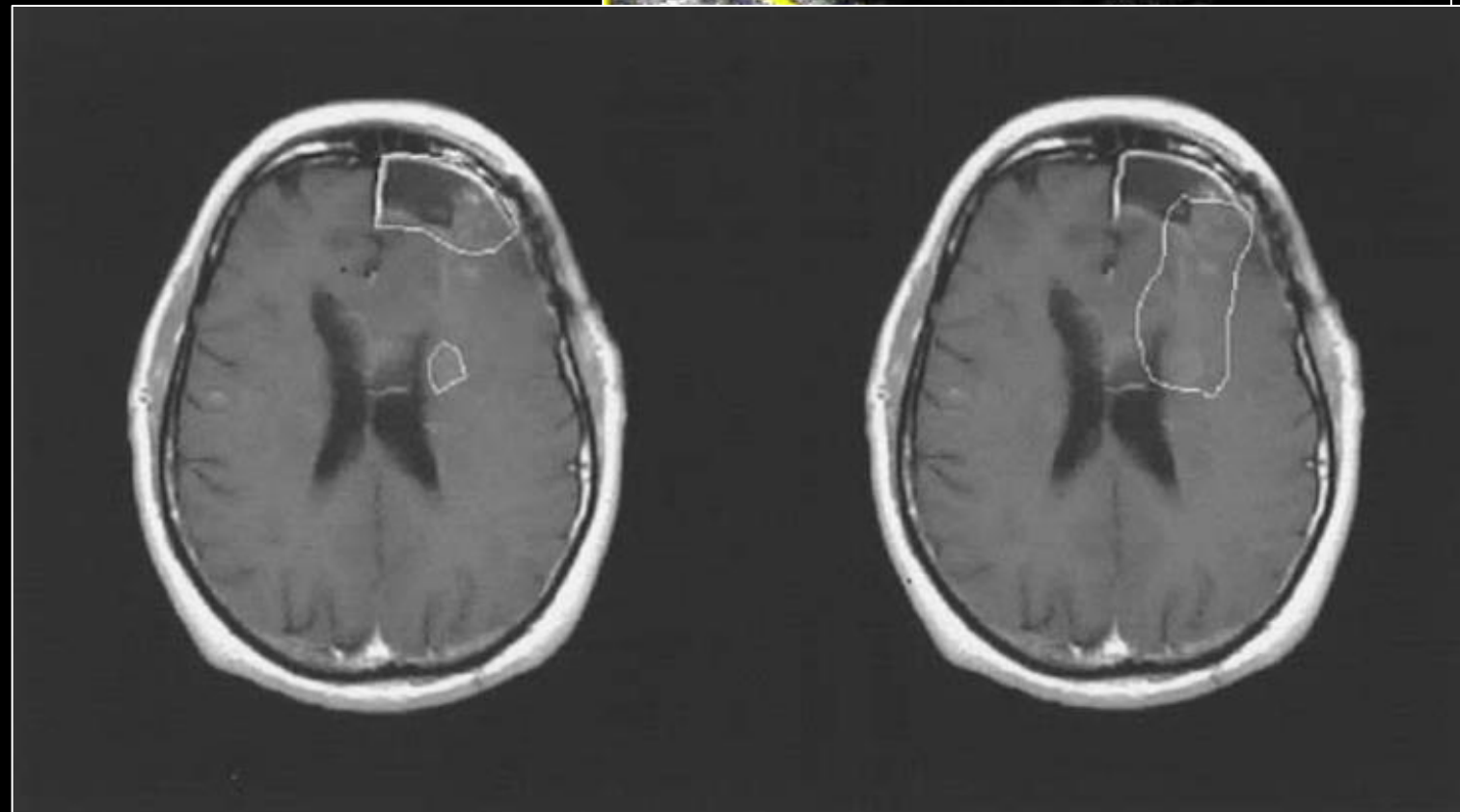


5 μm

Challenges

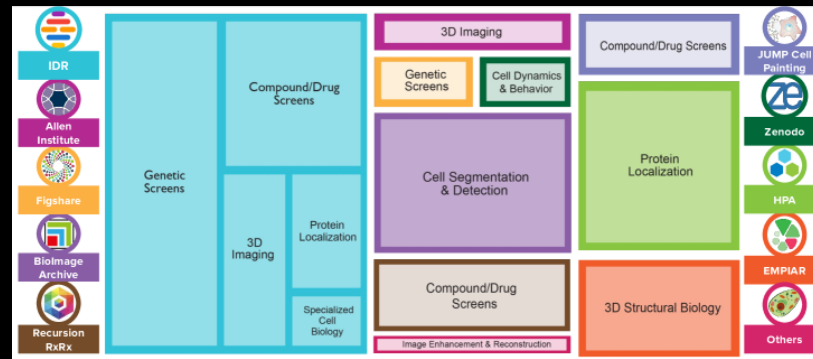
Expert variability

Mazzara *et al.*, J Rad Onco (2004);

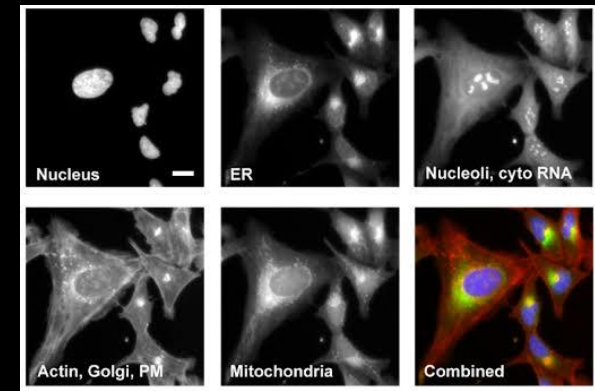


Challenges

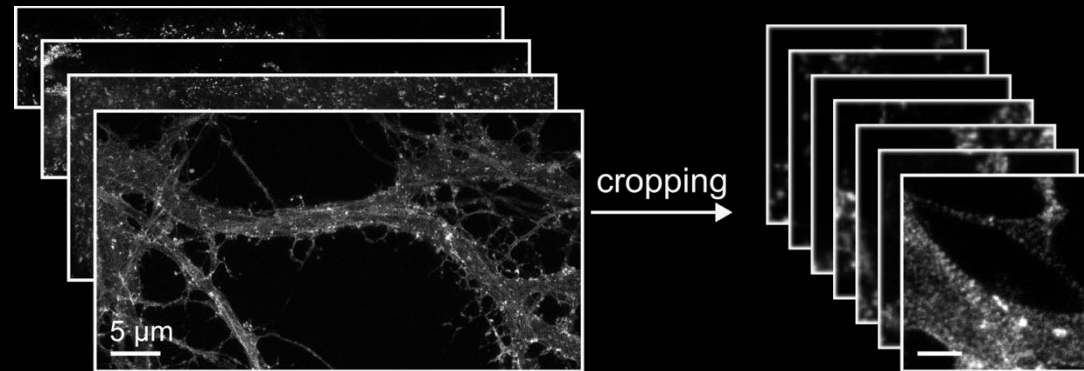
Datasets



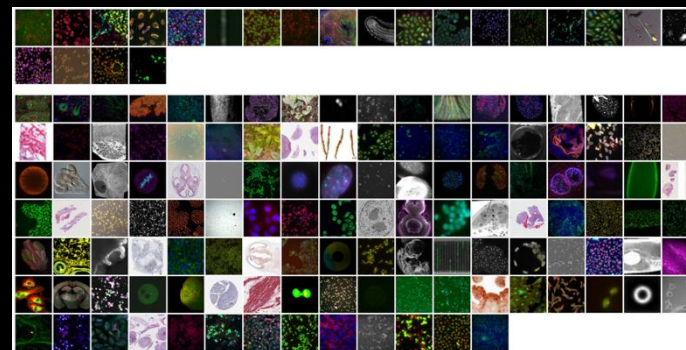
2026 : CHAMMI75 - Initial dataset of 26.7M images



Since 2023 : JUMP-CP
8M images



2025 : STED-FM 1M STED crops



Since 2017 : IDR – now around 14M images

Challenges

Generalization properties

